

Variational Boltzmann Calculation

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COLLISION INTEGRAL AND ITS PROPERTIES FOR THE ELECTRON-PHONON PROBLEM

For the collision integral we have

$$C_{\mathbf{k}} = -\frac{2\pi}{\hbar} \sum_{\mathbf{k}'} |M(\mathbf{k}, \mathbf{k}')|^2 f_{\mathbf{k}} (1 - f_{\mathbf{k}'}) (\delta(\epsilon_{\mathbf{k}} - \epsilon_{\mathbf{k}'} - \omega_{\mathbf{k}-\mathbf{k}'})(n_B(\omega_{\mathbf{k}-\mathbf{k}'} + 1) + \delta(\epsilon_{\mathbf{k}} - \epsilon_{\mathbf{k}'} + \omega_{\mathbf{k}-\mathbf{k}'})) n_B(\omega_{\mathbf{k}-\mathbf{k}'})) \\ + \frac{2\pi}{\hbar} |M(\mathbf{k}, \mathbf{k}')|^2 f_{\mathbf{k}'} (1 - f_{\mathbf{k}}) (\delta(\epsilon_{\mathbf{k}} - \epsilon_{\mathbf{k}'} - \omega_{\mathbf{k}-\mathbf{k}'})) n_B(\omega_{\mathbf{k}-\mathbf{k}'} + \delta(\epsilon_{\mathbf{k}} - \epsilon_{\mathbf{k}'} + \omega_{\mathbf{k}-\mathbf{k}'})(1 + n_B(\omega_{\mathbf{k}-\mathbf{k}'})))$$

simplifying

$$C_{\mathbf{k}} = \frac{2\pi}{\hbar} \sum_{\mathbf{k}'} |M(\mathbf{k}, \mathbf{k}')|^2 (\delta(\epsilon_{\mathbf{k}} - \epsilon_{\mathbf{k}'} - \omega_{\mathbf{k}-\mathbf{k}'})((f_{\mathbf{k}'} - f_{\mathbf{k}}) n_B(\omega_{\mathbf{k}-\mathbf{k}'} - f_{\mathbf{k}}(1 - f_{\mathbf{k}'}))) \\ + \frac{2\pi}{\hbar} |M(\mathbf{k}, \mathbf{k}')|^2 \delta(\epsilon_{\mathbf{k}} - \epsilon_{\mathbf{k}'} + \omega_{\mathbf{k}-\mathbf{k}'})((f_{\mathbf{k}'} - f_{\mathbf{k}}) n_B(\omega_{\mathbf{k}-\mathbf{k}'} + f_{\mathbf{k}'}(1 - f_{\mathbf{k}})))$$

Using the delta functions

$$C_{\mathbf{k}} = \frac{2\pi}{\hbar} \sum_{\mathbf{k}'} |M(\mathbf{k}, \mathbf{k}')|^2 (\delta(\epsilon_{\mathbf{k}} - \epsilon_{\mathbf{k}'} - \omega_{\mathbf{k}-\mathbf{k}'})((f_{\mathbf{k}'} - f_{\mathbf{k}}) n_B(\epsilon_{\mathbf{k}} - \epsilon_{\mathbf{k}'} - f_{\mathbf{k}}(1 - f_{\mathbf{k}'}))) \\ + \frac{2\pi}{\hbar} |M(\mathbf{k}, \mathbf{k}')|^2 \delta(\epsilon_{\mathbf{k}} - \epsilon_{\mathbf{k}'} + \omega_{\mathbf{k}-\mathbf{k}'}(-(f_{\mathbf{k}'} - f_{\mathbf{k}}) n_B(\epsilon_{\mathbf{k}} - \epsilon_{\mathbf{k}'} - f_{\mathbf{k}'} + f_{\mathbf{k}} + f_{\mathbf{k}'}(1 - f_{\mathbf{k}}))) \\ = \frac{2\pi}{\hbar} \sum_{\mathbf{k}'} |M(\mathbf{k}, \mathbf{k}')|^2 (\delta(\epsilon_{\mathbf{k}} - \epsilon_{\mathbf{k}'} - \omega_{\mathbf{k}-\mathbf{k}'})((f_{\mathbf{k}'} - f_{\mathbf{k}}) n_B(\epsilon_{\mathbf{k}} - \epsilon_{\mathbf{k}'} - f_{\mathbf{k}}(1 - f_{\mathbf{k}'}))) \\ + \frac{2\pi}{\hbar} |M(\mathbf{k}, \mathbf{k}')|^2 \delta(\epsilon_{\mathbf{k}} - \epsilon_{\mathbf{k}'} + \omega_{\mathbf{k}-\mathbf{k}'}(-(f_{\mathbf{k}'} - f_{\mathbf{k}}) n_B(\epsilon_{\mathbf{k}} - \epsilon_{\mathbf{k}'} + f_{\mathbf{k}}(1 - f_{\mathbf{k}}))) \\ = \frac{2\pi}{\hbar} \sum_{\mathbf{k}'} |M(\mathbf{k}, \mathbf{k}')|^2 (\delta(\epsilon_{\mathbf{k}} - \epsilon_{\mathbf{k}'} - \omega_{\mathbf{k}-\mathbf{k}'} - \delta(\epsilon_{\mathbf{k}} - \epsilon_{\mathbf{k}'} + \omega_{\mathbf{k}-\mathbf{k}'})) (n_B(\epsilon_{\mathbf{k}} - \epsilon_{\mathbf{k}'})(f_{\mathbf{k}'} - f_{\mathbf{k}}) - f_{\mathbf{k}}(1 - f_{\mathbf{k}'})))$$

The collision integral can be compactly expressed as

$$C_{\mathbf{k}} = \frac{2\pi}{\hbar} \sum_{\mathbf{k}'} |M(\mathbf{k}, \mathbf{k}')|^2 (\delta(\epsilon_{\mathbf{k}} - \epsilon_{\mathbf{k}'} - \omega_{\mathbf{k}-\mathbf{k}'} - \delta(\epsilon_{\mathbf{k}} - \epsilon_{\mathbf{k}'} + \omega_{\mathbf{k}-\mathbf{k}'})) (n_B(\epsilon_{\mathbf{k}} - \epsilon_{\mathbf{k}'})(f_{\mathbf{k}'} - f_{\mathbf{k}}) - f_{\mathbf{k}}(1 - f_{\mathbf{k}'})))$$

Even more compactly, we have

$$B(\mathbf{k}, \nu) = \frac{\pi \hbar}{2M\omega_{\mathbf{k}}} (\delta(\nu + \omega_{\mathbf{k}}) - \delta(\nu - \omega_{\mathbf{k}}))$$

so that

$$D(\mathbf{q}, \omega) = \int_{-\infty}^{\infty} \frac{B(\mathbf{q}, \omega')}{\omega' - \omega} \frac{d\omega'}{\pi} \quad (1)$$

where

$$D''(\mathbf{q}, \omega + i\delta) = \int_{-\infty}^{\infty} \pi \delta(\omega' - \omega) B(\mathbf{q}, \omega') \frac{d\omega'}{\pi} \\ = B(\mathbf{q}, \omega)$$

And we verify for the free function that

$$D_{ret}(\mathbf{q}, \omega) = \frac{\hbar}{2M\omega_{\mathbf{k}}} \left(\frac{2\omega_{\mathbf{k}}}{\omega^2 - \omega_{\mathbf{k}}^2} \right) \quad (2)$$

$$= \frac{\hbar}{2M\omega_{\mathbf{k}}} \left(\frac{1}{\omega + i\delta - \omega_{\mathbf{k}}} - \frac{1}{\omega + i\delta + \omega_{\mathbf{k}}} \right) \quad (3)$$

$$= \frac{\pi\hbar}{2M\omega_{\mathbf{k}}} \int \frac{d\nu}{\pi} \left(\frac{\delta(\nu - \omega_{\mathbf{k}})}{\omega + i\delta - \nu} - \frac{\delta(\nu + \omega_{\mathbf{k}})}{\omega + i\delta - \nu} \right) \quad (4)$$

$$= \frac{\pi\hbar}{2M\omega_{\mathbf{k}}} \int \frac{d\nu}{\pi} \frac{1}{\omega + i\delta - \nu} (\delta(\nu - \omega_{\mathbf{k}}) - \delta(\nu + \omega_{\mathbf{k}})) \quad (5)$$

$$= \frac{\pi\hbar}{2M\omega_{\mathbf{k}}} \int \frac{d\nu}{\pi} \frac{1}{\nu - (\omega + i\delta)} (\delta(\nu + \omega_{\mathbf{k}}) - \delta(\nu - \omega_{\mathbf{k}})) \quad (6)$$

$$= \int \frac{d\nu}{\pi} \frac{B(\mathbf{q}, \nu)}{\nu - (\omega + i\delta)} \quad (7)$$

this imaginary time propagator is the one that results when we take the definition of the retarded greens function obtained by analytically extending the imaginary time greens function

$$D(\mathbf{q}, \tau) = -\langle T_{\tau} \varphi_{\mathbf{q}}(\tau) \varphi_{-\mathbf{q}}(0) \rangle$$

$$B(\mathbf{q}, \nu) = \frac{\pi\hbar}{2M\omega_{\mathbf{k}}} (\delta(\nu + \omega_{\mathbf{k}}) - \delta(\nu - \omega_{\mathbf{k}}))$$

where D_{ret} corresponds to the comutator of the phonon fields and not the direct correlation function per se. we also have

$$|M(\mathbf{k}, \mathbf{k}')|^2 = \frac{\hbar\alpha^2 (\mathbf{k} - \mathbf{k}')^2}{2M\omega_{\mathbf{k}-\mathbf{k}'}}$$

if we define

$$g(\mathbf{k}, \mathbf{k}') = \alpha |\mathbf{k} - \mathbf{k}'|$$

and

$$C_{\mathbf{k}} = \frac{2}{\hbar} \sum_{\mathbf{k}'} |g(\mathbf{k}, \mathbf{k}')|^2 \mathcal{B}(\mathbf{k} - \mathbf{k}', \epsilon_{\mathbf{k}} - \epsilon_{\mathbf{k}'}) (n_B(\epsilon_{\mathbf{k}} - \epsilon_{\mathbf{k}'})(f_{\mathbf{k}'} - f_{\mathbf{k}}) - f_{\mathbf{k}}(1 - f_{\mathbf{k}'}))$$

Collision integral vanishes at equilibrium

The cancellation must happen between the in and out parts of the collision integral since there is on average zero changes on the state.

$$\begin{aligned} C_{\mathbf{k}} &= \frac{2\pi}{\hbar} \sum_{\mathbf{k}'} |M(\mathbf{k}, \mathbf{k}')|^2 (\delta(\epsilon_{\mathbf{k}} - \epsilon_{\mathbf{k}'} - \omega_{\mathbf{k}-\mathbf{k}'})((f_{\mathbf{k}'} - f_{\mathbf{k}}) n_B(\omega_{\mathbf{k}-\mathbf{k}'} - f_{\mathbf{k}}(1 - f_{\mathbf{k}'}))) \\ &\quad + \frac{2\pi}{\hbar} |M(\mathbf{k}, \mathbf{k}')|^2 \delta(\epsilon_{\mathbf{k}} - \epsilon_{\mathbf{k}'} + \omega_{\mathbf{k}-\mathbf{k}'})((f_{\mathbf{k}'} - f_{\mathbf{k}}) n_B(\omega_{\mathbf{k}-\mathbf{k}'} + f_{\mathbf{k}'}(1 - f_{\mathbf{k}}))) \\ &= \frac{2\pi}{\hbar} \sum_{\mathbf{k}'} |M(\mathbf{k}, \mathbf{k}')|^2 (\delta(\epsilon_{\mathbf{k}} - \epsilon_{\mathbf{k}'} - \omega_{\mathbf{k}-\mathbf{k}'})((f_{\mathbf{k}'} - f_{\mathbf{k}}) n_B(\epsilon_{\mathbf{k}} - \epsilon_{\mathbf{k}'} - f_{\mathbf{k}}(1 - f_{\mathbf{k}'}))) \\ &\quad + \frac{2\pi}{\hbar} |M(\mathbf{k}, \mathbf{k}')|^2 \delta(\epsilon_{\mathbf{k}} - \epsilon_{\mathbf{k}'} + \omega_{\mathbf{k}-\mathbf{k}'})((f_{\mathbf{k}'} - f_{\mathbf{k}}) n_B(\epsilon_{\mathbf{k}'} - \epsilon_{\mathbf{k}} + f_{\mathbf{k}'}(1 - f_{\mathbf{k}}))) \end{aligned}$$

We start by simplifying

$$\begin{aligned}
(f_y - f_z) n_{z-y} &= \left(\frac{1}{e^y + 1} - \frac{1}{e^z + 1} \right) \frac{1}{e^{z-y} - 1} \\
&= f_y f_z \frac{e^z - e^y}{e^{z-y} - 1} \\
&= e^y f_y f_z \\
&= (1 - f_y) f_z
\end{aligned}$$

so we have

$$(f_{\mathbf{k}'} - f_{\mathbf{k}}) n_B (\epsilon_{\mathbf{k}} - \epsilon_{\mathbf{k}'}) = f_{\mathbf{k}} (1 - f_{\mathbf{k}'})$$

at equilibrium.

Furthermore, we also have

$$\begin{aligned}
(f_y - f_z) n_{z-y} &= \left(\frac{1}{e^y + 1} - \frac{1}{e^z + 1} \right) \frac{1}{e^{y-z} - 1} \\
&= f_y f_z \frac{e^z - e^y}{e^{y-z} - 1} \\
&= -e^z f_y f_z \\
&= -(1 - f_z) f_y
\end{aligned}$$

so we have

$$(f_{\mathbf{k}'} - f_{\mathbf{k}}) n_B (\epsilon_{\mathbf{k}'} - \epsilon_{\mathbf{k}}) = -f_{\mathbf{k}'} (1 - f_{\mathbf{k}})$$

so we verify that the collision integral vanishes at equilibrium, since the coefficients in front of the delta functions cancel. We curiously have the following statistical transmutation

$$n_B (\epsilon_{\mathbf{k}'} - \epsilon_{\mathbf{k}}) = \frac{f_{\mathbf{k}'} (1 - f_{\mathbf{k}})}{f_{\mathbf{k}} (1 - f_{\mathbf{k}'}) - f_{\mathbf{k}'} (1 - f_{\mathbf{k}})}$$

Linearized collision integral for phonons

Linearizing the collision integral we have

$$f_{\mathbf{k}} = n_{F\mathbf{k}} - \frac{\partial n_{F\mathbf{k}}}{\partial \epsilon_{\mathbf{k}}} \phi_{\mathbf{k}}$$

then

$$\begin{aligned}
f_{\mathbf{k}'} (1 - f_{\mathbf{k}}) &= \left(n_{F\mathbf{k}'} - \frac{\partial n_{F\mathbf{k}'}}{\partial \epsilon_{\mathbf{k}'}} \phi_{\mathbf{k}'} \right) \left(1 - n_{F\mathbf{k}} + \frac{\partial n_{F\mathbf{k}}}{\partial \epsilon_{\mathbf{k}}} \phi_{\mathbf{k}} \right) \\
&= n_{F\mathbf{k}'} (1 - n_{F\mathbf{k}}) - \frac{\partial n_{F\mathbf{k}'}}{\partial \epsilon_{\mathbf{k}'}} \phi_{\mathbf{k}'} (1 - n_{F\mathbf{k}}) + n_{F\mathbf{k}'} \frac{\partial n_{F\mathbf{k}}}{\partial \epsilon_{\mathbf{k}}} \phi_{\mathbf{k}}
\end{aligned}$$

with

$$f_{\mathbf{k}} (1 - f_{\mathbf{k}'}) = n_{F\mathbf{k}} (1 - n_{F\mathbf{k}'}) - \frac{\partial n_{F\mathbf{k}}}{\partial \epsilon_{\mathbf{k}}} \phi_{\mathbf{k}} (1 - n_{F\mathbf{k}'}) + n_{F\mathbf{k}} \frac{\partial n_{F\mathbf{k}'}}{\partial \epsilon_{\mathbf{k}'}} \phi_{\mathbf{k}'}$$

and

$$(f_{\mathbf{k}'} - f_{\mathbf{k}}) = n_{F\mathbf{k}'} - n_{F\mathbf{k}} + \frac{\partial n_{F\mathbf{k}}}{\partial \epsilon_{\mathbf{k}}} \phi_{\mathbf{k}} - \frac{\partial n_{F\mathbf{k}'}}{\partial \epsilon_{\mathbf{k}'}} \phi_{\mathbf{k}'}$$

Since the collision integral vanishes at equilibrium, we have

$$C_{\mathbf{k}}^0 = \frac{2\pi}{\hbar} \sum_{\mathbf{k}'} |M(\mathbf{k}, \mathbf{k}')|^2 \delta(\epsilon_{\mathbf{k}} - \epsilon_{\mathbf{k}'} - \omega_{\mathbf{k}-\mathbf{k}'}) \left(\left(\frac{\partial n_{F\mathbf{k}}}{\partial \epsilon_{\mathbf{k}}} \phi_{\mathbf{k}} - \frac{\partial n_{F\mathbf{k}'}}{\partial \epsilon_{\mathbf{k}'}} \phi_{\mathbf{k}'} \right) n_B(\omega_{\mathbf{k}-\mathbf{k}'}) + \frac{\partial n_{F\mathbf{k}}}{\partial \epsilon_{\mathbf{k}}} \phi_{\mathbf{k}} (1 - n_{F\mathbf{k}'}) - n_{F\mathbf{k}} \frac{\partial n_{F\mathbf{k}'}}{\partial \epsilon_{\mathbf{k}'}} \phi_{\mathbf{k}'} \right) \\ + \frac{2\pi}{\hbar} |M(\mathbf{k}, \mathbf{k}')|^2 \delta(\epsilon_{\mathbf{k}} - \epsilon_{\mathbf{k}'} + \omega_{\mathbf{k}-\mathbf{k}'}) \left(\left(\frac{\partial n_{F\mathbf{k}}}{\partial \epsilon_{\mathbf{k}}} \phi_{\mathbf{k}} - \frac{\partial n_{F\mathbf{k}'}}{\partial \epsilon_{\mathbf{k}'}} \phi_{\mathbf{k}'} \right) n_B(\omega_{\mathbf{k}-\mathbf{k}'}) - \frac{\partial n_{F\mathbf{k}'}}{\partial \epsilon_{\mathbf{k}'}} \phi_{\mathbf{k}'} (1 - n_{F\mathbf{k}}) + n_{F\mathbf{k}'} \frac{\partial n_{F\mathbf{k}}}{\partial \epsilon_{\mathbf{k}}} \phi_{\mathbf{k}} \right)$$

Expressing things in terms of the distribution function

$$C_{\mathbf{k}}^0 = \frac{2\pi}{\hbar} \sum_{\mathbf{k}'} |M(\mathbf{k}, \mathbf{k}')|^2 \delta(\epsilon_{\mathbf{k}} - \epsilon_{\mathbf{k}'} - \omega_{\mathbf{k}-\mathbf{k}'}) \left(\left(\frac{\partial n_{F\mathbf{k}}}{\partial \epsilon_{\mathbf{k}}} \phi_{\mathbf{k}} - \frac{\partial n_{F\mathbf{k}'}}{\partial \epsilon_{\mathbf{k}'}} \phi_{\mathbf{k}'} \right) n_B(\epsilon_{\mathbf{k}} - \epsilon_{\mathbf{k}'}) + \frac{\partial n_{F\mathbf{k}}}{\partial \epsilon_{\mathbf{k}}} \phi_{\mathbf{k}} (1 - n_{F\mathbf{k}'}) - n_{F\mathbf{k}} \frac{\partial n_{F\mathbf{k}'}}{\partial \epsilon_{\mathbf{k}'}} \phi_{\mathbf{k}'} \right) \\ + \frac{2\pi}{\hbar} |M(\mathbf{k}, \mathbf{k}')|^2 \delta(\epsilon_{\mathbf{k}} - \epsilon_{\mathbf{k}'} + \omega_{\mathbf{k}-\mathbf{k}'}) \left(- \left(\frac{\partial n_{F\mathbf{k}}}{\partial \epsilon_{\mathbf{k}}} \phi_{\mathbf{k}} - \frac{\partial n_{F\mathbf{k}'}}{\partial \epsilon_{\mathbf{k}'}} \phi_{\mathbf{k}'} \right) n_B(\epsilon_{\mathbf{k}} - \epsilon_{\mathbf{k}'}) - \left(\frac{\partial n_{F\mathbf{k}}}{\partial \epsilon_{\mathbf{k}}} \phi_{\mathbf{k}} - \frac{\partial n_{F\mathbf{k}'}}{\partial \epsilon_{\mathbf{k}'}} \phi_{\mathbf{k}'} \right) - \frac{\partial n_{F\mathbf{k}'}}{\partial \epsilon_{\mathbf{k}'}} \phi_{\mathbf{k}'} (1 - n_{F\mathbf{k}}) \right) \\ \text{property of } n(-x) \\ = \frac{2\pi}{\hbar} \sum_{\mathbf{k}'} |M(\mathbf{k}, \mathbf{k}')|^2 \delta(\epsilon_{\mathbf{k}} - \epsilon_{\mathbf{k}'} - \omega_{\mathbf{k}-\mathbf{k}'}) \left(\left(\frac{\partial n_{F\mathbf{k}}}{\partial \epsilon_{\mathbf{k}}} \phi_{\mathbf{k}} - \frac{\partial n_{F\mathbf{k}'}}{\partial \epsilon_{\mathbf{k}'}} \phi_{\mathbf{k}'} \right) n_B(\epsilon_{\mathbf{k}} - \epsilon_{\mathbf{k}'}) + \frac{\partial n_{F\mathbf{k}}}{\partial \epsilon_{\mathbf{k}}} \phi_{\mathbf{k}} (1 - n_{F\mathbf{k}'}) - n_{F\mathbf{k}} \frac{\partial n_{F\mathbf{k}'}}{\partial \epsilon_{\mathbf{k}'}} \phi_{\mathbf{k}'} \right) \\ + \frac{2\pi}{\hbar} |M(\mathbf{k}, \mathbf{k}')|^2 \delta(\epsilon_{\mathbf{k}} - \epsilon_{\mathbf{k}'} + \omega_{\mathbf{k}-\mathbf{k}'}) \left(- \left(\frac{\partial n_{F\mathbf{k}}}{\partial \epsilon_{\mathbf{k}}} \phi_{\mathbf{k}} - \frac{\partial n_{F\mathbf{k}'}}{\partial \epsilon_{\mathbf{k}'}} \phi_{\mathbf{k}'} \right) n_B(\epsilon_{\mathbf{k}} - \epsilon_{\mathbf{k}'}) + \frac{\partial n_{F\mathbf{k}'}}{\partial \epsilon_{\mathbf{k}'}} \phi_{\mathbf{k}'} n_{F\mathbf{k}} - (1 - n_{F\mathbf{k}'}) \frac{\partial n_{F\mathbf{k}}}{\partial \epsilon_{\mathbf{k}}} \phi_{\mathbf{k}} \right)$$

factorizing the coefficient of deltas

$$= \frac{2\pi}{\hbar} \sum_{\mathbf{k}'} |M(\mathbf{k}, \mathbf{k}')|^2 [\delta(\epsilon_{\mathbf{k}} - \epsilon_{\mathbf{k}'} - \omega_{\mathbf{k}-\mathbf{k}'}) - \delta(\epsilon_{\mathbf{k}} - \epsilon_{\mathbf{k}'} + \omega_{\mathbf{k}-\mathbf{k}'})] \left(\left(\frac{\partial n_{F\mathbf{k}}}{\partial \epsilon_{\mathbf{k}}} \phi_{\mathbf{k}} - \frac{\partial n_{F\mathbf{k}'}}{\partial \epsilon_{\mathbf{k}'}} \phi_{\mathbf{k}'} \right) n_B(\epsilon_{\mathbf{k}} - \epsilon_{\mathbf{k}'}) + \frac{\partial n_{F\mathbf{k}}}{\partial \epsilon_{\mathbf{k}}} \phi_{\mathbf{k}} (1 - n_{F\mathbf{k}'}) - n_{F\mathbf{k}} \frac{\partial n_{F\mathbf{k}'}}{\partial \epsilon_{\mathbf{k}'}} \phi_{\mathbf{k}'} \right)$$

using def of phonon spectral funct

$$= \frac{2\pi}{\hbar} \sum_{\mathbf{k}'} |g(\mathbf{k}, \mathbf{k}')|^2 \mathcal{B}(\mathbf{k} - \mathbf{k}', \epsilon_{\mathbf{k}} - \epsilon_{\mathbf{k}'}) \left(\left(\frac{\partial n_{F\mathbf{k}}}{\partial \epsilon_{\mathbf{k}}} \phi_{\mathbf{k}} - \frac{\partial n_{F\mathbf{k}'}}{\partial \epsilon_{\mathbf{k}'}} \phi_{\mathbf{k}'} \right) n_B(\epsilon_{\mathbf{k}} - \epsilon_{\mathbf{k}'}) + \frac{\partial n_{F\mathbf{k}}}{\partial \epsilon_{\mathbf{k}}} \phi_{\mathbf{k}} (1 - n_{F\mathbf{k}'}) - n_{F\mathbf{k}} \frac{\partial n_{F\mathbf{k}'}}{\partial \epsilon_{\mathbf{k}'}} \phi_{\mathbf{k}'} \right)$$

taking derivatives explicitly (minus in front)

$$= -\beta \frac{2}{\hbar} \sum_{\mathbf{k}'} |g(\mathbf{k}, \mathbf{k}')|^2 \mathcal{B}(\mathbf{k} - \mathbf{k}', \epsilon_{\mathbf{k}} - \epsilon_{\mathbf{k}'}) ((n_{F\mathbf{k}} (1 - n_{F\mathbf{k}}) \phi_{\mathbf{k}} - n_{F\mathbf{k}'} (1 - n_{F\mathbf{k}'}) \phi_{\mathbf{k}'}) n_B(\epsilon_{\mathbf{k}} - \epsilon_{\mathbf{k}'}) + n_{F\mathbf{k}} (1 - n_{F\mathbf{k}}) \phi_{\mathbf{k}} (1 - n_{F\mathbf{k}'}) - n_{F\mathbf{k}'} (1 - n_{F\mathbf{k}'}) \phi_{\mathbf{k}'} (1 - n_{F\mathbf{k}}))$$

factoring out the specific combination of occ funcs

$$= -\beta \frac{2}{\hbar} \sum_{\mathbf{k}'} |g(\mathbf{k}, \mathbf{k}')|^2 \mathcal{B}(\mathbf{k} - \mathbf{k}', \epsilon_{\mathbf{k}} - \epsilon_{\mathbf{k}'}) n_{F\mathbf{k}'} (1 - n_{F\mathbf{k}}) n_B(\epsilon_{\mathbf{k}} - \epsilon_{\mathbf{k}'}) \left(\frac{n_{F\mathbf{k}}}{n_{F\mathbf{k}'}} \phi_{\mathbf{k}} - \frac{(1 - n_{F\mathbf{k}'})}{(1 - n_{F\mathbf{k}})} \phi_{\mathbf{k}'} + \phi_{\mathbf{k}} \frac{n_{F\mathbf{k}} (1 - n_{F\mathbf{k}'})}{n_{F\mathbf{k}'} n_B(\epsilon_{\mathbf{k}} - \epsilon_{\mathbf{k}'})} - \frac{(1 - n_{F\mathbf{k}'})}{(1 - n_{F\mathbf{k}})} \phi_{\mathbf{k}'} \right)$$

factorizing ϕ' s

$$= -\beta \frac{2}{\hbar} \sum_{\mathbf{k}'} |g(\mathbf{k}, \mathbf{k}')|^2 \mathcal{B}(\mathbf{k} - \mathbf{k}', \epsilon_{\mathbf{k}} - \epsilon_{\mathbf{k}'}) n_{F\mathbf{k}'} (1 - n_{F\mathbf{k}}) n_B(\epsilon_{\mathbf{k}} - \epsilon_{\mathbf{k}'}) \left(\frac{n_{F\mathbf{k}}}{n_{F\mathbf{k}'}} \left(1 + \frac{(1 - n_{F\mathbf{k}'})}{n_B(\epsilon_{\mathbf{k}} - \epsilon_{\mathbf{k}'})} \right) \phi_{\mathbf{k}} - \frac{(1 - n_{F\mathbf{k}'})}{(1 - n_{F\mathbf{k}})} \left(1 + \frac{(1 - n_{F\mathbf{k}'})}{n_B(\epsilon_{\mathbf{k}} - \epsilon_{\mathbf{k}'})} \right) \phi_{\mathbf{k}'} \right)$$

Now we simplify each of these terms individually

$$1 + \frac{(1 - n_{F\mathbf{k}'})}{n_B(\epsilon_{\mathbf{k}} - \epsilon_{\mathbf{k}'})} = 1 + \frac{e^y (e^{z-y} - 1)}{1 + e^y} \\ = 1 + \frac{(e^z - e^y)}{1 + e^y} \\ = \frac{1 + e^y + (e^z - e^y)}{1 + e^y} \\ = \frac{1 + e^z}{1 + e^y} = \frac{n_{f\mathbf{k}'}}{n_{F\mathbf{k}}}$$

Similarly, we have

$$\begin{aligned}
1 + \frac{n_{F\mathbf{k}}}{n_B(\epsilon_{\mathbf{k}} - \varepsilon_{\mathbf{k}'})} &= 1 + \frac{(e^{z-y} - 1)}{1 + e^z} \\
&= \frac{e^z + e^{z-y}}{1 + e^z} \\
&= \frac{e^z}{e^y} \frac{1 + e^y}{1 + e^z} = \frac{(1 - n_{F\mathbf{k}})}{(1 - n_{F\mathbf{k}'})}
\end{aligned}$$

Overall, we have then a linearize collision integral given by

$$C_{\mathbf{k}}^0 = \beta \frac{2}{\hbar} \sum_{\mathbf{k}'} |g(\mathbf{k}, \mathbf{k}')|^2 \mathcal{B}(\mathbf{k} - \mathbf{k}', \epsilon_{\mathbf{k}} - \varepsilon_{\mathbf{k}'}) n_{F\mathbf{k}'} (1 - n_{F\mathbf{k}}) n_B(\epsilon_{\mathbf{k}} - \varepsilon_{\mathbf{k}'}) (\phi_{\mathbf{k}'} - \phi_{\mathbf{k}})$$

In terms of a convolution over frequencies we have

$$C_{\mathbf{k}}^0 = \frac{2}{\hbar k T} \sum_{\mathbf{k}'} |g(\mathbf{k}, \mathbf{k}')|^2 \int d\omega \mathcal{B}(\mathbf{k} - \mathbf{k}', \omega) n_{F\mathbf{k}'} (1 - n_{F\mathbf{k}}) n_B(\omega) (\phi_{\mathbf{k}'} - \phi_{\mathbf{k}}) \delta(\epsilon_{\mathbf{k}} - \varepsilon_{\mathbf{k}'} - \omega)$$

which reproduces the Hlubina rice result of

$$W_{\mathbf{k}'\mathbf{k}} = \frac{2}{\hbar k T} |g(\mathbf{k}, \mathbf{k}')|^2 \mathcal{B}(\mathbf{k} - \mathbf{k}', \epsilon_{\mathbf{k}} - \varepsilon_{\mathbf{k}'}) n_{F\mathbf{k}'} (1 - n_{F\mathbf{k}}) n_B(\epsilon_{\mathbf{k}} - \varepsilon_{\mathbf{k}'})$$

Single particle rate from the collision integral

Starting with the single particle rate, we have

$$\tau_{\mathbf{k}}^{-1} = - \left(\frac{\delta C_{\mathbf{k}}}{\delta f_{\mathbf{k}}} \right)_{eq}$$

The explanation for this comes from the linearized collision integral for the Boltzmann equation

$$\begin{aligned}
\frac{dF_{\mathbf{k}}}{dt} &= C_{\mathbf{k}} \\
&= \sum_{\mathbf{q}} \frac{\delta C_{\mathbf{k}}}{\delta F_{\mathbf{q}}} \delta F_{\mathbf{q}}
\end{aligned}$$

considering an imbalance in a single momentum state we want to ask how fast it relaxes

$$\begin{aligned}
\frac{d\delta F_{\mathbf{k}}}{dt} &= C_{\mathbf{k}} \\
&= \frac{\delta C_{\mathbf{k}}}{\delta f_{\mathbf{q}}} \delta F_{\mathbf{k}}
\end{aligned}$$

so we have exponential relaxation given by

$$\tau_{\mathbf{k}}^{-1} = - \left(\frac{\delta C_{\mathbf{k}}}{\delta f_{\mathbf{k}}} \right)_{eq}$$

The single particle rate is

$$\begin{aligned}
\left(\frac{\delta C_{\mathbf{k}}}{\delta f_{\mathbf{k}}}\right)_{eq} &= \frac{2\pi}{\hbar} \sum_{\mathbf{k}'} |M(\mathbf{k}, \mathbf{k}')|^2 (\delta(\epsilon_{\mathbf{k}} - \varepsilon_{\mathbf{k}'} - \omega_{\mathbf{k}-\mathbf{k}'})) ((\delta_{\mathbf{k}'\mathbf{k}} - 1) n_B(\omega_{\mathbf{k}-\mathbf{k}'} - (1 - f_{\mathbf{k}'})) + f_{\mathbf{k}} \delta_{\mathbf{k}\mathbf{k}'}) \\
&\quad + \frac{2\pi}{\hbar} |M(\mathbf{k}, \mathbf{k}')|^2 \delta(\epsilon_{\mathbf{k}} - \varepsilon_{\mathbf{k}'} + \omega_{\mathbf{k}-\mathbf{k}'}) ((\delta_{\mathbf{k}'\mathbf{k}} - 1) n_B(\omega_{\mathbf{k}-\mathbf{k}'} - f_{\mathbf{k}'} + (1 - f_{\mathbf{k}}) \delta_{\mathbf{k}\mathbf{k}'})) \\
&= -\frac{2\pi}{\hbar} \sum_{\mathbf{k}'} |M(\mathbf{k}, \mathbf{k}')|^2 (\delta(\epsilon_{\mathbf{k}} - \varepsilon_{\mathbf{k}'} - \omega_{\mathbf{k}-\mathbf{k}'})) (n_B(\omega_{\mathbf{k}-\mathbf{k}'} + 1 - f_{\mathbf{k}'}) \\
&\quad - \frac{2\pi}{\hbar} \sum_{\mathbf{k}'} |M(\mathbf{k}, \mathbf{k}')|^2 \delta(\epsilon_{\mathbf{k}} - \varepsilon_{\mathbf{k}'} + \omega_{\mathbf{k}-\mathbf{k}'}) (n_B(\omega_{\mathbf{k}-\mathbf{k}'} + f_{\mathbf{k}'}) \\
&\quad + \sum_{\mathbf{k}'} \frac{2\pi}{\hbar} \delta_{\mathbf{k}'\mathbf{k}} |M(\mathbf{k}, \mathbf{k}')|^2 (\delta(\epsilon_{\mathbf{k}} - \varepsilon_{\mathbf{k}'} - \omega_{\mathbf{k}-\mathbf{k}'})) (n_B(\epsilon_{\mathbf{k}} - \varepsilon_{\mathbf{k}'} + f_{\mathbf{k}})) \\
&\quad + \sum_{\mathbf{k}'} \frac{2\pi}{\hbar} \delta_{\mathbf{k}'\mathbf{k}} |M(\mathbf{k}, \mathbf{k}')|^2 \delta(\epsilon_{\mathbf{k}} - \varepsilon_{\mathbf{k}'} + \omega_{\mathbf{k}-\mathbf{k}'}) (n_B(\varepsilon_{\mathbf{k}'} - \epsilon_{\mathbf{k}} + (1 - f_{\mathbf{k}})) \\
&= -\tau_{\mathbf{k}}^{-1} \\
&\quad + \sum_{\mathbf{k}'} \frac{2\pi}{\hbar} \delta_{\mathbf{k}'\mathbf{k}} |M(\mathbf{k}, \mathbf{k}')|^2 (\delta(\epsilon_{\mathbf{k}} - \varepsilon_{\mathbf{k}'} - \omega_{\mathbf{k}-\mathbf{k}'})) (n_B(\epsilon_{\mathbf{k}} - \varepsilon_{\mathbf{k}'} + f_{\mathbf{k}})) \\
&\quad + \sum_{\mathbf{k}'} \frac{2\pi}{\hbar} \delta_{\mathbf{k}'\mathbf{k}} |M(\mathbf{k}, \mathbf{k}')|^2 \delta(\epsilon_{\mathbf{k}} - \varepsilon_{\mathbf{k}'} + \omega_{\mathbf{k}-\mathbf{k}'}) (-n_B(\epsilon_{\mathbf{k}} - \varepsilon_{\mathbf{k}'} - f_{\mathbf{k}})) \\
&= -\tau_{\mathbf{k}}^{-1} \\
&\quad + \sum_{\mathbf{k}'} \frac{2\pi}{\hbar} \delta_{\mathbf{k}'\mathbf{k}} |M(\mathbf{k}, \mathbf{k}')|^2 \delta(\omega_{\mathbf{k}-\mathbf{k}'}) (n_B(0^+) + f_{\mathbf{k}}) \\
&\quad + \sum_{\mathbf{k}'} \frac{2\pi}{\hbar} \delta_{\mathbf{k}'\mathbf{k}} |M(\mathbf{k}, \mathbf{k}')|^2 \delta(\omega_{\mathbf{k}-\mathbf{k}'}) (-n_B(0^+) - f_{\mathbf{k}}) \\
&= -\tau_{\mathbf{k}}^{-1}
\end{aligned}$$

Such that

$$\begin{aligned}
\tau_{\mathbf{k}}^{-1} &= \frac{2\pi}{\hbar} \sum_{\mathbf{k}'} |M(\mathbf{k}, \mathbf{k}')|^2 (\delta(\epsilon_{\mathbf{k}} - \varepsilon_{\mathbf{k}'} - \omega_{\mathbf{k}-\mathbf{k}'})) (n_B(\omega_{\mathbf{k}-\mathbf{k}'} + 1 - f_{\mathbf{k}'}) \\
&\quad + \frac{2\pi}{\hbar} \sum_{\mathbf{k}'} |M(\mathbf{k}, \mathbf{k}')|^2 \delta(\epsilon_{\mathbf{k}} - \varepsilon_{\mathbf{k}'} + \omega_{\mathbf{k}-\mathbf{k}'}) (n_B(\omega_{\mathbf{k}-\mathbf{k}'} + f_{\mathbf{k}'}))
\end{aligned}$$

Another calculation of the single particle rate starting from the reduced form of the collision integral

$$C_{\mathbf{k}} = \frac{2}{\hbar} \sum_{\mathbf{k}'} |g(\mathbf{k}, \mathbf{k}')|^2 \mathcal{B}(\mathbf{k} - \mathbf{k}', \epsilon_{\mathbf{k}} - \varepsilon_{\mathbf{k}'}) (n_B(\epsilon_{\mathbf{k}} - \varepsilon_{\mathbf{k}'})) (f_{\mathbf{k}'} - f_{\mathbf{k}}) - f_{\mathbf{k}} (1 - f_{\mathbf{k}'})$$

taking

$$\begin{aligned}
\left(\frac{\delta C_{\mathbf{k}}}{\delta f_{\mathbf{k}}}\right)_{eq} &= \frac{2}{\hbar} \sum_{\mathbf{k}'} |g(\mathbf{k}, \mathbf{k}')|^2 \mathcal{B}(\mathbf{k} - \mathbf{k}', \epsilon_{\mathbf{k}} - \epsilon_{\mathbf{k}'}) (n_B(\epsilon_{\mathbf{k}} - \epsilon_{\mathbf{k}'}) (\delta_{\mathbf{k}\mathbf{k}'} - 1) - (1 - f_{\mathbf{k}'}) + f_{\mathbf{k}} \delta_{\mathbf{k}\mathbf{k}'}) \\
&= \frac{2}{\hbar} \sum_{\mathbf{k}'} |g(\mathbf{k}, \mathbf{k}')|^2 \mathcal{B}(\mathbf{k} - \mathbf{k}', \epsilon_{\mathbf{k}} - \epsilon_{\mathbf{k}'}) (n_B(\epsilon_{\mathbf{k}} - \epsilon_{\mathbf{k}'}) (-1) - (1 - f_{\mathbf{k}'})) \\
&\quad + \frac{2}{\hbar} \sum_{\mathbf{k}'} |g(\mathbf{k}, \mathbf{k}')|^2 \mathcal{B}(\mathbf{k} - \mathbf{k}', \epsilon_{\mathbf{k}} - \epsilon_{\mathbf{k}'}) (n_B(\epsilon_{\mathbf{k}} - \epsilon_{\mathbf{k}'}) \delta_{\mathbf{k}\mathbf{k}'} + f_{\mathbf{k}} \delta_{\mathbf{k}\mathbf{k}'})
\end{aligned}$$

since the spectral function has to vanish at zero frequency, we have

$$\left(\frac{\delta C_{\mathbf{k}}}{\delta f_{\mathbf{k}}}\right)_{eq} = -\frac{2}{\hbar} \sum_{\mathbf{k}'} |g(\mathbf{k}, \mathbf{k}')|^2 \mathcal{B}(\mathbf{k} - \mathbf{k}', \epsilon_{\mathbf{k}} - \epsilon_{\mathbf{k}'}) (n_B(\epsilon_{\mathbf{k}} - \epsilon_{\mathbf{k}'}) + f(-\epsilon_{\mathbf{k}'}))$$

which coincides with the diagram calculation. The 2 comes from the relation between self energy and qp rate and the $\hbar$ comes from the conversion from energy to rate.

RANDOM RELATIONS BETWEEN KERNELS AND STUFF

Entropy generation Kernel near minima

asdf

We have

$$W = kT \sum_{\mathbf{k}} \log\left(\frac{1}{F_{\mathbf{k}}} - 1\right) C_{\mathbf{k}}$$

Taking one functional derivative

$$\begin{aligned}
\frac{\delta W}{\delta F_{\mathbf{p}}} &= kT \sum_{\mathbf{k}} \log\left(\frac{1}{F_{\mathbf{k}}} - 1\right) \frac{\delta C_{\mathbf{k}}}{\delta F_{\mathbf{p}}} - \frac{\delta F_{\mathbf{k}}}{\delta F_{\mathbf{p}}} \frac{1}{F_{\mathbf{k}}^2} \frac{1}{\frac{1}{F_{\mathbf{k}}} - 1} C_{\mathbf{k}} \\
&= kT \sum_{\mathbf{k}} \log\left(\frac{1}{F_{\mathbf{k}}} - 1\right) \frac{\delta C_{\mathbf{k}}}{\delta F_{\mathbf{p}}} - \delta_{\mathbf{k}\mathbf{p}} \frac{1}{F_{\mathbf{k}}} \frac{1}{1 - F_{\mathbf{k}}} C_{\mathbf{k}}
\end{aligned}$$

if we evaluate at the fermi distribution

$$\begin{aligned}
\frac{\delta W}{\delta F_{\mathbf{p}}} &= kT \sum_{\mathbf{k}} \beta(\epsilon_{\mathbf{k}} - \mu) \frac{\delta C_{\mathbf{k}}}{\delta F_{\mathbf{p}}} \\
&= \frac{\delta}{\delta F_{\mathbf{p}}} \sum_{\mathbf{k}} (\epsilon_{\mathbf{k}} - \mu) C_{\mathbf{k}} \\
&= 0
\end{aligned}$$

since collision locally conserve energy and particle number

$$\sum_{\mathbf{k}} C_{\mathbf{k}} = 0$$

and

$$\sum_{\mathbf{k}} \epsilon_{\mathbf{k}} C_{\mathbf{k}} = 0$$

this follows from the total rate of change of the total number of particles and the total rate of change of the total energy of the system.

Now we take the second derivative

$$\begin{aligned} \frac{\delta W}{\delta F_p \delta F_q} &= kT \sum_k \log \left(\frac{1}{F_k} - 1 \right) \frac{\delta C}{\delta F_p \delta F_q} - \delta_{kp} \frac{1}{F_k} \frac{1}{1 - F_k} \frac{\delta C_k}{\delta F_p} \\ &\quad - kT \sum_k \delta_{kp} \frac{1}{F_k} \frac{1}{1 - F_k} \frac{\delta C_k}{\delta F_q} - \delta_{kp} \left(\frac{\delta}{\delta F_q} \frac{1}{F_k} \frac{1}{1 - F_k} \right) C_k \end{aligned}$$

Evaluating at equilibrium the first and the last term vanish for the same reason as above, we are left with

$$-\frac{1}{kT} \frac{\delta W}{\delta F_p \delta F_q} = \frac{1}{f_p} \frac{1}{1 - f_p} \frac{\delta C_p}{\delta f_q} + \frac{1}{f_q} \frac{1}{1 - f_q} \frac{\delta C_q}{\delta f_p}$$

So that

$$\begin{aligned} \frac{1}{2} \frac{\delta W}{\delta F_p \delta F_q} &= \frac{1}{2} \left(\frac{1}{\partial f_p / \partial \epsilon_p} \frac{\delta C_p}{\delta f_q} + \frac{1}{\partial f_q / \partial \epsilon_q} \frac{\delta C_q}{\delta f_p} \right) \\ &\equiv \frac{1}{2} S(\mathbf{q}, \mathbf{p}) \end{aligned}$$

General results for the Linearized Boltzman eqn

we start by defining the functional only to second order

$$\begin{aligned} W &= \frac{1}{2} \sum_{qp} \delta F_q \frac{\delta W}{\delta F_p \delta F_q} \delta F_p \\ &= \frac{1}{2} \sum_{qp} \delta F_q S(\mathbf{q}, \mathbf{p}) \delta F_p \end{aligned}$$

Now we want to minimize this functional at fixed current, which minimizes the rate of entropy production. At the same time would maximize the rate of entropy production at fixed E field (not very clear why to me). Optimizing the functional and fixing the current, we have

$$\frac{\delta}{\delta F_k} (W - \boldsymbol{\lambda} \cdot \mathbf{J}) = 0$$

where the definition of the current in terms of the distribution function is given by

$$\mathbf{J} = \frac{e}{V} \sum_q \mathbf{v}(\mathbf{q}) F_q$$

this is a key step conceptually. Without this additional current term, we know the optimum of the functional is the equilibrium distribution. The addition of this linear term shifts the minimum slightly.

Taking the derivative we get

$$\frac{\delta}{\delta F_k} \mathbf{J} = \frac{-e}{V} \mathbf{v}(\mathbf{k})$$

$$\begin{aligned} \frac{\delta W}{\delta F_k} &= \frac{1}{2} \sum_{qp} \delta_{qk} S(\mathbf{q}, \mathbf{p}) \delta F_p + \frac{1}{2} \sum_{qp} \delta F_q S(\mathbf{q}, \mathbf{p}) \delta_{pk} \\ &= \frac{1}{2} \sum_p S(\mathbf{k}, \mathbf{p}) \delta F_p + \frac{1}{2} \sum_{qp} \delta F_q S(\mathbf{q}, \mathbf{k}) \\ &= \sum_p S(\mathbf{k}, \mathbf{p}) \delta F_p \end{aligned}$$

Overall we find

$$\sum_{\mathbf{p}} S(\mathbf{k}, \mathbf{p}) \delta F_{\mathbf{p}}^* = \frac{-e}{V} \boldsymbol{\lambda} \cdot \mathbf{v}(\mathbf{k})$$

where the star denotes the value at the saddle point. Inserting this back in the functional

$$\begin{aligned} W^* &= \frac{1}{2} \sum_{\mathbf{q}\mathbf{p}} \delta F_{\mathbf{q}}^* S(\mathbf{q}, \mathbf{p}) \delta F_{\mathbf{p}}^* \\ &= \frac{1}{2} \sum_{\mathbf{q}} \delta F_{\mathbf{q}}^* \left(\frac{-e}{V} \boldsymbol{\lambda} \cdot \mathbf{v}(\mathbf{q}) \right) \\ &= \frac{1}{2} \frac{-e}{V} \boldsymbol{\lambda} \cdot \sum_{\mathbf{q}} \delta F_{\mathbf{q}}^* \mathbf{v}(\mathbf{q}) \\ &= \frac{1}{2} \boldsymbol{\lambda} \cdot \mathbf{J}^* \end{aligned}$$

Since this functional is supposed to match the rate of entropy production, given by

$$T\dot{S} = V \mathbf{E} \cdot \mathbf{J}$$

we have the expression for the lagrange multiplier.

$$\boldsymbol{\lambda} = 2V \mathbf{E}.$$

We can then relate the kernel of the functional to the Boltzmann equation

$$\frac{dF_{\mathbf{k}}}{dt} = C_{\mathbf{k}}$$

by linearizing this equation, we find

$$\frac{dF_{\mathbf{k}}}{dt} \sim -e \mathbf{E} \cdot \mathbf{v}_{\mathbf{k}} \frac{\partial f}{\partial \epsilon_{\mathbf{k}}}$$

where we only kept terms linear in the field, there is no explicit time or space dependence, and because $\dot{\mathbf{k}} = -e \mathbf{E}$, we only need to retain $\nabla_{\mathbf{k}} F_{\mathbf{k}} = \mathbf{v}_{\mathbf{k}} \frac{\partial f}{\partial \epsilon_{\mathbf{k}}} + \mathcal{O}(\mathbf{E})$.

Linearizing the collision operator as well, we find

$$-e \mathbf{E} \cdot \mathbf{v}_{\mathbf{k}} \frac{\partial f}{\partial \epsilon_{\mathbf{k}}} = \sum_{\mathbf{q}} \frac{\delta C_{\mathbf{k}}}{\delta F_{\mathbf{q}}} \delta F_{\mathbf{q}}$$

comparing with

$$-e \mathbf{E} \cdot \mathbf{v}_{\mathbf{k}} = \frac{1}{2} \sum_{\mathbf{p}} S(\mathbf{k}, \mathbf{p}) \delta F_{\mathbf{p}}$$

we find

$$\frac{1}{2} S(\mathbf{k}, \mathbf{p}) = \frac{1}{\partial f_{\mathbf{k}} / \partial \epsilon_{\mathbf{k}}} \frac{\delta C_{\mathbf{k}}}{\delta F_{\mathbf{q}}}$$

however, we already had

$$\begin{aligned} S(\mathbf{q}, \mathbf{p}) &= \frac{1}{\partial f_{\mathbf{k}} / \partial \epsilon_{\mathbf{k}}} \frac{\delta C_{\mathbf{k}}}{\delta F_{\mathbf{q}}} + \frac{1}{\partial f_{\mathbf{p}} / \partial \epsilon_{\mathbf{p}}} \frac{\delta C_{\mathbf{p}}}{\delta f_{\mathbf{q}}} \\ &= \frac{1}{2} S(\mathbf{q}, \mathbf{p}) + \frac{1}{\partial f_{\mathbf{p}} / \partial \epsilon_{\mathbf{p}}} \frac{\delta C_{\mathbf{p}}}{\delta f_{\mathbf{q}}} \end{aligned}$$

From which we conclude that

$$\frac{1}{2}S(\mathbf{q}, \mathbf{p}) = \frac{1}{\partial f_{\mathbf{q}}/\partial \epsilon_{\mathbf{q}}} \frac{\delta C_{\mathbf{q}}}{\delta f_{\mathbf{p}}} = \frac{1}{\partial f_{\mathbf{p}}/\partial \epsilon_{\mathbf{p}}} \frac{\delta C_{\mathbf{p}}}{\delta f_{\mathbf{q}}}$$

From the symmetry of $S(\mathbf{q}, \mathbf{p})$ we conclude that the expression

$$\frac{1}{\partial f_{\mathbf{q}}/\partial \epsilon_{\mathbf{q}}} \frac{\delta C_{\mathbf{q}}}{\delta f_{\mathbf{p}}}$$

is also symmetric. We then define

$$\begin{aligned} Q_{\mathbf{k}\mathbf{p}} &= \frac{1}{2} \frac{\partial f_{\mathbf{k}}}{\partial \epsilon_{\mathbf{k}}} \frac{\partial f_{\mathbf{p}}}{\partial \epsilon_{\mathbf{p}}} S(\mathbf{k}, \mathbf{p}) \\ &= \frac{1}{2} \frac{\partial f_{\mathbf{k}}}{\partial \epsilon_{\mathbf{k}}} \frac{\delta C_{\mathbf{p}}}{\delta f_{\mathbf{k}}} \end{aligned}$$

Phonon kernel

starting from the collision integral, we have

$$C_{\mathbf{k}} = \frac{2}{\hbar} \sum_{\mathbf{k}'} |g(\mathbf{k}, \mathbf{k}')|^2 \mathcal{B}(\mathbf{k} - \mathbf{k}', \epsilon_{\mathbf{k}} - \epsilon_{\mathbf{k}'}) (n_B(\epsilon_{\mathbf{k}} - \epsilon_{\mathbf{k}'})(f_{\mathbf{k}'} - f_{\mathbf{k}}) - f_{\mathbf{k}}(1 - f_{\mathbf{k}'}))$$

For the kernel we have

$$\begin{aligned} \frac{\delta C_{\mathbf{k}}}{\delta f_{\mathbf{p}}} &= \frac{2}{\hbar} \sum_{\mathbf{k}'} |g(\mathbf{k}, \mathbf{k}')|^2 \mathcal{B}(\mathbf{k} - \mathbf{k}', \epsilon_{\mathbf{k}} - \epsilon_{\mathbf{k}'}) (n_B(\epsilon_{\mathbf{k}} - \epsilon_{\mathbf{k}'})(\delta_{\mathbf{p}\mathbf{k}'} - \delta_{\mathbf{p}\mathbf{k}}) - \delta_{\mathbf{p}\mathbf{k}}(1 - f_{\mathbf{k}'})) + f_{\mathbf{k}}\delta_{\mathbf{p}\mathbf{k}'} \\ &= \frac{2}{\hbar} |g(\mathbf{k}, \mathbf{p})|^2 \mathcal{B}(\mathbf{k} - \mathbf{p}, \epsilon_{\mathbf{k}} - \epsilon_{\mathbf{p}}) (n_B(\epsilon_{\mathbf{k}} - \epsilon_{\mathbf{p}}) + f_{\mathbf{k}}) \\ &\quad - \delta_{\mathbf{p}\mathbf{k}} \frac{2\pi}{\hbar} \sum_{\mathbf{k}'} |g(\mathbf{k}, \mathbf{k}')|^2 \mathcal{B}(\mathbf{k} - \mathbf{k}', \epsilon_{\mathbf{k}} - \epsilon_{\mathbf{k}'}) (n_B(\epsilon_{\mathbf{k}} - \epsilon_{\mathbf{k}'} + 1 - f_{\mathbf{k}'})) \\ &= \frac{2}{\hbar} |g(\mathbf{k}, \mathbf{p})|^2 \mathcal{B}(\mathbf{k} - \mathbf{p}, \epsilon_{\mathbf{k}} - \epsilon_{\mathbf{p}}) (n_B(\epsilon_{\mathbf{k}} - \epsilon_{\mathbf{p}}) + f(\epsilon_{\mathbf{k}})) \\ &\quad - \delta_{\mathbf{p}\mathbf{k}} \frac{2}{\hbar} \sum_{\mathbf{k}'} |g(\mathbf{k}, \mathbf{k}')|^2 \mathcal{B}(\mathbf{k} - \mathbf{k}', \epsilon_{\mathbf{k}} - \epsilon_{\mathbf{k}'}) (n_B(\epsilon_{\mathbf{k}} - \epsilon_{\mathbf{k}'} + f(-\epsilon_{\mathbf{k}'})) \end{aligned}$$

Continuing

$$\begin{aligned}
\frac{\delta C_{\mathbf{k}}}{\delta f_{\mathbf{p}}} &= \frac{2}{\hbar} |g(\mathbf{k}, \mathbf{p})|^2 \mathcal{B}(\mathbf{k} - \mathbf{p}, \epsilon_{\mathbf{k}} - \varepsilon_{\mathbf{p}}) (-1 - n_B(\varepsilon_{\mathbf{p}} - \epsilon_{\mathbf{k}}) + f(\epsilon_{\mathbf{k}})) \\
&\quad - \frac{2}{\hbar} \delta_{\mathbf{p}\mathbf{k}} \sum_{\mathbf{k}'} |g(\mathbf{k}, \mathbf{k}')|^2 \mathcal{B}(\mathbf{k} - \mathbf{k}', \epsilon_{\mathbf{k}} - \varepsilon_{\mathbf{k}'})(n_B(\epsilon_{\mathbf{k}} - \varepsilon_{\mathbf{k}'} + f(-\varepsilon_{\mathbf{k}'})) \\
&= -\frac{2}{\hbar} |g(\mathbf{k}, \mathbf{p})|^2 \mathcal{B}(\mathbf{k} - \mathbf{p}, \epsilon_{\mathbf{k}} - \varepsilon_{\mathbf{p}})(n_B(\varepsilon_{\mathbf{p}} - \epsilon_{\mathbf{k}}) + f(-\epsilon_{\mathbf{k}})) \\
&\quad - \delta_{\mathbf{p}\mathbf{k}} \frac{2}{\hbar} \sum_{\mathbf{k}'} |g(\mathbf{k}, \mathbf{k}')|^2 \mathcal{B}(\mathbf{k} - \mathbf{k}', \epsilon_{\mathbf{k}} - \varepsilon_{\mathbf{k}'})(n_B(\epsilon_{\mathbf{k}} - \varepsilon_{\mathbf{k}'} + f(-\varepsilon_{\mathbf{k}'})) \\
&= \frac{2}{\hbar} |g(\mathbf{k}, \mathbf{p})|^2 \mathcal{B}(\mathbf{p} - \mathbf{k}, \varepsilon_{\mathbf{p}} - \epsilon_{\mathbf{k}})(n_B(\varepsilon_{\mathbf{p}} - \epsilon_{\mathbf{k}}) + f(-\epsilon_{\mathbf{k}})) \\
&\quad - \delta_{\mathbf{p}\mathbf{k}} \frac{2}{\hbar} \sum_{\mathbf{k}'} |g(\mathbf{k}, \mathbf{k}')|^2 \mathcal{B}(\mathbf{k} - \mathbf{k}', \epsilon_{\mathbf{k}} - \varepsilon_{\mathbf{k}'})(n_B(\epsilon_{\mathbf{k}} - \varepsilon_{\mathbf{k}'} + f(-\varepsilon_{\mathbf{k}'}))
\end{aligned}$$

Now we multiply by the factor

$$\frac{\partial f_{\mathbf{p}}}{\partial \epsilon_{\mathbf{p}}}$$

which is supposed to make this expression for the functional derivative of the collision integral symmetric

$$\begin{aligned}
Q_{\mathbf{k}\mathbf{p}} &= \left(\frac{\delta C_{\mathbf{k}}}{\delta f_{\mathbf{p}}} \right) \frac{\partial f_{\mathbf{p}}}{\partial \epsilon_{\mathbf{p}}} \\
&= \frac{\partial f_{\mathbf{p}}}{\partial \epsilon_{\mathbf{p}}} \frac{2}{\hbar} |g(\mathbf{k}, \mathbf{p})|^2 \mathcal{B}(\mathbf{p} - \mathbf{k}, \varepsilon_{\mathbf{p}} - \epsilon_{\mathbf{k}})(n_B(\varepsilon_{\mathbf{p}} - \epsilon_{\mathbf{k}}) + f(-\epsilon_{\mathbf{k}})) \\
&\quad - \delta_{\mathbf{p}\mathbf{k}} \frac{2}{\hbar} \frac{\partial f_{\mathbf{p}}}{\partial \epsilon_{\mathbf{p}}} \sum_{\mathbf{k}'} |g(\mathbf{p}, \mathbf{k}')|^2 \mathcal{B}(\mathbf{p} - \mathbf{k}', \epsilon_{\mathbf{p}} - \varepsilon_{\mathbf{k}'})(n_B(\epsilon_{\mathbf{p}} - \varepsilon_{\mathbf{k}'} + f(-\varepsilon_{\mathbf{k}'}))
\end{aligned}$$

We now define

$$P_{\mathbf{p}\mathbf{k}} = -kT \frac{2}{\hbar} \frac{\partial f_{\mathbf{p}}}{\partial \epsilon_{\mathbf{p}}} |g(\mathbf{k}, \mathbf{p})|^2 \mathcal{B}(\mathbf{p} - \mathbf{k}, \varepsilon_{\mathbf{p}} - \epsilon_{\mathbf{k}})(n_B(\varepsilon_{\mathbf{p}} - \epsilon_{\mathbf{k}}) + f(-\epsilon_{\mathbf{k}}))$$

leaving the diagonal untouched since it is always symmetric under the exchange of $\mathbf{k}$ and $\mathbf{p}$ we want to verify two things: (i) that the off diagonal is symmetric (ii) that the off diagonal is related to the sum appearing in the diagonal. With this two facts proven, we then would have made contact with the sum rules that appear in the transport notes.

Let's start by proving the first statement. To establish this we first assume inversion symmetry such that

$$\mathcal{B}(\mathbf{k} - \mathbf{p}, \epsilon_{\mathbf{k}} - \varepsilon_{\mathbf{p}}) = \mathcal{B}(\mathbf{p} - \mathbf{k}, \epsilon_{\mathbf{k}} - \varepsilon_{\mathbf{p}})$$

also from the analytic properties of the spectral function

$$\mathcal{B}(\mathbf{k} - \mathbf{p}, \epsilon_{\mathbf{k}} - \varepsilon_{\mathbf{p}}) = -\mathcal{B}(\mathbf{k} - \mathbf{p}, \varepsilon_{\mathbf{p}} - \epsilon_{\mathbf{k}})$$

in the end, we only need to prove that

$$\frac{\partial f}{\partial \epsilon_{\mathbf{p}}} (n_B(\varepsilon_{\mathbf{p}} - \epsilon_{\mathbf{k}}) + f(-\epsilon_{\mathbf{k}}))$$

is antisymmetric under the exchange of $\mathbf{p}$ and $\mathbf{k}$. First, we can simplify the above in the following way

$$\begin{aligned}
\frac{\partial f}{\partial \epsilon_{\mathbf{p}}} (n_B(\epsilon_{\mathbf{p}} - \epsilon_{\mathbf{k}}) + f(-\epsilon_{\mathbf{k}})) &= -\frac{1}{kT} f(\epsilon_{\mathbf{p}}) (1 - f(\epsilon_{\mathbf{p}})) (n_B(\epsilon_{\mathbf{p}} - \epsilon_{\mathbf{k}}) + f(-\epsilon_{\mathbf{k}})) \\
&= -\frac{1}{kT} f(\epsilon_{\mathbf{p}}) (1 - f(\epsilon_{\mathbf{p}})) (-1 - n_B(\epsilon_{\mathbf{k}} - \epsilon_{\mathbf{p}}) + f(-\epsilon_{\mathbf{k}})) \\
&= \frac{1}{kT} n_B(\epsilon_{\mathbf{k}} - \epsilon_{\mathbf{p}}) f(\epsilon_{\mathbf{p}}) (1 - f(\epsilon_{\mathbf{p}})) \left(\frac{1 - f(-\epsilon_{\mathbf{k}})}{n_B(\epsilon_{\mathbf{k}} - \epsilon_{\mathbf{p}})} + 1 \right) \\
&= \frac{1}{kT} n_B(\epsilon_{\mathbf{k}} - \epsilon_{\mathbf{p}}) f(\epsilon_{\mathbf{p}}) (1 - f(\epsilon_{\mathbf{p}})) \left(\frac{f(\epsilon_{\mathbf{k}})}{n_B(\epsilon_{\mathbf{k}} - \epsilon_{\mathbf{p}})} + 1 \right) \\
&\text{multiplying and dividing by } 1 - f(\epsilon_{\mathbf{k}}) \\
&= \frac{1}{kT} n_B(\epsilon_{\mathbf{k}} - \epsilon_{\mathbf{p}}) f(\epsilon_{\mathbf{p}}) (1 - f(\epsilon_{\mathbf{k}})) (1 - f(\epsilon_{\mathbf{p}})) \frac{\left(\frac{f(\epsilon_{\mathbf{k}})}{n_B(\epsilon_{\mathbf{k}} - \epsilon_{\mathbf{p}})} + 1 \right)}{(1 - f(\epsilon_{\mathbf{k}}))}
\end{aligned}$$

Now focusing on the last part, we have

$$\begin{aligned}
\frac{f(\epsilon_{\mathbf{k}})}{n_B(\epsilon_{\mathbf{k}} - \epsilon_{\mathbf{p}})} + 1 &= \frac{e^{y-z} - 1}{1 + e^y} + 1 \\
&= \frac{e^{y-z} - 1 + 1 + e^y}{1 + e^y} \\
&= \frac{(1 + e^{-z}) e^y}{1 + e^y} \\
&= \frac{(1 + e^{-z})}{1 + e^{-y}} \\
&= \frac{1 - f(\epsilon_{\mathbf{k}})}{1 - f(\epsilon_{\mathbf{p}})}
\end{aligned}$$

So that

$$\frac{\partial f}{\partial \epsilon_{\mathbf{p}}} (n_B(\epsilon_{\mathbf{p}} - \epsilon_{\mathbf{k}}) + f(-\epsilon_{\mathbf{k}})) = \frac{1}{kT} n_B(\epsilon_{\mathbf{k}} - \epsilon_{\mathbf{p}}) f(\epsilon_{\mathbf{p}}) (1 - f(\epsilon_{\mathbf{k}}))$$

Which is a way more intuitive expression, $\mathbf{p}$ is occupied, absorbs a boson and transitions to $\mathbf{k}$, which is unoccupied, the energy of the boson has to match the difference in energies and all is well. Furthermore we can prove now that

$$n_B(\epsilon_{\mathbf{k}} - \epsilon_{\mathbf{p}}) f(\epsilon_{\mathbf{p}}) (1 - f(\epsilon_{\mathbf{k}})) = -n_B(\epsilon_{\mathbf{p}} - \epsilon_{\mathbf{k}}) f(\epsilon_{\mathbf{k}}) (1 - f(\epsilon_{\mathbf{p}}))$$

Starting from

$$\begin{aligned}
n_B(\epsilon_{\mathbf{k}} - \epsilon_{\mathbf{p}}) f(\epsilon_{\mathbf{p}}) (1 - f(\epsilon_{\mathbf{k}})) &= -(1 + n_B(\epsilon_{\mathbf{p}} - \epsilon_{\mathbf{k}})) f(\epsilon_{\mathbf{p}}) (1 - f(\epsilon_{\mathbf{k}})) \\
&= -\left(\frac{1}{n_B(\epsilon_{\mathbf{p}} - \epsilon_{\mathbf{k}})} + 1 \right) n_B(\epsilon_{\mathbf{p}} - \epsilon_{\mathbf{k}}) f(\epsilon_{\mathbf{p}}) (1 - f(\epsilon_{\mathbf{k}}))
\end{aligned}$$

Now focusing on the first term

$$\begin{aligned}
\left(\frac{1}{n_B(\epsilon_{\mathbf{p}} - \epsilon_{\mathbf{k}})} + 1 \right) &= e^{y-z} - 1 + 1 \\
&= e^{\beta \epsilon_{\mathbf{p}} - \beta \epsilon_{\mathbf{k}}}
\end{aligned}$$

This immediately gives

$$e^{\beta \epsilon_{\mathbf{p}} - \beta \epsilon_{\mathbf{k}}} f(\epsilon_{\mathbf{p}}) (1 - f(\epsilon_{\mathbf{k}})) = f(\epsilon_{\mathbf{k}}) (1 - f(\epsilon_{\mathbf{p}}))$$

proving that

$$n_B(\epsilon_{\mathbf{k}} - \epsilon_{\mathbf{p}}) f(\epsilon_{\mathbf{p}}) (1 - f(\epsilon_{\mathbf{k}})) = -n_B(\epsilon_{\mathbf{p}} - \epsilon_{\mathbf{k}}) f(\epsilon_{\mathbf{k}}) (1 - f(\epsilon_{\mathbf{p}})).$$

With this, we conclude that

$$\begin{aligned} P_{\mathbf{k}\mathbf{p}} &= -kT \frac{2}{\hbar} \frac{\partial f_{\mathbf{p}}}{\partial \epsilon_{\mathbf{p}}} |g(\mathbf{k}, \mathbf{p})|^2 \mathcal{B}(\mathbf{p} - \mathbf{k}, \epsilon_{\mathbf{p}} - \epsilon_{\mathbf{k}}) (n_B(\epsilon_{\mathbf{p}} - \epsilon_{\mathbf{k}}) + f(-\epsilon_{\mathbf{k}})) \\ &= \frac{2}{\hbar} |g(\mathbf{k}, \mathbf{p})|^2 \mathcal{B}(\mathbf{k} - \mathbf{p}, \epsilon_{\mathbf{k}} - \epsilon_{\mathbf{p}}) n_B(\epsilon_{\mathbf{k}} - \epsilon_{\mathbf{p}}) f(\epsilon_{\mathbf{p}}) (1 - f(\epsilon_{\mathbf{k}})) \\ &= kTW_{\mathbf{k}\mathbf{p}} \end{aligned}$$

Is symmetric with respect to the momentum interchange. This is a general feature given the general properties given above from the functional derivatives of the heat kernel. This transition probability is the fundamental object in the Boltzmann theory as we will see briefly. Probably a better convention would have been to switch W and P since P is actually a rate with our definitions and W is actually a probability. Where we would usually expect a factor of $\frac{2\pi}{\hbar}$ we get a $\frac{2}{\hbar}$ but we packaged the factor of π in the spectral function which has the normalization with

$$\int \frac{d\omega}{\pi}$$

used consistently throughout to relate the spectral with the retarded function. Which means that

Transition rate

We then have the definition

$$Q_{\mathbf{k}\mathbf{p}} = \left(\frac{\delta C_{\mathbf{k}}}{\delta f_{\mathbf{p}}} \right) \frac{\partial f}{\partial \epsilon_{\mathbf{p}}}$$

which is symmetric under the exchange of $\mathbf{k}$ and $\mathbf{p}$. For the off diagonal elements of this kernel we have the definition

$$Q_{\mathbf{k}\mathbf{p}} = -\frac{1}{kT} P_{\mathbf{k}\mathbf{p}}$$

where P is supposed to be symmetric, owing to the symmetry of Q . Similarly, we have

$$Q_{\mathbf{k}\mathbf{k}} = -\tau_{\mathbf{k}} \frac{\partial f}{\partial \epsilon_{\mathbf{k}}}$$

With recalling the definition of τ :

$$\tau_{\mathbf{k}}^{-1} = -\left(\frac{\delta C_{\mathbf{k}}}{\delta f_{\mathbf{k}}} \right)$$

Where everything is evaluated at equilibrium since we are working with the linearized equation and all the kernels will be used

As such we have overall

$$Q_{\mathbf{k}\mathbf{p}} = \frac{1}{kT} (f_{\mathbf{k}} (1 - f_{\mathbf{k}}) \tau_{\mathbf{k}}^{-1} \delta_{\mathbf{k}\mathbf{p}} - P_{\mathbf{k}\mathbf{p}})$$

If we have

$$\tau_{\mathbf{k}}^{-1} f_{\mathbf{k}} (1 - f_{\mathbf{k}}) = \sum_{\mathbf{k}'} P_{\mathbf{k}\mathbf{k}'}$$

Or in other words

$$\tau_{\mathbf{k}}^{-1} = \frac{1}{f_{\mathbf{k}} (1 - f_{\mathbf{k}})} \sum_{\mathbf{k}'} P_{\mathbf{k}\mathbf{k}'}$$

then we can write

$$Q_{\mathbf{k}\mathbf{p}} = \frac{1}{kT} \left(\delta_{\mathbf{k}\mathbf{p}} \sum_{\mathbf{k}'} P_{\mathbf{k}\mathbf{k}'} - P_{\mathbf{k}\mathbf{p}} \right)$$

We also have to assume for this that

$$P_{\mathbf{k}\mathbf{k}} = 0$$

The definition of Q is useful by choosing a specific parametrization of the distribution function

$$\begin{aligned} F_{\mathbf{k}} &= f_{\mathbf{k}} + \delta F_{\mathbf{k}} \\ &= f_{\mathbf{k}} - \frac{\partial f_{\mathbf{k}}}{\partial \epsilon_{\mathbf{k}}} \phi_{\mathbf{k}} \end{aligned}$$

Going back to the linearized Boltzmann equation

$$\begin{aligned} -e\mathbf{E} \cdot \mathbf{v}_{\mathbf{k}} \frac{\partial f}{\partial \epsilon_{\mathbf{k}}} &= C_{\mathbf{k}}^0 \\ &= \sum_{\mathbf{q}} \frac{\delta C_{\mathbf{k}}}{\delta F_{\mathbf{q}}} \delta F_{\mathbf{q}} \end{aligned}$$

we see that this entails

$$e\mathbf{E} \cdot \mathbf{v}_{\mathbf{k}} \frac{\partial f}{\partial \epsilon_{\mathbf{k}}} = \sum_{\mathbf{q}} Q_{\mathbf{k}\mathbf{q}} \phi_{\mathbf{q}}$$

Where the linearized collision integral can be expanded as

$$\begin{aligned} C_{\mathbf{k}}^0 &= \sum_{\mathbf{q}} \frac{\delta C_{\mathbf{k}}}{\delta F_{\mathbf{q}}} \delta F_{\mathbf{q}} \\ &= - \sum_{\mathbf{q}} \frac{\delta C_{\mathbf{k}}}{\delta F_{\mathbf{q}}} \frac{\partial f_{\mathbf{q}}}{\partial \epsilon_{\mathbf{q}}} \phi_{\mathbf{q}} \\ &= - \sum_{\mathbf{q}} Q_{\mathbf{k}\mathbf{q}} \phi_{\mathbf{q}} \end{aligned}$$

here we see why the parametrization of $\delta F_{\mathbf{k}}$ is more convenient, it leads to the symmetric kernel $Q_{\mathbf{k}\mathbf{q}}$. Now we use the relation between the kernel Q and the transition probability to simplify the linearized collision integral:

$$\begin{aligned} C_{\mathbf{k}}^0 &= - \sum_{\mathbf{q}} Q_{\mathbf{k}\mathbf{q}} \phi_{\mathbf{q}} \\ &= - \sum_{\mathbf{q}} \frac{1}{kT} \left(\delta_{\mathbf{k}\mathbf{q}} \sum_{\mathbf{k}'} P_{\mathbf{k}\mathbf{k}'} - P_{\mathbf{k}\mathbf{q}} \right) \phi_{\mathbf{q}} \\ &= - \frac{1}{kT} \left(\sum_{\mathbf{k}'} P_{\mathbf{k}\mathbf{k}'} \phi_{\mathbf{k}} - \sum_{\mathbf{q}} P_{\mathbf{k}\mathbf{q}} \phi_{\mathbf{q}} \right) \\ &= - \frac{1}{kT} \sum_{\mathbf{q}} P_{\mathbf{k}\mathbf{q}} (\phi_{\mathbf{k}} - \phi_{\mathbf{q}}) \end{aligned}$$

Therefore, we have the equation

$$e\mathbf{E} \cdot \mathbf{v}_{\mathbf{k}} \frac{\partial f}{\partial \epsilon_{\mathbf{k}}} = \frac{1}{kT} \sum_{\mathbf{q}} P_{\mathbf{k}\mathbf{q}} (\phi_{\mathbf{k}} - \phi_{\mathbf{q}})$$

VARIATIONAL PRINCIPLE

The integral equation above can be solved numerically on a grid, however we also have an analytical tool at hand. We can use the fact that the functional $W[F]$ is extremal. We will continue to work with the linearized equation in what follows. We can do the following, starting with the definition

$$X_k = e\mathbf{E} \cdot \mathbf{v}_k \frac{\partial f}{\partial \epsilon_k}$$

and the L_2 inner product in momentum space, we have that the rate of entropy production, will be given by

$$\begin{aligned} \mathbf{J} \cdot \mathbf{E} &= \frac{-e}{V} \left(\sum_{\mathbf{q}} \delta F_{\mathbf{q}}^* \mathbf{v}(\mathbf{q}) \right) \cdot \mathbf{E} \\ &= \frac{e}{V} \left(\sum_{\mathbf{q}} \frac{\partial f}{\partial \epsilon_{\mathbf{q}}} \phi_{\mathbf{q}} \mathbf{v}(\mathbf{q}) \right) \cdot \mathbf{E} \\ &= \frac{1}{V} \sum_{\mathbf{q}} \phi_{\mathbf{q}} X_{\mathbf{q}} \\ &= \frac{1}{V} (\phi | X) \end{aligned}$$

Where we defined the inner product in momentum space similar to with Dirac notation. When ϕ is a solution to the Boltzmann equation we also have

$$X_k = \sum_{\mathbf{p}} Q_{k\mathbf{p}} \phi_{\mathbf{p}}$$

so we have the natural bilinear

$$\begin{aligned} \mathbf{J} \cdot \mathbf{E} &= \frac{1}{V} \sum_{\mathbf{qp}} \phi_{\mathbf{q}} Q_{\mathbf{qp}} \phi_{\mathbf{p}} \\ &= \frac{(\phi | Q | \phi)}{V} \end{aligned}$$

Because of the special symmetric structure of Q and its relation to the transition probability, we also have

$$\begin{aligned} \mathbf{J} \cdot \mathbf{E} &= \frac{1}{VkT} \sum_{\mathbf{qp}} \phi_{\mathbf{q}} P_{\mathbf{pq}} (\phi_{\mathbf{q}} - \phi_{\mathbf{p}}) \\ &= \frac{1}{2VkT} \left(\sum_{\mathbf{qp}} \phi_{\mathbf{q}} P_{\mathbf{pq}} (\phi_{\mathbf{q}} - \phi_{\mathbf{p}}) + \sum_{\mathbf{pq}} P_{\mathbf{qp}} (\phi_{\mathbf{p}} - \phi_{\mathbf{q}}) \phi_{\mathbf{p}} \right) \\ &= \frac{1}{2VkT} \sum_{\mathbf{qp}} P_{\mathbf{pq}} (\phi_{\mathbf{q}} - \phi_{\mathbf{p}})^2 \end{aligned}$$

where we just relabeled the momentum indices and used the symmetry of P in the second line. Sometimes in the literature things are written such that

$$\frac{1}{2VkT} \sum_{\mathbf{qp}} P_{\mathbf{pq}} (\phi_{\mathbf{q}} - \phi_{\mathbf{p}})^2 = \frac{(\phi | P | \phi)}{V}$$

to avoid explaining what Q is and save space. Now we use that the rate of entropy production at fixed field is maximal for solutions to the Boltzmann equation, this generates a variational principle for the conductivity since:

$$\begin{aligned} \mathbf{J} \cdot \mathbf{E} &= \sigma E^2 \\ &= \frac{(\phi | Q | \phi)}{V} \\ &\geq \frac{|(\psi | Q | \psi)|^2}{V |(\psi | Q | \psi)|} \end{aligned}$$

This last inequality is just the Cauchy-Schwartz inequality reexpressed where we define

$$\sqrt{Q}|\phi\rangle = |v\rangle$$

$$\sqrt{Q}|\psi\rangle = |u\rangle$$

then

$$|(u|v)|^2 \leq (v|v)(u|u)$$

for an arbitrary function of momentum $\psi_{\mathbf{k}}$. Using the fact that ϕ is a solution to the Boltzmann equation, we have

$$\frac{(\phi|Q|\phi)}{V} \geq \frac{|(\psi|X)|^2}{V|(\psi|Q|\psi)|}.$$

Using Ohm's law we have the variational principle

$$\sigma \geq \frac{|(\psi|X)|^2}{E^2 V |(\psi|Q|\psi)|}$$

In terms of the resistivity, we have

$$\rho \leq \frac{E^2 V |(\psi|Q|\psi)|}{|(\psi|X)|^2}.$$

So the resistivity minimizes the rate of entropy production at fixed current. Plugging everything explicitly, we have

$$\begin{aligned} \rho &\leq \frac{\frac{1}{2kT} \sum_{\mathbf{qp}} P_{\mathbf{pq}} (\psi_{\mathbf{q}} - \psi_{\mathbf{p}})^2}{\frac{1}{E^2 V} \left| \sum_{\mathbf{q}} e \frac{\partial f}{\partial \epsilon_{\mathbf{q}}} \psi_{\mathbf{q}} \mathbf{v}(\mathbf{q}) \cdot \mathbf{E} \right|^2} \\ &= \frac{\frac{V}{2kT} \sum_{\mathbf{qp}} P_{\mathbf{pq}} (\psi_{\mathbf{q}} - \psi_{\mathbf{p}})^2}{\left| \sum_{\mathbf{q}} e \frac{\partial f}{\partial \epsilon_{\mathbf{q}}} \psi_{\mathbf{q}} \mathbf{v}(\mathbf{q}) \cdot \mathbf{n} \right|^2} = \rho_{\text{trial}} \end{aligned}$$

where $\mathbf{n}$ is the unit vector which determines the direction of the electric field.

relaxation time approximation

for the relaxation time approximation we chose the ansatz $\psi_{\mathbf{q}} = -e\mathbf{v}_{\mathbf{q}} \cdot \mathbf{E}\tau_{\text{tr}}$. The physical interpretation of this is that the distribution equilibrates to a shifted distribution in momentum by a constant given by

$$\delta\mathbf{k} = \dot{\mathbf{k}}\tau$$

such that τ is a momentum relaxation rate. Then, if we assume that the shifted distribution in the presence of the field is given by the original distribution displaced in momentum space, we have

$$f(\mathbf{k} - \dot{\mathbf{k}}\tau_{\text{tr}}) \sim f(\mathbf{k}) - \frac{\partial f}{\partial \epsilon_{\mathbf{k}}} \mathbf{v}_{\mathbf{k}} \cdot \dot{\mathbf{k}}\tau$$

with the acceleration determined by the field

$$\dot{\mathbf{k}} = -e\mathbf{E}$$

so we have

$$\begin{aligned} f(\mathbf{k} - \dot{\mathbf{k}}\tau_{\text{tr}}) &\sim f(\mathbf{k}) - \frac{\partial f}{\partial \epsilon_{\mathbf{k}}} (-e\mathbf{v}_{\mathbf{k}} \cdot \mathbf{E}\tau) \\ &= f(\mathbf{k}) - \frac{\partial f}{\partial \epsilon_{\mathbf{k}}} \psi_{\mathbf{k}}. \end{aligned}$$

With this ansatz, we find

$$\rho_{\text{trial}} = \frac{\frac{V}{2kT} \sum_{\mathbf{qp}} P_{\mathbf{pq}} (\mathbf{v}_{\mathbf{q}} \cdot \mathbf{n} - \mathbf{v}_{\mathbf{p}} \cdot \mathbf{n})^2}{e^2 \left| \sum_{\mathbf{q}} \frac{\partial f}{\partial \epsilon_{\mathbf{q}}} (\mathbf{v}(\mathbf{q}) \cdot \mathbf{n})^2 \right|^2}$$

If we define the ratio

$$\left(\frac{n}{m} \right)_{nn} = -\frac{1}{V} \sum_{\mathbf{q}} \frac{\partial f}{\partial \epsilon_{\mathbf{q}}} (\mathbf{v}(\mathbf{q}) \cdot \mathbf{n})^2$$

the ratio of the density to the band mass in the direction of the unit vector $\mathbf{n}$, then we have

$$\rho_{\text{trial}} = \frac{\frac{1}{kT} \sum_{\mathbf{qp}} P_{\mathbf{pq}} (\mathbf{v}_{\mathbf{q}} \cdot \mathbf{n} - \mathbf{v}_{\mathbf{p}} \cdot \mathbf{n})^2}{e^2 \left(\frac{n}{m} \right)_{nn} \sum_{\mathbf{q}} 2 \left(-\frac{\partial f}{\partial \epsilon_{\mathbf{q}}} \right) (\mathbf{v}(\mathbf{q}) \cdot \mathbf{n})^2}$$

comparing to the drude formula, we have

$$\tau_{\text{tr}}^{-1} = \frac{\frac{1}{kT} \sum_{\mathbf{qp}} P_{\mathbf{pq}} (\mathbf{v}_{\mathbf{q}} \cdot \mathbf{n} - \mathbf{v}_{\mathbf{p}} \cdot \mathbf{n})^2}{\sum_{\mathbf{q}} 2 \left(-\frac{\partial f}{\partial \epsilon_{\mathbf{q}}} \right) (\mathbf{v}(\mathbf{q}) \cdot \mathbf{n})^2}$$

in the special case of cubic symmetry and a field applied along the x direction we have

$$\tau_{\text{tr}}^{-1} = \frac{\frac{1}{kT} \sum_{\mathbf{qp}} P_{\mathbf{pq}} (v_{qx} - v_{px})^2}{\sum_{\mathbf{q}} 2 \left(-\frac{\partial f}{\partial \epsilon_{\mathbf{q}}} \right) v_{qx}^2}$$

and

$$\rho_{\text{trial}} = \frac{\tau_{\text{tr}}^{-1}}{e^2 \left(\frac{n}{m} \right)_{xx}}$$

Averaged single particle time

We previously had the result

$$\tau_{\mathbf{k}}^{-1} = \frac{1}{f_{\mathbf{k}}(1-f_{\mathbf{k}})} \sum_{\mathbf{k}'} P_{\mathbf{k}\mathbf{k}'}$$

when discussing the transition rate. This, again, is the time scale on which the deviation of the distribution on a single momentum state relaxes back to equilibrium instead of the transport time discussed above which is the time scale on which momentum itself relaxes back to equilibrium. It is common to then average this time by a thermal distribution as follows

$$\begin{aligned} \tau_{qp}^{-1} &= \frac{\sum_{\mathbf{k}} \tau_{\mathbf{k}}^{-1} \left(-\frac{\partial f}{\partial \epsilon_{\mathbf{k}}} \right)}{\sum_{\mathbf{k}} \left(-\frac{\partial f}{\partial \epsilon_{\mathbf{k}}} \right)} \\ &= \frac{\frac{1}{kT} \sum_{\mathbf{k}'} P_{\mathbf{k}\mathbf{k}'}}{\sum_{\mathbf{k}} \left(-\frac{\partial f}{\partial \epsilon_{\mathbf{k}}} \right)} \end{aligned}$$

note here the similarities and differences with the transport time defined above. This can also be related to the single particle rate averaged over the density of states

$$\tau_{qp}^{-1}(\epsilon) = \frac{\sum_{\mathbf{k}} \tau_{\mathbf{k}}^{-1} \delta(\epsilon - \epsilon_{\mathbf{k}})}{\sum_{\mathbf{k}} \left(-\frac{\partial f}{\partial \epsilon_{\mathbf{k}}} \right)}$$

We can relate the two definitions by

$$\tau_{qp}^{-1} = \int_{-\infty}^{\infty} d\epsilon \tau_{qp}^{-1}(\epsilon) \left(-\frac{\partial f}{\partial \epsilon} \right).$$

Near zero temperature

$$\tau_{qp}^{-1}(0) \sim \tau_{qp}^{-1}.$$

PHONON TIMES

quasiparticle time

We will now calculate the phonon times from the definitions above. This will be done using our previous calculation of $P_{\mathbf{k}\mathbf{k}'}$ so that

$$\begin{aligned} \tau_{qp}^{-1}(\epsilon) &= \frac{\frac{2\pi}{\hbar} \sum_{\mathbf{k}\mathbf{k}'} |g(\mathbf{k}, \mathbf{k}')|^2 \mathcal{B}(\mathbf{k} - \mathbf{k}', \epsilon_{\mathbf{k}} - \epsilon_{\mathbf{k}'}) (n_B(\epsilon_{\mathbf{k}} - \epsilon_{\mathbf{k}'} + f(-\epsilon_{\mathbf{k}'})) \delta(\epsilon - \epsilon_{\mathbf{k}}))}{\sum_{\mathbf{k}} \left(-\frac{\partial f}{\partial \epsilon_{\mathbf{k}}} \right)} \\ &= \frac{\frac{2\pi}{\hbar} \sum_{\mathbf{k}\mathbf{k}'} |g(\mathbf{k}, \mathbf{k}')|^2 \mathcal{B}(\mathbf{k} - \mathbf{k}', \epsilon - \epsilon_{\mathbf{k}'}) (n_B(\epsilon - \epsilon_{\mathbf{k}'} + f(-\epsilon_{\mathbf{k}'})) \delta(\epsilon - \epsilon_{\mathbf{k}}))}{\sum_{\mathbf{k}} \left(-\frac{\partial f}{\partial \epsilon_{\mathbf{k}}} \right)} \\ &= \frac{\frac{2\pi}{\hbar} \int_{-\infty}^{\infty} d\nu \sum_{\mathbf{k}\mathbf{k}'} |g(\mathbf{k}, \mathbf{k}')|^2 \delta(\epsilon - \epsilon_{\mathbf{k}}) \delta(\nu - \epsilon_{\mathbf{k}'}) \mathcal{B}(\mathbf{k} - \mathbf{k}', \epsilon - \nu) (n_B(\epsilon - \nu) + f(-\nu))}{\sum_{\mathbf{k}} \left(-\frac{\partial f}{\partial \epsilon_{\mathbf{k}}} \right)} \end{aligned}$$

Using some of the usual tricks with

$$\mathcal{B}(\mathbf{k} - \mathbf{k}', \epsilon - \nu) = \mathcal{B}^-(\mathbf{k} - \mathbf{k}', \epsilon - \nu) - \mathcal{B}^+(\mathbf{k} - \mathbf{k}', \epsilon - \nu)$$

we find

$$\tau_{qp}^{-1}(\epsilon) = \frac{\frac{2\pi}{\hbar} \int_{-\infty}^{\infty} d\nu \sum_{\mathbf{k}\mathbf{k}'} |g(\mathbf{k}, \mathbf{k}')|^2 \delta(\epsilon - \epsilon_{\mathbf{k}}) \delta(\epsilon - \nu - \epsilon_{\mathbf{k}'}) \mathcal{B}^-(\mathbf{k} - \mathbf{k}', \nu) (2n_B(\nu) + f(\nu - \epsilon) + f(\nu + \epsilon))}{\sum_{\mathbf{k}} \left(-\frac{\partial f}{\partial \epsilon_{\mathbf{k}}} \right)}$$

Defining

$$\frac{\sum_{\mathbf{k}'} |g(\mathbf{k}, \mathbf{k}')|^2 \delta(\epsilon - \epsilon_{\mathbf{k}}) \delta(\epsilon - \nu - \epsilon_{\mathbf{k}'}) \mathcal{B}^-(\mathbf{k} - \mathbf{k}', \nu)}{\sum_{\mathbf{k}} \left(-\frac{\partial f}{\partial \epsilon_{\mathbf{k}}} \right)} = \alpha_{\mathbf{k}} F(\epsilon, \nu)$$

and

$$\sum_{\mathbf{k}} \alpha_{\mathbf{k}} F(\epsilon, \nu) = \alpha F(\epsilon, \nu)$$

$$\tau_{qp}^{-1}(\epsilon) = \frac{2\pi}{\hbar} \int_0^{\infty} d\nu \alpha F(\epsilon, \nu) (2n_B(\nu) + f(\nu - \epsilon) + f(\nu + \epsilon))$$

neglecting the ϵ dependence in $\alpha F(\epsilon, \nu)$, by setting $\alpha F(\nu) = \alpha F(0, \nu)$ we finally get

$$\tau_{qp}^{-1}(\epsilon) = \frac{2\pi}{\hbar} \int_0^{\infty} d\nu \alpha F(\nu) (2n_B(\nu) + f(\nu - \epsilon) + f(\nu + \epsilon))$$

To get the averaged quasiparticle relaxation rate, we then perform

$$\tau_{qp}^{-1} = \int_{-\infty}^{\infty} d\epsilon \tau_{qp}^{-1}(\epsilon) \left(-\frac{\partial f}{\partial \epsilon} \right)$$

In the end, because the dependence is only in the occupation functions we only worry about the integral

$$\int_{-\infty}^{\infty} d\epsilon \left(-\frac{\partial f}{\partial \epsilon} \right) (2n_B(\nu) + f(\nu - \epsilon) + f(\nu + \epsilon))$$

There are two types of integrals here

$$\int_{-\infty}^{\infty} d\epsilon \left(-\frac{\partial f}{\partial \epsilon} \right) = 1$$

and

$$\int_{-\infty}^{\infty} d\epsilon \left(-\frac{\partial f}{\partial \epsilon} \right) f(\nu \pm \epsilon) = n_B(\nu)^2 + n_B(\nu)(n_B(\nu) + 1)(\nu - 1)$$

Such that we have

$$\begin{aligned} \int_{-\infty}^{\infty} d\epsilon \left(-\frac{\partial f}{\partial \epsilon} \right) (2n_B(\nu) + f(\nu - \epsilon) + f(\nu + \epsilon)) &= 2 \left(n_B(\nu) + n_B(\nu)^2 + n_B(\nu)(n_B(\nu) + 1)(\nu - 1) \right) \\ &= 2 \frac{\beta\nu}{4} \operatorname{csch} \left(\frac{\beta\nu}{2} \right)^2 \\ &= \frac{2}{\beta\nu} \left[\frac{\beta\nu}{2} \operatorname{csch} \left(\frac{\beta\nu}{2} \right) \right]^2 \end{aligned}$$

in the end, we have

$$\begin{aligned} \tau_{qp}^{-1} &= \int_{-\infty}^{\infty} d\epsilon \tau_{qp}^{-1}(\epsilon) \left(-\frac{\partial f}{\partial \epsilon} \right) \\ &= \frac{4\pi}{\hbar\beta} \int_0^{\infty} d\nu \frac{\alpha F(\nu)}{\nu} \left[\frac{\beta\nu}{2} \operatorname{csch} \left(\frac{\beta\nu}{2} \right) \right]^2 \end{aligned}$$

Transport time

we can use again the formulas derived above with very small modifications due to the similar structure of the quasiparticle time and the transport time in the relaxation time approximation

$$\begin{aligned} \tilde{\tau}_{tr}^{-1}(\epsilon) &= \frac{\frac{2\pi}{\hbar} \sum_{\mathbf{k}\mathbf{k}'} \left(-\frac{\partial f}{\partial \epsilon_{\mathbf{k}}} \right) |g(\mathbf{k}, \mathbf{k}')|^2 (v_{\mathbf{k}x} - v_{\mathbf{k}'x})^2 \mathcal{B}(\mathbf{k} - \mathbf{k}', \epsilon_{\mathbf{k}} - \epsilon_{\mathbf{k}'}) (n_B(\epsilon_{\mathbf{k}} - \epsilon_{\mathbf{k}'} + f(-\epsilon_{\mathbf{k}'})) \delta(\epsilon - \epsilon_{\mathbf{k}})}{\sum_{\mathbf{k}} \left(-\frac{\partial f}{\partial \epsilon_{\mathbf{k}}} \right) 2v_{\mathbf{k}x}^2} \\ &= \frac{\frac{2\pi}{\hbar} \int_{-\infty}^{\infty} d\nu \sum_{\mathbf{k}\mathbf{k}'} |g(\mathbf{k}, \mathbf{k}')|^2 (v_{\mathbf{k}x} - v_{\mathbf{k}'x})^2 \delta(\epsilon - \epsilon_{\mathbf{k}}) \delta(\epsilon - \nu - \epsilon_{\mathbf{k}'}) \mathcal{B}(\mathbf{k} - \mathbf{k}', \epsilon - \nu) \left(-\frac{\partial f}{\partial \epsilon} \right) (n_B(\epsilon - \nu) + f(-\nu))}{\sum_{\mathbf{k}} \left(-\frac{\partial f}{\partial \epsilon_{\mathbf{k}}} \right) 2v_{\mathbf{k}x}^2} \end{aligned}$$

using then the tricks above we get

$$\tilde{\tau}_{tr}^{-1}(\epsilon) = \frac{\frac{2\pi}{\hbar} \int_{-\infty}^{\infty} d\nu \sum_{\mathbf{k}\mathbf{k}'} |g(\mathbf{k}, \mathbf{k}')|^2 (v_{\mathbf{k}x} - v_{\mathbf{k}'x})^2 \delta(\epsilon - \epsilon_{\mathbf{k}}) \delta(\epsilon - \nu - \epsilon_{\mathbf{k}'}) \mathcal{B}^-(\mathbf{k} - \mathbf{k}', \nu) \left(-\frac{\partial f}{\partial \epsilon} \right) (2n_B(\nu) + f(\nu - \epsilon) + f(\nu + \epsilon))}{\sum_{\mathbf{k}} 2v_{\mathbf{k}x}^2 \left(-\frac{\partial f}{\partial \epsilon_{\mathbf{k}}} \right)}$$

Defining

$$\frac{\sum_{\mathbf{k}} |g(\mathbf{k}, \mathbf{k}')|^2 (v_{\mathbf{k}x} - v_{\mathbf{k}'x})^2 \delta(\epsilon - \epsilon_{\mathbf{k}}) \delta(\epsilon - \nu - \epsilon_{\mathbf{k}'}) \mathcal{B}^-(\mathbf{k} - \mathbf{k}', \nu)}{\sum_{\mathbf{k}} 2v_{\mathbf{k}z}^2 \left(-\frac{\partial f}{\partial \epsilon_{\mathbf{k}}} \right)} = \alpha_{tr\mathbf{k}}^2 F(\epsilon, \nu)$$

and

$$\sum_{\mathbf{k}} \alpha_{tr\mathbf{k}}^2 F(\epsilon, \nu) = \alpha_{tr}^2 F(\epsilon, \nu)$$

neglecting the ϵ dependence as usual in the $\alpha_{tr}^2 F(\epsilon, \nu)$ function, we get we get

$$\tilde{\tau}_{tr}^{-1}(\epsilon) = \frac{2\pi}{\hbar} \int_0^\infty d\nu \alpha_{tr}^2 F(\nu) (2n_B(\nu) + f(\nu - \epsilon) + f(\nu + \epsilon)) \left(-\frac{\partial f}{\partial \epsilon} \right)$$

using now

$$\tau_{tr}^{-1} = \int_{-\infty}^\infty d\epsilon \tilde{\tau}_{tr}^{-1}(\epsilon).$$

finally, averaging over the derivative of the fermi distribution, we get

$$\tau_{tr}^{-1} = \frac{4\pi}{\hbar\beta} \int_0^\infty d\nu \frac{\alpha_{tr}^2 F(\nu)}{\nu} \left[\frac{\beta\nu}{2} \text{csch}\left(\frac{\beta\nu}{2}\right) \right]^2$$

From this, we can then continue by calculating the $\alpha_{tr}^2 F(\epsilon, \nu)$ function for instance in the Debye model with a spherical fermi surface, this would give in the end the Bloch-Grüneisen law. Where the transport time transitions to a T^5 temperature dependence in 3D below $T_{BG} = v_s k_F$ when the phonons are not able to be scattered.

Optical mode

We start by calculating the $\alpha_{tr}^2 F(\nu)$ function

$$\frac{\sum_{\mathbf{k}'} |g(\mathbf{k}, \mathbf{k}')|^2 (v_{\mathbf{k}x} - v_{\mathbf{k}'x})^2 \delta(\epsilon_{\mathbf{k}}) \delta(\epsilon_{\mathbf{k}'}) \mathcal{B}^-(\mathbf{k} - \mathbf{k}', \nu)}{\sum_{\mathbf{k}} 2v_{\mathbf{k}z}^2 \left(-\frac{\partial f}{\partial \epsilon_{\mathbf{k}}} \right)} = \alpha_{tr\mathbf{k}}^2 F(\nu)$$

Modeling the dispersion as

$$\epsilon_{\mathbf{k}} = \epsilon_{\parallel\mathbf{k}} - 2t_{\perp} \cos k_z a_z$$

which gives the velocity

$$v_{\mathbf{k}z} = 2t_{\perp} a_z \sin k_z a_z$$

and regarding a dispersionless phonon, we get

$$\frac{\sum_{\mathbf{k}} |g(\mathbf{k}, \mathbf{k}')|^2 v_{\mathbf{k}z}^2 \delta(\epsilon_{\mathbf{k}}) \delta(\epsilon_{\mathbf{k}'}) \mathcal{B}^-(\nu)}{\sum_{\mathbf{k}} v_{\mathbf{k}z}^2 \left(-\frac{\partial f}{\partial \epsilon_{\mathbf{k}}} \right)} = \alpha_{tr\mathbf{k}'}^2 F(\nu)$$

With the coupling

$$g(\mathbf{k}, \mathbf{k}') = \alpha [2 \cos(k_z a_z)] [2 \cos(k'_z a_z)]$$

and the spectral function

$$\begin{aligned} B(\nu) &= \frac{\pi\hbar}{2M\omega_0} (\delta(\nu - \omega_0) - \delta(\nu + \omega_0)) \\ &= B^-(\nu) - B^+(\nu) \end{aligned}$$

we get

$$\frac{\pi \hbar \alpha^2}{2M\omega_0} \frac{\sum_{\mathbf{k}'} |g(\mathbf{k}, \mathbf{k}')|^2 v_{\mathbf{k}z}^2 \delta(\epsilon_{\mathbf{k}}) \delta(\epsilon_{\mathbf{k}'})}{\sum_{\mathbf{k}} v_{\mathbf{k}z}^2 \left(-\frac{\partial f}{\partial \epsilon_{\mathbf{k}}}\right)} \delta(\nu - \omega_0) = \alpha_{tr\mathbf{k}}^2 F(\nu)$$

using

$$a_z \int \frac{dk_z}{2\pi} [2 \sin(k_z a_z)]^2 [2 \cos(k_z a_z)]^2 = a_z \int \frac{dk_z}{2\pi} [2 \sin(k_z a_z)]^2 = a_z \int \frac{dp_z}{2\pi} [2 \cos(p_z a_z)]^2 = 2$$

$$2\nu_0 \frac{\pi \hbar \alpha^2}{2M\omega_0} \delta(\nu - \omega_0) = \alpha_{tr}^2 F(\nu)$$

Further accounting for two polarizations

$$4\nu_0 \frac{\pi \hbar \alpha^2}{2M\omega_0} \delta(\nu - \omega_0) = \alpha_{tr}^2 F(\nu)$$

which has to be compared with

$$\frac{\lambda_{E,tr}}{2} \omega_0 \delta(\nu - \omega_0) = \alpha_{tr}^2 F(\nu)$$

making

$$\lambda_{E,tr} = 4\nu_0 \frac{\pi \hbar \alpha^2}{M\omega_0^2}.$$

we then have

$$\begin{aligned} \tau_{tr}^{-1} &= \frac{4\pi}{\hbar\beta} \int_0^\infty d\nu \frac{\alpha_{tr}^2 F(\nu)}{\nu} \left[\frac{\beta\nu}{2} \operatorname{csch}\left(\frac{\beta\nu}{2}\right) \right]^2 \\ &= \frac{4\pi}{\hbar\beta} 4\nu_0 \frac{\pi \hbar \alpha^2}{2M\omega_0} \frac{1}{\omega_0} \left[\frac{\beta\omega_0}{2} \operatorname{csch}\left(\frac{\beta\omega_0}{2}\right) \right]^2 \\ &= \frac{2\pi}{\hbar} \frac{\lambda_E}{\beta} \left[\left(\frac{\beta\omega_0}{2}\right) \operatorname{csch}\left(\frac{\beta\omega_0}{2}\right) \right]^2 \end{aligned}$$

with

$$\lambda_{E,tr} = 4\pi \frac{\hbar \alpha^2 \nu_0}{M\omega_0^2}$$

here we have a factor of π from the spectral function of the phonons, a factor of 2 for polarization and a factor of 2 for the integrals over the velocity and coupling constant

So that at high temperatures we have

$$\tau_{tr}^{-1} = \frac{2\pi \lambda_{E,tr}}{\hbar} k_b T$$

this matches the form of the

MAGNETOELASTIC TIMES

The derivation of the magnetoelastic transport time and resistivity in the relaxation time approximation is very similar to what has been done so far for the electron-phonon problem. We can first define

$$\tilde{\tau}_{tr}^{-1}(\epsilon) = \frac{2\pi}{\hbar k T} \frac{\sum_{\mathbf{k}\mathbf{k}'} (v_{\mathbf{k}z} - v_{\mathbf{k}'z})^2 P_{\mathbf{k}\mathbf{k}'} \delta(\epsilon - \epsilon_{\mathbf{k}})}{\sum_{\mathbf{k}} \left(-\frac{\partial f}{\partial \epsilon_{\mathbf{k}}}\right) 2v_{\mathbf{k}z}^2}$$

and then we can integrate to get the transport rate

$$\tau_{tr}^{-1} = \int_{-\infty}^{\infty} d\epsilon \tau_{tr}^{-1}(\epsilon).$$

The only missing piece is the probability $P_{\mathbf{k}\mathbf{k}'}$ this can be extracted from the self energy

$$\Sigma''_{ME}(\mathbf{k}, \omega) = -\frac{2\lambda^2}{N^2} \sum_{\mathbf{q}\mathbf{p}} \int \frac{d\nu}{\pi} \chi''(\mathbf{p}, \omega - \nu) [n_B(\omega - \nu) + n_F(-\nu)] \int \frac{d\epsilon}{\pi} |g(\mathbf{k} + \mathbf{q} + \mathbf{p}, \mathbf{k})|^2 [A(\mathbf{k} + \mathbf{q} + \mathbf{p}, \nu + \epsilon) + A(\mathbf{k} + \mathbf{q} + \mathbf{p}, \nu - \epsilon)]$$

by comparing with the quasiparticle time

$$\tau_{\mathbf{k}}^{-1} = \frac{1}{f_{\mathbf{k}}(1 - f_{\mathbf{k}})} \sum_{\mathbf{k}'} P_{\mathbf{k}\mathbf{k}'}$$

$$\Sigma''_{ME}(\mathbf{k}, \omega) = -\frac{1}{N^2} \sum_{\mathbf{q}\mathbf{p}} \int \frac{d\nu}{\pi} \int \frac{d\epsilon}{\pi} \sum_{\ell} |\lambda_{\ell}(\mathbf{k} + \mathbf{q} + \mathbf{p}, \mathbf{k})|^2 A(\mathbf{k} + \mathbf{q} + \mathbf{p}, \epsilon) B_{\ell}(\mathbf{q}, \epsilon - \omega - \nu) \chi''(\mathbf{p}, \nu) [n_B(\epsilon - \omega - \nu) + n_F(\epsilon)] [n_B(\nu - \omega) + n_F(-\nu)]$$

Changing variables $\omega + \nu = u$, $d\nu = du$

$$\Sigma''_{ME}(\mathbf{k}, \omega) = -\frac{1}{N^2} \sum_{\mathbf{q}\mathbf{p}} \int \frac{d\nu}{\pi} \int \frac{d\epsilon}{\pi} \sum_{\ell} |\lambda_{\ell}(\mathbf{k} + \mathbf{q} + \mathbf{p}, \mathbf{k})|^2 A(\mathbf{k} + \mathbf{q} + \mathbf{p}, \epsilon) B_{\ell}(\mathbf{q}, \epsilon - \nu) \chi''(\mathbf{p}, \nu - \omega) [n_B(\epsilon - \nu) + n_F(\epsilon)] [n_B(\nu - \omega) + n_F(-\nu)]$$

reorganizing using the symmetry properties of the occupation functions and the spectral functions

$$\Sigma''_{ME}(\mathbf{k}, \omega) = -\frac{1}{N^2} \sum_{\mathbf{q}\mathbf{p}} \int \frac{d\nu}{\pi} \int \frac{d\epsilon}{\pi} \sum_{\ell} |\lambda_{\ell}(\mathbf{k} + \mathbf{q} + \mathbf{p}, \mathbf{k})|^2 A(\mathbf{k} + \mathbf{q} + \mathbf{p}, \epsilon) B_{\ell}(\mathbf{q}, \nu - \epsilon) \chi''(\mathbf{p}, \omega - \nu) [n_B(\nu - \epsilon) + n_F(-\epsilon)] [n_B(\omega - \nu) + n_F(-\omega)]$$

using the same phonon dispersion for all branches

$$\begin{aligned} \Sigma''_{ME}(\mathbf{k}, \omega) &= -\frac{1}{N^2} \sum_{\mathbf{q}\mathbf{p}} \int \frac{d\nu}{\pi} \int \frac{d\epsilon}{\pi} \left| \tilde{\lambda}(\mathbf{k} + \mathbf{q} + \mathbf{p}, \mathbf{k}) \right|^2 A(\mathbf{k} + \mathbf{q} + \mathbf{p}, \epsilon) B(\mathbf{q}, \nu - \epsilon) \chi''(\mathbf{p}, \omega - \nu) [n_B(\nu - \epsilon) + n_F(-\epsilon)] [n_B(\omega - \nu) + n_F(-\omega)] \\ &= -\frac{1}{N^2} \sum_{\mathbf{q}\mathbf{p}} \left| \tilde{\lambda}(\mathbf{k} + \mathbf{q} + \mathbf{p}, \mathbf{k}) \right|^2 \int \frac{d\nu}{\pi} \chi''(\mathbf{p}, \omega - \nu) [n_B(\omega - \nu) + n_F(-\nu)] \int \frac{d\epsilon}{\pi} A(\mathbf{k} + \mathbf{q} + \mathbf{p}, \epsilon) B(\mathbf{q}, \nu - \epsilon) [n_B(\nu - \epsilon) + n_F(-\epsilon)] \end{aligned}$$

Performing $\mathbf{k} + \mathbf{q} + \mathbf{p} \rightarrow \mathbf{p}$

$$\Sigma''_{ME}(\mathbf{k}, \omega) = -\frac{1}{N^2} \sum_{\mathbf{q}\mathbf{p}} \left| \tilde{\lambda}(\mathbf{p}, \mathbf{k}) \right|^2 \int \frac{d\nu}{\pi} \chi''(\mathbf{p} - \mathbf{k} - \mathbf{q}, \omega - \nu) [n_B(\omega - \nu) + n_F(-\nu)] \int \frac{d\epsilon}{\pi} A(\mathbf{p}, \epsilon) B(\mathbf{q}, \nu - \epsilon) [n_B(\nu - \epsilon) + n_F(-\epsilon)]$$

Asumptions start

$$\tau^{-1}(\mathbf{k}) = \sum_{\mathbf{qp}} \left| \tilde{\lambda}(\mathbf{p}, \mathbf{k}) \right|^2 \int \frac{d\nu}{\pi} \chi''(\mathbf{p} - \mathbf{k} - \mathbf{q}, \epsilon_{\mathbf{k}} - \nu) [n_B(\epsilon_{\mathbf{k}} - \nu) + n_F(-\nu)] \int \frac{d\epsilon}{\pi} A(\mathbf{p}, \epsilon) B(\mathbf{q}, \nu - \epsilon) [n_B(\nu - \epsilon) + n_F(-\epsilon)]$$

where $\left| \tilde{\lambda}(\mathbf{p}, \mathbf{k}) \right|^2 = \sum_{\ell} |\lambda_{\ell}(\mathbf{p}, \mathbf{k})|^2$ and for the time itself we absorbed an extra factor of 2 from the relation between the rate and the self energy. We also obviate the factors of N which are required to transform sums to integrals.

Now, since

$$kT \left(-\frac{\partial f}{\partial \epsilon_{\mathbf{k}}} \right) \tau^{-1}(\mathbf{k}) = \sum_{\mathbf{p}} P_{\mathbf{kp}}$$

Assuming that

$$P_{\mathbf{kp}} = kT \left(-\frac{\partial f}{\partial \epsilon_{\mathbf{k}}} \right) \left| \tilde{\lambda}(\mathbf{p}, \mathbf{k}) \right|^2 \sum_{\mathbf{q}} \int \frac{d\nu}{\pi} \chi''(\mathbf{p} - \mathbf{k} - \mathbf{q}, \epsilon_{\mathbf{k}} - \nu) [n_B(\epsilon_{\mathbf{k}} - \nu) + n_F(-\nu)] \int \frac{d\epsilon}{\pi} A(\mathbf{p}, \epsilon) B(\mathbf{q}, \nu - \epsilon) [n_B(\nu - \epsilon) + n_F(-\epsilon)]$$

Now taking the integral over the electron spectral function

$$P_{\mathbf{kp}} = \frac{\partial f}{\partial \epsilon_{\mathbf{k}}} \left| \tilde{\lambda}(\mathbf{p}, \mathbf{k}) \right|^2 \sum_{\mathbf{q}} \int \frac{d\nu}{\pi} \chi''(\mathbf{p} - \mathbf{k} - \mathbf{q}, \epsilon_{\mathbf{k}} - \nu) [n_B(\epsilon_{\mathbf{k}} - \nu) + n_F(-\nu)] B(\mathbf{q}, \nu - \epsilon_{\mathbf{p}}) [n_B(\nu - \epsilon_{\mathbf{p}}) + n_F(-\epsilon_{\mathbf{p}})]$$

we want to check whether this is symmetric with respect to the interchange of $\mathbf{p}$ and $\mathbf{k}$. From our previous calculations, we have

$$P_{\mathbf{kp}} = \left| \tilde{\lambda}(\mathbf{p}, \mathbf{k}) \right|^2 \sum_{\mathbf{q}} \int \frac{d\nu}{\pi} \chi''(\mathbf{p} - \mathbf{k} - \mathbf{q}, \nu - \epsilon_{\mathbf{k}}) B(\mathbf{q}, \nu - \epsilon_{\mathbf{p}}) n_B(\nu - \epsilon_{\mathbf{k}}) n_F(\epsilon_{\mathbf{k}}) (1 - n_F(\nu)) [n_B(\nu - \epsilon_{\mathbf{p}}) + n_F(-\epsilon_{\mathbf{p}})]$$

This form is vaguely intuitive as it is the convolution of the transition rate for scattering off of spins convolved with the one for phonons. Now we change variables so that the dispersion only appears in a single spectral function

$$P_{\mathbf{kp}} = \left| \tilde{\lambda}(\mathbf{p}, \mathbf{k}) \right|^2 \sum_{\mathbf{q}} \int \frac{d\nu}{\pi} \chi''(\mathbf{p} - \mathbf{k} - \mathbf{q}, \nu + \epsilon_{\mathbf{p}} - \epsilon_{\mathbf{k}}) B(\mathbf{q}, \nu) n_B(\nu + \epsilon_{\mathbf{p}} - \epsilon_{\mathbf{k}}) n_F(\epsilon_{\mathbf{k}}) (1 - n_F(\nu + \epsilon_{\mathbf{p}})) [n_B(\nu) + n_F(-\epsilon_{\mathbf{p}})]$$

Now we interchange the momenta

$$P_{\mathbf{kp}} = \left| \tilde{\lambda}(\mathbf{k}, \mathbf{p}) \right|^2 \sum_{\mathbf{q}} \int \frac{d\nu}{\pi} \chi''(\mathbf{k} - \mathbf{p} - \mathbf{q}, \nu + \epsilon_{\mathbf{k}} - \epsilon_{\mathbf{p}}) B(\mathbf{q}, \nu) n_B(\nu + \epsilon_{\mathbf{k}} - \epsilon_{\mathbf{p}}) n_F(\epsilon_{\mathbf{p}}) (1 - n_F(\nu + \epsilon_{\mathbf{k}})) [n_B(\nu) + n_F(-\epsilon_{\mathbf{k}})]$$

using the symmetry of χ'' and B in the momentum argument, the symmetry of the bare vertex, and changing the sum over $\mathbf{q}$ we have

$$P_{\mathbf{kp}} = \left| \tilde{\lambda}(\mathbf{p}, \mathbf{k}) \right|^2 \sum_{\mathbf{q}} \int \frac{d\nu}{\pi} \chi''(\mathbf{p} - \mathbf{k} - \mathbf{q}, \nu + \epsilon_{\mathbf{k}} - \epsilon_{\mathbf{p}}) B(\mathbf{q}, \nu) n_B(\nu + \epsilon_{\mathbf{k}} - \epsilon_{\mathbf{p}}) n_F(\epsilon_{\mathbf{p}}) (1 - n_F(\nu + \epsilon_{\mathbf{k}})) [n_B(\nu) + n_F(-\epsilon_{\mathbf{k}})]$$

using the antisymmetry of χ'' and B we have

$$P_{\mathbf{kp}} = \left| \tilde{\lambda}(\mathbf{p}, \mathbf{k}) \right|^2 \sum_{\mathbf{q}} \int \frac{d\nu}{\pi} \chi''(\mathbf{p} - \mathbf{k} - \mathbf{q}, -\nu + \epsilon_{\mathbf{p}} - \epsilon_{\mathbf{k}}) B(\mathbf{q}, -\nu) n_B(\nu + \epsilon_{\mathbf{k}} - \epsilon_{\mathbf{p}}) n_F(\epsilon_{\mathbf{p}}) (1 - n_F(\nu + \epsilon_{\mathbf{k}})) [n_B(\nu) + n_F(-\epsilon_{\mathbf{k}})]$$

Changing variables, we then have

$$P_{\mathbf{k}\mathbf{p}} = \left| \tilde{\lambda}(\mathbf{p}, \mathbf{k}) \right|^2 \sum_{\mathbf{q}} \int \frac{d\nu}{\pi} \chi''(\mathbf{p} - \mathbf{k} - \mathbf{q}, \nu + \epsilon_{\mathbf{p}} - \epsilon_{\mathbf{k}}) B(\mathbf{q}, \nu) n_B(-\nu + \epsilon_{\mathbf{k}} - \epsilon_{\mathbf{p}}) n_F(\epsilon_{\mathbf{p}}) (1 - n_F(-\nu + \epsilon_{\mathbf{k}})) [n_B(-\nu) + n_F(-\epsilon_{\mathbf{k}})]$$

Therefore, we only need to verify that

$$\begin{aligned} n_B(-\nu + \epsilon_{\mathbf{k}} - \epsilon_{\mathbf{p}}) n_F(\epsilon_{\mathbf{p}}) (1 - n_F(-\nu + \epsilon_{\mathbf{k}})) [n_B(-\nu) + n_F(-\epsilon_{\mathbf{k}})] = \\ n_B(\nu + \epsilon_{\mathbf{p}} - \epsilon_{\mathbf{k}}) n_F(\epsilon_{\mathbf{k}}) (1 - n_F(\nu + \epsilon_{\mathbf{p}})) [n_B(\nu) + n_F(-\epsilon_{\mathbf{p}})] \end{aligned}$$

and the proof of the symmetry of $P_{\mathbf{k}\mathbf{p}}$ would be complete. Since mathematica says this is true, we gain confidence that

$$P_{\mathbf{k}\mathbf{p}} = \left| \tilde{\lambda}(\mathbf{p}, \mathbf{k}) \right|^2 \sum_{\mathbf{q}} \int \frac{d\nu}{\pi} \chi''(\mathbf{p} - \mathbf{k} - \mathbf{q}, \nu + \epsilon_{\mathbf{p}} - \epsilon_{\mathbf{k}}) B(\mathbf{q}, \nu) n_B(\nu + \epsilon_{\mathbf{p}} - \epsilon_{\mathbf{k}}) n_F(\epsilon_{\mathbf{k}}) (1 - n_F(\nu + \epsilon_{\mathbf{p}})) [n_B(\nu) + n_F(-\epsilon_{\mathbf{p}})]$$

is the correct expression for the transition probability in Boltzmann theory.

Quasiparticle time consistency check

Now just to be on the safe side and verify that we get what we put in, we calculate the quasiparticle time from this expression

$$P_{\mathbf{k}\mathbf{p}} = kT \left(-\frac{\partial f}{\partial \epsilon_{\mathbf{k}}} \right) \left| \tilde{\lambda}(\mathbf{p}, \mathbf{k}) \right|^2 \sum_{\mathbf{q}} \int \frac{d\nu}{\pi} \chi''(\mathbf{p} - \mathbf{k} - \mathbf{q}, \epsilon_{\mathbf{k}} - \nu) [n_B(\epsilon_{\mathbf{k}} - \nu) + n_F(-\nu)] \int \frac{d\epsilon}{\pi} A(\mathbf{p}, \epsilon) B(\mathbf{q}, \nu - \epsilon) [n_B(\nu - \epsilon) + n_F(\epsilon)]$$

First, we have

$$\tau^{-1}(\mathbf{k}) = \frac{1}{kT} \frac{1}{-\frac{\partial f}{\partial \epsilon_{\mathbf{k}}}} \sum_{\mathbf{p}} P_{\mathbf{k}\mathbf{p}}$$

$$\begin{aligned} \tau^{-1}(\mathbf{k}) &= \frac{kT \left(-\frac{\partial f}{\partial \epsilon_{\mathbf{k}}} \right)}{kT \left(-\frac{\partial f}{\partial \epsilon_{\mathbf{k}}} \right)} \sum_{\mathbf{p}} \left| \tilde{\lambda}(\mathbf{p}, \mathbf{k}) \right|^2 \sum_{\mathbf{q}} \int \frac{d\nu}{\pi} \chi''(\mathbf{p} - \mathbf{k} - \mathbf{q}, \epsilon_{\mathbf{k}} - \nu) [n_B(\epsilon_{\mathbf{k}} - \nu) + n_F(-\nu)] \int \frac{d\epsilon}{\pi} A(\mathbf{p}, \epsilon) B(\mathbf{q}, \nu - \epsilon) [n_B(\nu - \epsilon) + n_F(\epsilon)] \\ &= -2\Sigma''(\mathbf{k}, \omega = \epsilon_{\mathbf{k}}) \end{aligned}$$

Transport time

going back to the master variational formula for the transport time,

$$\tilde{\tau}_{tr}^{-1}(\omega) = \frac{2\pi}{\hbar kT} \frac{\sum_{\mathbf{k}\mathbf{k}'} (v_{\mathbf{k}z} - v_{\mathbf{k}'z})^2 P_{\mathbf{k}\mathbf{k}'} \delta(\omega - \epsilon_{\mathbf{k}})}{\sum_{\mathbf{k}} \left(-\frac{\partial f}{\partial \epsilon_{\mathbf{k}}} \right) 2v_{\mathbf{k}z}^2},$$

we now plug in our expression for the transition rate. This can be done straight forwardly

$$\tilde{\tau}_{tr}^{-1}(\omega) = \frac{2\pi \sum_{\mathbf{k}\mathbf{k}'} (v_{\mathbf{k}z} - v_{\mathbf{p}z})^2 \left(-\frac{\partial f}{\partial \epsilon_{\mathbf{k}}} \right) \left| \tilde{\lambda}(\mathbf{p}, \mathbf{k}) \right|^2 \sum_{\mathbf{q}} \int \frac{d\nu}{\pi} \chi''(\mathbf{p} - \mathbf{k} - \mathbf{q}, \epsilon_{\mathbf{k}} - \nu) [n_B(\epsilon_{\mathbf{k}} - \nu) + n_F(-\nu)] \int \frac{d\epsilon}{\pi} A(\mathbf{p}, \epsilon) B(\mathbf{q}, \nu - \epsilon)}{\sum_{\mathbf{k}} \left(-\frac{\partial f}{\partial \epsilon_{\mathbf{k}}} \right) 2v_{\mathbf{k}z}^2}$$

Simplifying by enforcing the delta function

$$\tilde{\tau}_{tr}^{-1}(\omega) = \frac{2\pi \sum_{\mathbf{k}\mathbf{p}} (v_{\mathbf{k}z} - v_{\mathbf{p}z})^2 \left(-\frac{\partial f}{\partial \omega}\right) \left|\tilde{\lambda}(\mathbf{p}, \mathbf{k})\right|^2 \sum_{\mathbf{q}} \int \frac{d\nu}{\pi} \chi''(\mathbf{p} - \mathbf{k} - \mathbf{q}, \omega - \nu) [n_B(\omega - \nu) + n_F(-\nu)] \int \frac{d\epsilon}{\pi} A(\mathbf{p}, \epsilon) B(\mathbf{q}, \nu - \epsilon) [n_B(\nu - \epsilon) + n_F(-\epsilon)]}{\sum_{\mathbf{k}} \left(-\frac{\partial f}{\partial \epsilon_{\mathbf{k}}}\right) 2v_{\mathbf{k}z}^2}$$

Focusing for a moment on the last portion of the numerator

$$\begin{aligned} & \int \frac{d\epsilon}{\pi} A(\mathbf{p}, \epsilon) B(\mathbf{q}, \nu - \epsilon) [n_B(\nu - \epsilon) + n_F(-\epsilon)] \\ &= \int \frac{d\epsilon}{\pi} A(\mathbf{p}, \nu - \epsilon) (B^-(\mathbf{q}, \epsilon) - B^+(\mathbf{q}, \epsilon)) [n_B(\epsilon) + n_F(\epsilon + \nu)] \\ &= \int \frac{d\epsilon}{\pi} A(\mathbf{p}, \nu - \epsilon) B^-(\mathbf{q}, \epsilon) [n_B(\epsilon) + n_F(\epsilon + \nu)] \\ &\quad - \int \frac{d\epsilon}{\pi} A(\mathbf{p}, \nu - \epsilon) B^+(\mathbf{q}, \epsilon) [n_B(\epsilon) + n_F(\epsilon + \nu)] \\ &= \int \frac{d\epsilon}{\pi} A(\mathbf{p}, \nu - \epsilon) B^-(\mathbf{q}, \epsilon) [n_B(\epsilon) + n_F(\epsilon + \nu)] \\ &\quad - \int \frac{d\epsilon}{\pi} A(\mathbf{p}, \nu + \epsilon) B^+(\mathbf{q}, -\epsilon) [n_B(-\epsilon) + n_F(-\epsilon + \nu)] \\ &= \int \frac{d\epsilon}{\pi} A(\mathbf{p}, \nu - \epsilon) B^-(\mathbf{q}, \epsilon) [n_B(\epsilon) + n_F(\epsilon + \nu)] \\ &\quad + \int \frac{d\epsilon}{\pi} A(\mathbf{p}, \nu + \epsilon) B^-(\mathbf{q}, \epsilon) [n_B(\epsilon) + n_F(\epsilon - \nu)] \\ &\approx \int \frac{d\epsilon}{\pi} A(\mathbf{p}, 0) B^-(\mathbf{q}, \epsilon) [2n_B(\epsilon) + n_F(\epsilon + \nu) + n_F(\epsilon - \nu)] \end{aligned}$$

where

$$B(\mathbf{q}, \epsilon) = B^-(\mathbf{q}, \epsilon) - B^+(\mathbf{q}, \epsilon)$$

We then have

$$\tilde{\tau}_{tr}^{-1}(\omega) = \frac{2\pi \sum_{\mathbf{k}\mathbf{p}\mathbf{q}} (v_{\mathbf{k}z} - v_{\mathbf{p}z})^2 \left|\tilde{\lambda}(\mathbf{p}, \mathbf{k})\right|^2 \delta(\omega - \epsilon_{\mathbf{k}}) A(\mathbf{p}, 0) \int \frac{d\nu}{\pi} \chi''(\mathbf{p} - \mathbf{k} - \mathbf{q}, \omega - \nu) [n_B(\omega - \nu) + n_F(-\nu)] \int \frac{d\epsilon}{\pi} B^-(\mathbf{q}, \epsilon) [2n_B(\epsilon) + n_F(\epsilon + \nu) + n_F(\epsilon - \nu)]}{\sum_{\mathbf{k}} \left(-\frac{\partial f}{\partial \epsilon_{\mathbf{k}}}\right) 2v_{\mathbf{k}z}^2}$$

Now we assume Einstein phonons and take the local approximation for χ'' . In this case, we have

$$\tilde{\tau}_{tr}^{-1}(\omega) \approx \frac{2\pi \sum_{\mathbf{k}\mathbf{p}\mathbf{q}} (v_{\mathbf{k}z} - v_{\mathbf{p}z})^2 \left|\tilde{\lambda}(\mathbf{p}, \mathbf{k})\right|^2 \delta(\omega - \epsilon_{\mathbf{k}}) \delta(\epsilon_{\mathbf{p}}) \int \frac{d\nu}{\pi} \int d\epsilon \chi''(\omega - \nu) B^-(\epsilon) [n_B(\omega - \nu) + n_F(-\nu)] [2n_B(\epsilon) + n_F(\nu + \epsilon) + n_F(\nu - \epsilon)]}{\sum_{\mathbf{k}} \left(-\frac{\partial f}{\partial \epsilon_{\mathbf{k}}}\right) 2v_{\mathbf{k}z}^2}$$

Now we approximate $\delta(\omega - \epsilon_{\mathbf{k}}) \sim \delta(\epsilon_{\mathbf{k}})$ and define

$$\begin{aligned} & \frac{\sum_{\mathbf{k}\mathbf{p}} (v_{\mathbf{k}z} - v_{\mathbf{p}z})^2 \left|\tilde{\lambda}(\mathbf{p}, \mathbf{k})\right|^2 \delta(\epsilon_{\mathbf{k}}) \delta(\epsilon_{\mathbf{p}})}{\sum_{\mathbf{k}} \left(-\frac{\partial f}{\partial \epsilon_{\mathbf{k}}}\right) 2v_{\mathbf{k}z}^2} \\ &\approx \frac{\sum_{\mathbf{k}\mathbf{p}} (v_{\mathbf{k}z} - v_{\mathbf{p}z})^2 \left|\tilde{\lambda}(\mathbf{p}, \mathbf{k})\right|^2 \delta(\epsilon_{\mathbf{k}}) \delta(\epsilon_{\mathbf{p}})}{2 \sum_{\mathbf{k}} \delta(\epsilon_{\mathbf{k}}) v_{\mathbf{k}z}^2} \\ &= \Lambda_{tr} \end{aligned}$$

we will omit the factor of

$$\frac{2\pi}{\hbar}$$

briefly to simplify notation

With this, we have

$$\tilde{\tau}_{tr}^{-1}(\omega) \approx \Lambda_{tr} \int \frac{d\nu}{\pi} \int d\epsilon \chi''(\omega - \nu) B^-(\epsilon) [n_B(\omega - \nu) + n_F(-\nu)] [2n_B(\epsilon) + n_F(\nu + \epsilon) + n_F(\epsilon - \nu)] \left(-\frac{\partial f}{\partial \omega} \right)$$

So that the transport time will be given by

$$\tau_{tr}^{-1} = \Lambda_{tr} \int \frac{d\nu}{\pi} \int d\epsilon \int d\omega \chi''(\omega - \nu) B^-(\epsilon) [n_B(\omega - \nu) + n_F(-\nu)] [2n_B(\epsilon) + n_F(\epsilon + \nu) + n_F(\epsilon - \nu)] \left(-\frac{\partial f}{\partial \omega} \right)$$

To simplify a bit the calculation we make the change of variables

$$\tau_{tr}^{-1} = \Lambda_{tr} \int \frac{d\nu}{\pi} \int d\epsilon \int d\omega \chi''(\nu) B^-(\epsilon) [n_B(\nu) + n_F(\nu - \omega)] [2n_B(\epsilon) + n_F(\omega - \nu + \epsilon) + n_F(\epsilon + \nu - \omega)] \left(-\frac{\partial f}{\partial \omega} \right)$$

We will use in what follows

$$\begin{aligned} B(\nu) &= \frac{\pi \hbar}{2M\omega_0} (\delta(\nu - \omega_0) - \delta(\nu + \omega_0)) \\ &= B^-(\nu) - B^+(\nu) \end{aligned}$$

Simplified transport time

In case we are at sufficiently low temperatures we have

$$\begin{aligned} \tau_{tr}^{-1} &= \Lambda_{tr} \int \frac{d\nu}{\pi} \int d\epsilon \int d\omega \chi''(\nu) B^-(\epsilon) [n_B(\nu) + n_F(\nu)] [2n_B(\epsilon) + n_F(-\nu + \epsilon) + n_F(\epsilon + \nu)] \\ &= \Lambda_{tr} \int \frac{d\nu}{\pi} \int d\epsilon \chi''(\nu) B^-(\epsilon) [n_B(\nu) + n_F(\nu)] [2n_B(\epsilon) + n_F(\nu + \epsilon) + n_F(\epsilon - \nu)] \\ &= \frac{\hbar}{2M\omega_0} \Lambda_{tr} \int d\nu \chi''(\nu) [n_B(\nu) + n_F(\nu)] [2n_B(\omega_0) + n_F(\nu + \omega_0) + n_F(\omega_0 - \nu)] \\ &= \frac{\hbar}{2M\omega_0} \Lambda_{tr} \int d\nu \chi''(\nu) \text{csch}(\beta\nu) [2n_B(\omega_0) + n_F(\nu + \omega_0) + n_F(\omega_0 - \nu)] \\ &= \frac{\hbar}{2M\omega_0} \Lambda_{tr} \int d\nu \chi''(\nu) \text{csch}(\beta\nu) [2n_B(\omega_0) + 1 + n_F(\omega_0 - \nu) - n_F(-\nu - \omega_0)] \end{aligned}$$

The integral in the end can be done analytically

$$\int \frac{d\nu}{\pi} \chi''(\nu) \text{csch}(\beta\nu) [2n_B(\omega_0) + n_F(\nu + \omega_0) + n_F(\nu - \omega_0)] = \kappa \alpha \pi \frac{1}{\pi \alpha \beta J} \int_{-\alpha \pi \beta J}^{\alpha \pi \beta J} d\nu \nu \text{csch}(\nu) \left[2n_B(\omega_0) + 1 + \frac{1}{e^{\beta \omega_0 - \nu} + 1} - \frac{1}{e^{-\beta \omega_0 - \nu} + 1} \right]$$

For the first integral above, we have

$$\frac{1}{y} \int_{-y}^y d\nu \nu \text{csch}(\nu) = Z_1(y)$$

$$Z_1(y) = -\frac{y}{2} - \log[(e^{-y} + 1)(e^y + 1)] + \frac{\text{Li}_2(-e^{-y}) - \text{Li}_2(-e^y) - 2\text{Li}_2(1 - e^y)}{y}$$

For the second and third integrals, we have

$$\begin{aligned} \pm \frac{1}{y} \int_{-y}^y d\nu \nu \text{csch}(\nu) \frac{1}{e^{\pm\beta\omega_0-\nu} + 1} &= \pm [n_F(\pm\omega_0) Z_2(\alpha\pi\beta J, \pm\beta\omega_0) + n_B(\pm\omega_0) Z_3(\alpha\pi\beta J, \pm\beta\omega_0)] \\ &= \pm [n_F(\pm\omega_0) Z_2(\alpha\pi\beta J, \beta\omega_0) + n_B(\pm\omega_0) Z_3(\alpha\pi\beta J, \beta\omega_0)] \end{aligned}$$

$$Z_1(y, z) = - \left[y + \frac{z^2}{2y} - \frac{\pi^2}{6y} + \log \left(\frac{\cosh(y) + \cosh(z)}{\cosh(y) - 1} \right) + \frac{\text{Li}_2(-e^{y-z}) + \text{Li}_2(-e^{y+z}) + 2\text{Li}_2(e^{-y})}{y} \right]$$

$$Z_2(y, z) = - \left[\frac{z^2}{2y} + \log \left(\frac{\cosh(y) + \cosh(z)}{\cosh(y) + 1} \right) + \frac{\text{Li}_2(-e^{y-z}) + \text{Li}_2(-e^{y+z}) - 2\text{Li}_2(-e^y)}{y} \right]$$

Overall we have

$$\begin{aligned} \frac{\hbar\kappa\alpha\pi}{2M\omega_0} \Lambda_{tr} [(2n_B(\omega_0) + 1) Z_1(\alpha\pi\beta J) + [n_F(\omega_0) - n_F(-\omega_0)] Z_2(\alpha\pi\beta J, \beta\omega_0) + [n_B(\omega_0) - n_B(-\omega_0)] Z_3(\alpha\pi\beta J, \beta\omega_0)] \\ \frac{\hbar\kappa\alpha\pi}{2M\omega_0} \Lambda_{tr} [(2n_B(\omega_0) + 1) Z_1(\alpha\pi\beta J) + [2n_F(\omega_0) - 1] Z_2(\alpha\pi\beta J, \beta\omega_0) + [2n_B(\omega_0) + 1] Z_3(\alpha\pi\beta J, \beta\omega_0)] \\ \frac{\hbar\kappa\alpha\pi}{2M\omega_0} \Lambda_{tr} [(2n_B(\omega_0) + 1) (Z_1(\alpha\pi\beta J) + Z_3(\alpha\pi\beta J, \beta\omega_0)) + [2n_F(\omega_0) - 1] Z_2(\alpha\pi\beta J, \beta\omega_0)] \end{aligned}$$

We then conclude that

$$\tau_{tr}^{-1} = \frac{\hbar\kappa\alpha\pi}{2M\omega_0} \Lambda_{tr} [(2n_B(\omega_0) + 1) (Z_1(\alpha\pi\beta J) + Z_3(\alpha\pi\beta J, \beta\omega_0)) + [2n_F(\omega_0) - 1] Z_2(\alpha\pi\beta J, \beta\omega_0)]$$

More physical average

taking the transport time from before we now calculate the integral over ω first

$$\tau_{tr}^{-1} = \Lambda_{tr} \int \frac{d\nu}{\pi} \int d\epsilon \int d\omega \chi''(\nu) B^-(\epsilon) [n_B(\nu) + n_F(\nu - \omega)] [2n_B(\epsilon) + n_F(\omega - \nu + \epsilon) + n_F(\epsilon + \nu - \omega)] \left(-\frac{\partial f}{\partial \omega} \right)$$

However, it is still possible for an optical mode to calculate the integral over ϵ , this again yields

$$\tau_{tr}^{-1} = \frac{\hbar}{2M\omega_0} \Lambda_{tr} \int d\nu \int d\omega \chi''(\nu) [n_B(\nu) + n_F(\nu - \omega)] [2n_B(\omega_0) + n_F(\omega + \omega_0 - \nu) + n_F(\omega_0 + \nu - \omega)] \left(-\frac{\partial f}{\partial \omega} \right)$$

The integral over ω can be performed since it only involves the occupation functions

$$\begin{aligned} \int d\omega [n_B(\nu) + n_F(\nu - \omega)] [2n_B(\omega_0) + n_F(\omega + \omega_0 - \nu) + n_F(\omega_0 + \nu - \omega)] \left(-\frac{\partial f}{\partial \omega} \right) \\ = \frac{1}{4} \text{csch} \left(\frac{\beta\nu}{2} \right) \text{csch} \left(\frac{\beta\omega_0}{2} \right) \left(\frac{\beta\nu - \beta\omega_0}{2} \text{csch} \left(\frac{\beta\nu - \beta\omega_0}{2} \right) + \frac{\beta\nu + \beta\omega_0}{2} \text{csch} \left(\frac{\beta\nu + \beta\omega_0}{2} \right) \right) \end{aligned}$$

The integral that we have to calculate is

$$\tau_{tr}^{-1} = \frac{\hbar}{2M\omega_0} \Lambda_{tr} \int d\nu \chi''(\nu) \left[\frac{1}{4} \text{csch} \left(\frac{\beta\nu}{2} \right) \text{csch} \left(\frac{\beta\omega_0}{2} \right) \left(\frac{\beta\nu - \beta\omega_0}{2} \text{csch} \left(\frac{\beta\nu - \beta\omega_0}{2} \right) + \frac{\beta\nu + \beta\omega_0}{2} \text{csch} \left(\frac{\beta\nu + \beta\omega_0}{2} \right) \right) \right]$$

which can be expressed as

$$\begin{aligned}
\tau_{tr}^{-1} &= \frac{\hbar \kappa \alpha \pi}{2M\omega_0} \Lambda_{tr} \frac{1}{\pi \alpha \beta J} \int_{-\alpha \pi \beta J}^{\alpha \pi \beta J} d\nu \, \nu \left[\frac{1}{4} \operatorname{csch} \left(\frac{\nu}{2} \right) \operatorname{csch} \left(\frac{\beta \omega_0}{2} \right) \left(\frac{\nu - \beta \omega_0}{2} \operatorname{csch} \left(\frac{\nu - \beta \omega_0}{2} \right) + \frac{\nu + \beta \omega_0}{2} \operatorname{csch} \left(\frac{\nu + \beta \omega_0}{2} \right) \right) \right] \\
&= \frac{\hbar \kappa \alpha \pi}{2M\omega_0} \Lambda_{tr} I(\alpha \pi \beta J, \beta \omega_0)
\end{aligned}$$

So the relevant integral is

$$I(y, z) = \frac{1}{y} \int_{-y}^y dx \, x \left[\frac{1}{4} \operatorname{csch} \left(\frac{x}{2} \right) \operatorname{csch} \left(\frac{z}{2} \right) \left(\frac{x - z}{2} \operatorname{csch} \left(\frac{x - z}{2} \right) + \frac{\nu + z}{2} \operatorname{csch} \left(\frac{x + z}{2} \right) \right) \right]$$

This integral gives the result

$$I(y, z) = \frac{z}{\cosh(z) - 1} \mathcal{J}(y, z)$$

with

$$\begin{aligned}
\mathcal{J}(y, z) &= \frac{-2z \operatorname{Li}_2(e^{-y}) + (2y + z) \operatorname{Li}_2(e^{-y-z}) + (z - 2y) \operatorname{Li}_2(e^{z-y}) + 2 \operatorname{Li}_3(e^{-y-z}) - 2 \operatorname{Li}_3(e^{z-y})}{2yz} \\
&+ \frac{z^2}{12y} - \frac{1}{2} \log((e^y - e^z)(e^{y+z} - 1)) + \frac{y \log(\sinh(\frac{y-z}{2}) \operatorname{csch}(\frac{y+z}{2}))}{2z} + \frac{y}{2} + \frac{\pi^2}{3y} + \log(e^y - 1) + \frac{z}{2}
\end{aligned}$$

$$\tau_{tr}^{-1} = \frac{\hbar \kappa \alpha \pi}{M\omega_0^2} \Lambda_{tr} \frac{\beta \omega_0^2}{\cosh(\beta \omega_0) - 1} \mathcal{J}(\alpha \pi \beta J, \beta \omega_0)$$

where we have

$$\mathcal{J}(y \rightarrow 0^+, z) = 1$$

Which then has

$$\frac{\hbar \kappa \alpha \pi}{M\omega_0^2} \Lambda_{tr} = \lambda_{tr, ME}$$

then

$$\tau_{tr}^{-1} = \frac{2\pi}{\hbar} \lambda_{tr, ME} \frac{\beta \omega_0^2}{\cosh(\beta \omega_0) - 1} \mathcal{J}(\alpha \pi \beta J, \beta \omega_0)$$

where we reinstated the factor of

$$\frac{2\pi}{\hbar}$$

Calculating the magnetoelastic coupling

In the definition of the magnetoelastic coupling, we have several parts

$$\Lambda_{tr} = \frac{\sum_{\mathbf{k}\mathbf{p}} (v_{\mathbf{k}z} - v_{\mathbf{p}z})^2 \left| \tilde{\lambda}(\mathbf{p}, \mathbf{k}) \right|^2 \delta(\epsilon_{\mathbf{k}}) \delta(\epsilon_{\mathbf{p}})}{2 \sum_{\mathbf{k}} \delta(\epsilon_{\mathbf{k}}) v_{\mathbf{k}z}^2}$$

where we had

$$\Lambda_{tr} = 2\lambda^2 \frac{\sum_{\mathbf{k}\mathbf{p}} (v_{\mathbf{k}z} - v_{\mathbf{p}z})^2 |g(\mathbf{p}, \mathbf{k})|^2 \delta(\epsilon_{\mathbf{k}}) \delta(\epsilon_{\mathbf{p}})}{2 \sum_{\mathbf{k}} \delta(\epsilon_{\mathbf{k}}) v_{\mathbf{k}z}^2}$$

with

$$g(\mathbf{k}, \mathbf{k}') = J_K [2 \cos(k_z a_z)] [2 \cos(k'_z a_z)]$$

and

$$J_K = \frac{4g^2}{U}$$

so that we have

$$\Lambda_{tr} = 2\lambda^2 J_K^2 \frac{\sum_{\mathbf{k}\mathbf{p}} (v_{\mathbf{k}z} - v_{\mathbf{p}z})^2 [2 \cos(k_z a_z)]^2 [2 \cos(p_z a_z)]^2 \delta(\epsilon_{\mathbf{k}}) \delta(\epsilon_{\mathbf{p}})}{2 \sum_{\mathbf{k}} \delta(\epsilon_{\mathbf{k}}) v_{\mathbf{k}z}^2}$$

Now we model the dispersion

$$\epsilon_{\mathbf{k}} = \epsilon_{\parallel \mathbf{k}} - 2t_{\perp} \cos k_z a_z$$

Such that the velocity is given by

$$v_{\mathbf{k}z} = 2t_{\perp} a_z \sin k_z a_z$$

Since this is odd in momentum, we have

$$\Lambda_{tr} = 2\lambda^2 J_K^2 \frac{\sum_{\mathbf{k}\mathbf{p}} v_{\mathbf{k}z}^2 [2 \cos(k_z a_z)]^2 [2 \cos(p_z a_z)]^2 \delta(\epsilon_{\mathbf{k}}) \delta(\epsilon_{\mathbf{p}})}{\sum_{\mathbf{k}} \delta(\epsilon_{\mathbf{k}}) v_{\mathbf{k}z}^2}$$

Replacing our expression for the velocity, we have

$$\Lambda_{tr} = 2\lambda^2 J_K^2 \frac{\sum_{\mathbf{k}} [2 \sin(k_z a_z)]^2 [2 \cos(k_z a_z)]^2 \delta(\epsilon_{\mathbf{k}}) \sum_{\mathbf{p}} [2 \cos(p_z a_z)]^2 \delta(\epsilon_{\mathbf{p}})}{\sum_{\mathbf{k}} \delta(\epsilon_{\mathbf{k}}) [2 \sin(k_z a_z)]^2}$$

we see that the factor of the bandwidth cancels out explicitly, we have then to calculate three integrals

$$\frac{1}{N} \sum_{\mathbf{k}} [2 \sin(k_z a_z)]^2 \delta(\epsilon_{\mathbf{k}})$$

$$\frac{1}{N} \sum_{\mathbf{k}} [2 \sin(k_z a_z)]^2 [2 \cos(k_z a_z)]^2 \delta(\epsilon_{\mathbf{k}})$$

$$\frac{1}{N} \sum_{\mathbf{p}} [2 \cos(p_z a_z)]^2 \delta(\epsilon_{\mathbf{p}})$$

the sum over planes gives factors of the 2d density of states, while the remaining parts are only integrals over the k_z direction

$$\Lambda_{tr} = 2\lambda^2 J_K^2 \nu_0 \frac{a_z \int \frac{dk_z}{2\pi} [2 \sin(k_z a_z)]^2 [2 \cos(k_z a_z)]^2 a_z \int \frac{dp_z}{2\pi} [2 \cos(p_z a_z)]^2}{a_z \int \frac{dk_z}{2\pi} [2 \sin(k_z a_z)]^2}$$

the last part only gives a numerical factor that can be evaluated right away

$$a_z \int \frac{dk_z}{2\pi} [2 \sin(k_z a_z)]^2 [2 \cos(k_z a_z)]^2 = a_z \int \frac{dk_z}{2\pi} [2 \sin(k_z a_z)]^2 = a_z \int \frac{dp_z}{2\pi} [2 \cos(p_z a_z)]^2 = 2$$

$$\Lambda_{tr} = 4\lambda^2 J_K^2 \nu_0$$

With this, we find that

$$\begin{aligned} \lambda_{tr,ME} &= 2\pi\kappa\alpha \frac{J_K^2 \nu_0}{\hbar\omega_0} \frac{2\hbar\lambda^2}{M\omega_0} \\ &= 4\pi\kappa\alpha \frac{J_K^2 \nu_0}{\hbar\omega_0} \frac{\hbar\lambda^2}{M\omega_0} \end{aligned}$$

Or in terms of the notation in the paper

$$\lambda_{tr,ME} = 8\pi S(S+1) \frac{\tilde{\alpha}\nu_0}{\hbar\omega_0} \frac{\hbar}{M\omega_0}$$

If I use the transport time derived for the Kondo scattering I get

$$\lambda_{tr,ME} = 2\pi\lambda_{K,tr} \frac{J_K}{\hbar\omega_0} \frac{\hbar\lambda^2}{M\omega_0}$$

Comparison with the bare optical modes calculation

In the high temperature limit we have in one case

$$\tau_{tr}^{-1} = \frac{2\pi\lambda_{E,tr}}{\hbar} k_b T$$

and in the other

$$\tau_{tr}^{-1} = \frac{2\pi\lambda_{tr,ME}}{\hbar} k_b T$$

where we had previously defined

$$\lambda_{E,tr} = 4\pi \frac{\hbar\alpha_D^2 \nu_0}{M\omega_0^2}$$

$$\lambda_{tr,ME} = 4\pi\kappa\alpha \frac{J_K^2 \nu_0}{\omega_0} \frac{\hbar\lambda^2}{M\omega_0}$$

taking the ratio of this scattering rates and keeping the band masses and the frequencies the same we get

$$\begin{aligned} \frac{\lambda_{tr,ME}}{\lambda_{E,tr}} &= \frac{4\pi\kappa\alpha \frac{J_K^2 \nu_0}{\omega_0} \frac{\hbar\lambda^2}{M\omega_0}}{4\pi \frac{\hbar\alpha_D^2 \nu_0}{M\omega_0^2}} \\ &= \frac{\kappa\alpha J_K^2 \lambda^2}{\alpha_D^2} \end{aligned}$$

If we say that

$$t_\perp \lambda \sim \alpha_D$$

and

$$J_K = \frac{4g^2}{U}$$

we have

$$\frac{\lambda_{ME,tr}}{\lambda_{E,tr}} = \frac{\kappa\alpha J_k^2}{t_\perp^2}$$

since we estimate both J_K and $t_\perp$ to be in the order of mev , the enhancement is something of order 1 .

Sum rule for the spin correlation function

Let's start with the value of the integral

$$\begin{aligned} \int S(\nu) d\nu &= \int \frac{\chi''(\nu)}{\nu\beta} = \frac{T\kappa}{TJ} \int_{-\alpha\pi J}^{\alpha\pi J} \frac{\nu d\nu}{\nu} \\ &= \frac{\kappa}{J} 2\alpha\pi J \\ &= 2\pi\alpha\kappa \\ &= C \end{aligned}$$

Now we compare this with the sum rule In general, we have

$$C = \frac{\mu_0\mu_B^2}{3k_B} g^2 J(J+1),$$

with J being the angular momentum, g the lande g-factor, μ_0 the permeability of free space, μ_B the bohr magneton and k_B the boltzmann constant.

Full resistivity calculation

Now that we have the transport time, the only missing part of the calculation is the mass tensor since the full expresion for the resistivity is given by

$$\rho_{\text{trial}} = \frac{\tau_{\text{tr}}^{-1}}{e^2 \left(\frac{n}{m}\right)_{zz}}$$

The relevan calculation is then

$$\left(\frac{n}{m}\right)_{zz} = -\frac{1}{V} \sum_{\mathbf{q}} \frac{\partial f}{\partial \epsilon_{\mathbf{q}}} v_{\mathbf{kz}}^2$$

at temperatures below the degeneracy temperature we have

$$\begin{aligned} \left(\frac{n}{m}\right)_{zz} &= \frac{1}{a_{xy}^2 a_z} \frac{1}{N} \sum_{\mathbf{q}} \delta(\epsilon_{\mathbf{k}}) v_{\mathbf{kz}}^2 \\ &= \frac{2\nu_0 \left(\frac{2t_\perp a_z}{\hbar}\right)^2}{a_{xy}^2 a_z} \\ &= 2\nu_0 \left(\frac{2t_\perp}{\hbar}\right)^2 \frac{a_z}{a_{xy}^2} \end{aligned}$$

Putting things together

$$\begin{aligned}
\frac{\tau_{tr}^{-1}}{e^2 \left(\frac{n}{m}\right)_{zz}} &= \frac{\frac{2\pi\lambda_{E,tr}}{\hbar} k_b T}{2\nu_0 \left(\frac{2t_\perp}{\hbar}\right)^2 \frac{a_z}{a_{xy}^2}} \\
&= \frac{\hbar}{e^2} \frac{a_{xy}^2}{a_z} \left[\frac{2\pi\lambda_{E,tr} k_b T}{2\nu_0 (2t_\perp)^2} \right] \\
&= \frac{\hbar}{e^2} \frac{a_{xy}^2}{a_z} \left[\frac{\pi\lambda_{E,tr} k_b T}{\nu_0 (2t_\perp)^2} \right]
\end{aligned}$$

For the full transport rate we would instead have

$$\rho_{\text{trial}} = \frac{\tau_{tr}^{-1}}{e^2 \left(\frac{n}{m}\right)_{zz}} = \frac{\hbar}{e^2} \frac{a_{xy}^2}{a_z} \left[\frac{\pi\lambda_{tr,ME}}{(2t_\perp)^2 \nu_0} \frac{\beta\omega_0^2}{\cosh(\beta\omega_0) - 1} \mathcal{J}(\alpha\pi\beta J, \beta\omega_0) \right]$$

KONDO SCATTERING AT HIGH TEMPS

We start once again from

$$\tilde{\tau}_{tr}^{-1}(\omega) = \frac{2\pi}{\hbar k T} \frac{\sum_{\mathbf{k}\mathbf{k}'} (v_{\mathbf{k}z} - v_{\mathbf{k}'z})^2 P_{\mathbf{k}\mathbf{k}'} \delta(\omega - \epsilon_{\mathbf{k}})}{\sum_{\mathbf{k}} \left(-\frac{\partial f}{\partial \epsilon_{\mathbf{k}}}\right) 2v_{\mathbf{k}z}^2},$$

with

$$\begin{aligned}
P_{\mathbf{k}\mathbf{p}} &= -kT \frac{\partial f_{\mathbf{p}}}{\partial \epsilon_{\mathbf{p}}} |g(\mathbf{k}, \mathbf{p})|^2 \chi''(\mathbf{p} - \mathbf{k}, \epsilon_{\mathbf{p}} - \epsilon_{\mathbf{k}}) (n_B(\epsilon_{\mathbf{p}} - \epsilon_{\mathbf{k}}) + f(-\epsilon_{\mathbf{k}})) \\
&= |g(\mathbf{k}, \mathbf{p})|^2 \chi''(\mathbf{k} - \mathbf{p}, \epsilon_{\mathbf{k}} - \epsilon_{\mathbf{p}}) n_B(\epsilon_{\mathbf{k}} - \epsilon_{\mathbf{p}}) f(\epsilon_{\mathbf{p}}) (1 - f(\epsilon_{\mathbf{k}})) \\
&= kT W_{\mathbf{k}\mathbf{p}}
\end{aligned}$$

We then have

$$\tilde{\tau}_{tr}^{-1}(\omega) = \frac{2\pi}{\hbar} \frac{\sum_{\mathbf{p}\mathbf{k}} (v_{\mathbf{p}z} - v_{\mathbf{k}z})^2 \left(-\frac{\partial f_{\mathbf{p}}}{\partial \epsilon_{\mathbf{p}}}\right) |g(\mathbf{k}, \mathbf{p})|^2 \chi''(\mathbf{p} - \mathbf{k}, \epsilon_{\mathbf{p}} - \epsilon_{\mathbf{k}}) (n_B(\epsilon_{\mathbf{p}} - \epsilon_{\mathbf{k}}) + f(-\epsilon_{\mathbf{k}})) \delta(\omega - \epsilon_{\mathbf{p}})}{\sum_{\mathbf{k}} \left(-\frac{\partial f}{\partial \epsilon_{\mathbf{k}}}\right) 2v_{\mathbf{k}z}^2},$$

Enforcing the delta function and using the usual spectral function trick

$$\tilde{\tau}_{tr}^{-1}(\omega) = \frac{2\pi}{\hbar} \frac{\int d\epsilon \sum_{\mathbf{p}\mathbf{k}} (v_{\mathbf{p}z} - v_{\mathbf{k}z})^2 \left(-\frac{\partial f}{\partial \omega}\right) |g(\mathbf{k}, \mathbf{p})|^2 \chi''(\mathbf{p} - \mathbf{k}, \omega - \epsilon) (n_B(\omega - \epsilon) + f(-\epsilon)) \delta(\omega - \epsilon_{\mathbf{p}}) \delta(\epsilon - \epsilon_{\mathbf{k}})}{\sum_{\mathbf{k}} \left(-\frac{\partial f}{\partial \epsilon_{\mathbf{k}}}\right) 2v_{\mathbf{k}z}^2}$$

using the local approximation for the spin spectral function

$$\tilde{\tau}_{tr}^{-1}(\omega) = \frac{2\pi}{\hbar} \frac{\int d\epsilon \sum_{\mathbf{p}\mathbf{k}} (v_{\mathbf{p}z} - v_{\mathbf{k}z})^2 \left(-\frac{\partial f}{\partial \omega}\right) |g(\mathbf{k}, \mathbf{p})|^2 \chi''(\omega - \epsilon) (n_B(\omega - \epsilon) + f(-\epsilon)) \delta(\omega - \epsilon_{\mathbf{p}}) \delta(\epsilon - \epsilon_{\mathbf{k}})}{\sum_{\mathbf{k}} \left(-\frac{\partial f}{\partial \epsilon_{\mathbf{k}}}\right) 2v_{\mathbf{k}z}^2}$$

Defining

$$\frac{\sum_{\mathbf{p}\mathbf{k}} (v_{\mathbf{p}z} - v_{\mathbf{k}z})^2 |g(\mathbf{k}, \mathbf{p})|^2 \delta(\omega - \epsilon_{\mathbf{p}}) \delta(\epsilon - \epsilon_{\mathbf{k}})}{\sum_{\mathbf{k}} 2v_{\mathbf{k}z}^2 \left(-\frac{\partial f}{\partial \epsilon_{\mathbf{k}}}\right)} = \alpha_{tr}^2 F(\omega, \epsilon)$$

we have

$$\begin{aligned}\tilde{\tau}_{tr}^{-1} &= \frac{2\pi}{\hbar} \int d\epsilon d\omega \alpha_{tr}^2 F(\omega, \epsilon) \chi''(\omega - \epsilon) (n_B(\omega - \epsilon) + f(-\epsilon)) \left(-\frac{\partial f}{\partial \omega}\right) \\ &\sim \frac{2\pi}{\hbar} \alpha_{tr}^2 F(0, 0) \int d\epsilon d\omega \chi''(\omega - \epsilon) (n_B(\omega - \epsilon) + f(-\epsilon)) \left(-\frac{\partial f}{\partial \omega}\right)\end{aligned}$$

Changing variables, we have

$$= \frac{2\pi}{\hbar} \alpha_{tr}^2 F(0, 0) \int d\epsilon d\omega \chi''(\epsilon) (n_B(\epsilon) + f(\epsilon - \omega)) \left(-\frac{\partial f}{\partial \omega}\right)$$

Integrating over ω first,

$$\int d\omega f(\epsilon - \omega) \left(-\frac{\partial f}{\partial \omega}\right) = n_B(\epsilon) (n_B(\epsilon) + (1 + n_B(\epsilon))(\beta\epsilon - 1))$$

For the second integral, we have

$$\begin{aligned}&\frac{2\pi}{\hbar} \alpha_{tr}^2 F(0, 0) \int d\epsilon \chi''(\epsilon) (n_B(\epsilon) + n_B(\epsilon) (n_B(\epsilon) + (1 + n_B(\epsilon))(\beta\epsilon - 1))) \\ &= \frac{2\pi}{\hbar} \alpha_{tr}^2 F(0, 0) \int d\epsilon \chi''(\epsilon) n_B(\epsilon) (1 + n_B(\epsilon) + (1 + n_B(\epsilon))(\beta\epsilon - 1)) \\ &= \frac{2\pi}{\hbar} \alpha_{tr}^2 F(0, 0) \frac{\kappa}{TJ} \int_{-\alpha\pi J}^{\alpha\pi J} d\epsilon \epsilon n_B(\epsilon) (1 + n_B(\epsilon) + (1 + n_B(\epsilon))(\beta\epsilon - 1)) \\ &= \frac{2\pi}{\hbar} \alpha_{tr}^2 F(0, 0) \beta \frac{\kappa}{J} \int_{-\alpha\pi J}^{\alpha\pi J} d\epsilon \epsilon n_B(\epsilon) (1 + n_B(\epsilon) + (1 + n_B(\epsilon))(\beta\epsilon - 1)) \\ &= \frac{2\pi}{\hbar} \alpha_{tr}^2 F(0, 0) \beta \frac{\kappa}{J} \int_{-\alpha\pi J}^{\alpha\pi J} d\epsilon \epsilon n_B(\epsilon) (1 + n_B(\epsilon)) (1 + (\beta\epsilon - 1)) \\ &= \frac{2\pi}{\hbar} \alpha_{tr}^2 F(0, 0) \beta \frac{\kappa}{J} \int_{-\alpha\pi J}^{\alpha\pi J} d\epsilon \epsilon n_B(\epsilon) (1 + n_B(\epsilon)) (\beta\epsilon)\end{aligned}$$

Changing variables

$$\begin{aligned}&\beta \frac{\kappa}{J} \int_{-\alpha\pi J}^{\alpha\pi J} d\epsilon \epsilon n_B(\epsilon) (1 + n_B(\epsilon)) (\beta\epsilon) \\ &= \frac{\kappa}{\beta J} \int_{-\alpha\pi\beta J}^{\alpha\pi\beta J} d\epsilon \epsilon^2 \tilde{n}_B(\epsilon) (1 + \tilde{n}_B(\epsilon)) \\ &\quad \alpha\pi\kappa\mathcal{I}(\alpha\pi\beta J)\end{aligned}$$

where

$$\mathcal{I}(y) = \frac{1}{y} \int_{-y}^y d\epsilon \epsilon^2 \tilde{n}_B(\epsilon) (1 + \tilde{n}_B(\epsilon))$$

Performing this integral, we have

$$\mathcal{I}(y) = -\frac{4\text{Li}_2(e^{-y})}{y} - \frac{2y}{e^y - 1} - 4y + \frac{2\pi^2}{3y} + 4\log(e^y - 1)$$

Overall, we have

$$\tilde{\tau}_{tr}^{-1} = \frac{2\pi}{\hbar} \alpha_{tr}^2 F(0, 0) [\alpha\pi\kappa\mathcal{I}(\alpha\pi\beta J)]$$

We now evaluate

$$\frac{\sum_{\mathbf{p}\mathbf{k}} (v_{\mathbf{p}z} - v_{\mathbf{k}z})^2 |g(\mathbf{k}, \mathbf{p})|^2 \delta(\epsilon_{\mathbf{p}}) \delta(\epsilon_{\mathbf{k}})}{2 \sum_{\mathbf{k}} v_{\mathbf{k}z}^2 \left(-\frac{\partial f}{\partial \epsilon_{\mathbf{k}}}\right)} = \alpha_{tr}^2 F(0, 0)$$

Using our previous calculations we have

$$\alpha_{tr}^2 F(0, 0) = 2J_K^2 \nu_0$$

Then we have

$$\begin{aligned} \tilde{\tau}_{tr}^{-1} &= \frac{2\pi}{\hbar} [2\pi\alpha\kappa J_K^2 \nu_0 \mathcal{I}(\alpha\pi\beta J)] \\ &= \frac{2\pi}{\hbar} [\lambda_K J_K \mathcal{I}(\alpha\pi\beta J)] \end{aligned}$$

where we defined the dimensionless coupling

$$\lambda_K = 2\pi\alpha\kappa J_K \nu_0$$

Kondo resistivity

using the formula

$$\rho_{\text{trial}} = \frac{\tau_{\text{tr}}^{-1}}{e^2 \left(\frac{n}{m}\right)_{zz}}$$

The relevant calculation is then

$$\left(\frac{n}{m}\right)_{zz} = -\frac{1}{V} \sum_{\mathbf{q}} \frac{\partial f}{\partial \epsilon_{\mathbf{q}}} v_{\mathbf{k}z}^2$$

at temperatures below the degeneracy temperature we have

$$\begin{aligned} \left(\frac{n}{m}\right)_{zz} &= \frac{1}{a_{xy}^2 a_z} \frac{1}{N} \sum_{\mathbf{k}} \delta(\epsilon_{\mathbf{k}}) v_{\mathbf{k}z}^2 \\ &= \frac{2\nu_0 \left(\frac{t_{\perp} a_z}{\hbar}\right)^2}{a_{xy}^2 a_z} \\ &= 2\nu_0 \left(\frac{t_{\perp}}{\hbar}\right)^2 \frac{a_z}{a_{xy}^2} \end{aligned}$$

where we once again used

$$\frac{1}{N} \sum_{\mathbf{k}} \left[2 \frac{t_{\perp} a_z}{\hbar} \sin(k_z a_z) \right]^2 \delta(\epsilon_{\mathbf{k}}) = \nu_0 a_z \int_{-\pi/a_z}^{\pi/a_z} \frac{dk_z}{2\pi} \left[2 \frac{t_{\perp} a_z}{\hbar} \sin(k_z a_z) \right]^2 = 2\nu_0 \left(\frac{t_{\perp} a_z}{\hbar}\right)^2$$

if we include the sum over spins

$$\begin{aligned} \left(\frac{n}{m}\right)_{zz} &= 4\nu_0 \left(\frac{t_{\perp}}{\hbar}\right)^2 \frac{a_z}{a_{xy}^2} \\ &= 2\rho_0 \left(\frac{t_{\perp}}{\hbar}\right)^2 \frac{a_z}{a_{xy}^2} \end{aligned}$$

so that

$$\begin{aligned} \rho_{\text{trial}} &= \frac{\frac{2\pi}{\hbar} [\lambda_K J_K \mathcal{I}(\alpha\pi\beta J)]}{e^2 2\nu_0 \left(\frac{2t_{\perp}}{\hbar}\right)^2 \frac{a_z}{a_{xy}^2}} \\ &= \frac{\hbar a_{xy}^2}{e^2 a_z} \left[\frac{2\pi \lambda_K J_K \mathcal{I}(\alpha\pi\beta J)}{2\nu_0 (2t_{\perp})^2} \right] \end{aligned}$$

Full resistivity

we finally have

$$\begin{aligned}\rho_{\text{trial}} &= \frac{\tau_{\text{tr},ME}^{-1} + \tau_{\text{tr},K}^{-1}}{e^2 \left(\frac{n}{m}\right)_{zz}} = \frac{\hbar}{e^2} \frac{a_{xy}^2}{a_z} \left[\frac{2\pi \lambda_{tr,ME}}{2\nu_0 (2t_\perp)^2} \frac{\beta\omega_0^2}{\cosh(\beta\omega_0) - 1} \mathcal{J}(\alpha\pi\beta J, \beta\omega_0) + \frac{2\pi \lambda_K J_K \mathcal{I}(\alpha\pi\beta J)}{2\nu_0 (2t_\perp)^2} \right] \\ &= \frac{\hbar}{e^2} \frac{a_{xy}^2}{a_z} \left(\frac{1}{2\nu_0 (2t_\perp)^2} \right) \left[2\pi \frac{\lambda_{tr,ME}}{\beta} \mathcal{M}(\alpha\pi\beta J, \beta\omega_0) + 2\pi \lambda_K J_K \mathcal{I}(\alpha\pi\beta J) \right]\end{aligned}$$

with the new definition

$$\mathcal{M}(\alpha\pi\beta J, \beta\omega_0) = \frac{\beta^2 \omega_0^2}{\cosh(\beta\omega_0) - 1} \mathcal{J}(\alpha\pi\beta J, \beta\omega_0)$$

If we also added an optical mode, we would have

$$\begin{aligned}\tau_{tr}^{-1} &= \frac{1}{2} \frac{\pi}{\hbar} \lambda_E \beta \omega_0^2 \left[\text{csch}\left(\frac{\beta\omega_0}{2}\right) \right]^2 \\ &= \frac{2\pi}{\hbar} \frac{\lambda_E}{\beta} \left[\frac{\beta\omega_0}{2} \text{csch}\left(\frac{\beta\omega_0}{2}\right) \right]^2 \\ &= \frac{2\pi}{\hbar} \frac{\lambda_E}{\beta} \mathcal{K}(\beta\omega_0)\end{aligned}$$

with

$$\lambda_{E,tr} = 4\pi \frac{\hbar \alpha^2 \nu_0}{M \omega_0^2}$$

and

$$\mathcal{K}(\beta\omega_0) = \left[\frac{\beta\omega_0}{2} \text{csch}\left(\frac{\beta\omega_0}{2}\right) \right]^2$$

with this, the resistivity would be

$$= \frac{\hbar}{e^2} \frac{a_{xy}^2}{a_z} \left(\frac{1}{2\nu_0 (2t_\perp)^2} \right) \left[2\pi \lambda_K J_K \mathcal{I}(\alpha\pi\beta J) + 2\pi \frac{\lambda_{tr,ME}}{\beta} \mathcal{M}(\alpha\pi\beta J, \beta\omega_0) + 2\pi \frac{\lambda_{tr,E}}{\beta} \mathcal{K}(\beta\omega_0) \right]$$

the scaling functions

$$\mathcal{I}(y) = -\frac{4\text{Li}_2(e^{-y})}{y} - \frac{2y}{e^y - 1} - 4y + \frac{2\pi^2}{3y} + 4\log(e^y - 1)$$

$$\mathcal{K}(x) = \left[\frac{x}{2} \text{csch}\left(\frac{x}{2}\right) \right]^2$$

$$\mathcal{M}(y, z) = \frac{z^2}{\cosh(z) - 1} \mathcal{J}(y, z)$$

$$\begin{aligned}\mathcal{J}(y, z) &= \frac{-2z\text{Li}_2(e^{-y}) + (2y + z)\text{Li}_2(e^{-y-z}) + (z - 2y)\text{Li}_2(e^{z-y}) + 2\text{Li}_3(e^{-y-z}) - 2\text{Li}_3(e^{z-y})}{2yz} \\ &+ \frac{z^2}{12y} - \frac{1}{2} \log((e^y - e^z)(e^{y+z} - 1)) + \frac{y \log(\sinh(\frac{y-z}{2}) \text{csch}(\frac{y+z}{2}))}{2z} + \frac{y}{2} + \frac{\pi^2}{3y} + \log(e^y - 1) + \frac{z}{2}\end{aligned}$$

the constants

$$\begin{aligned}\lambda_K &= 2\pi\alpha\kappa J_K\nu_0 \\ &= 4\pi S(S+1)J_K\nu_0\end{aligned}$$

$$\lambda_{E,tr} = 4\pi \frac{\hbar\alpha^2\nu_0}{M\omega_0^2}$$

$$\begin{aligned}\lambda_{tr,ME} &= 4\pi\kappa\alpha \frac{J_K^2\nu_0}{\hbar\omega_0} \frac{\hbar\lambda^2}{M\omega_0} \\ &= 8\pi S(S+1) \frac{J_K^2\nu_0}{\hbar\omega_0} \frac{\hbar\lambda^2}{M\omega_0}\end{aligned}$$

and

$$J_K = \frac{4g^2}{U}$$

FOR THE SCATTERING RATES

$$\tau_{tr}^{-1} = \frac{2\pi}{\hbar} [\lambda_k J_K \mathcal{I}(\alpha\pi\beta J)]$$

$$\tau_{tr}^{-1} = \frac{2\pi}{\hbar} \frac{\lambda_{EME}}{\beta} \frac{\beta^2\omega_0^2}{\cosh(\beta\omega_0) - 1} \mathcal{J}(\alpha\pi\beta J, \beta\omega_0)$$

$$\tau_{tr}^{-1} = \frac{2\pi}{\hbar} \frac{\lambda_E}{\beta} \left[\frac{\beta\omega_0}{2} \text{csch}\left(\frac{\beta\omega_0}{2}\right) \right]^2$$

IN PLANE RESISTIVITIES

Having calculated the scattering rates we recover the in-plane resistivity from the expression

$$\rho_{\text{trial}} = \frac{\tau_{\text{tr}}^{-1}}{e^2 \left(\frac{n}{m}\right)_{xx}}$$

$$\left(\frac{n}{m}\right)_{nn} = -\frac{1}{V} \sum_{\mathbf{q}} \frac{\partial f}{\partial \epsilon_{\mathbf{q}}} (\mathbf{v}(\mathbf{q}) \cdot \mathbf{n})^2$$

So that we only need to calculate the mass tensor

$$\left(\frac{n}{m}\right)_{xx} = -\frac{1}{V} \sum_{\mathbf{q}} \frac{\partial f}{\partial \epsilon_{\mathbf{q}}} v_x(\mathbf{q})^2$$

if we assume an isotropic 2d system. We can perform this integral in a straightforward way in the low temperature limit

$$\left(\frac{n}{m}\right)_{xx} = \int \frac{d\mathbf{q}}{(2\pi)^3} \delta(\epsilon_{\mathbf{q}}) v_x(\mathbf{q})^2$$

for a circular FS we would have

$$\begin{aligned} \left(\frac{n}{m}\right)_{xx} &= \frac{1}{2} \frac{4\pi}{a_z} \int \frac{q dq d\theta}{(2\pi)^3} \delta\left(\frac{q^2 - p_F^2}{2m}\right) \frac{q^2}{m^2} \\ &= \frac{1}{2} \frac{4\pi}{a_z} \int \frac{q dq d\theta}{(2\pi)^3} \delta(q^2 - p_F^2) \frac{2mq^2}{m^2} \\ &= \frac{1}{2} \frac{4\pi}{a_z} \int \frac{dq d\theta}{(2\pi)^3} \delta(q^2 - p_F^2) \frac{2q^3}{m} \\ &= \frac{4\pi}{a_z} \int \frac{dq}{(2\pi)^3} \delta(q^2 - p_F^2) \frac{q^3}{m} \\ &= \frac{4\pi}{a_z} \int_0^\infty \frac{dq}{(2\pi)^3} \frac{\delta(q - p_F)}{2p_F} \frac{q^3}{m} \\ &= \frac{4\pi}{a_z} \frac{2\pi}{(2\pi)^3} \frac{p_F^2}{2m} \\ &= \frac{1}{m} \frac{1}{(2\pi)^3} \left(\frac{4\pi}{a_z}\right) \pi p_F^2 \\ &= \frac{1}{m} \frac{V_{BZ,occ}}{(2\pi)^3} \end{aligned}$$

If we include the sum over spins we have

$$\left(\frac{n}{m}\right)_{xx} = \frac{2}{m} \frac{V_{BZ,occ}}{(2\pi)^3}$$

Since we are at half filling in our setup, we have

$$\left(\frac{n}{m}\right)_{xx} = \frac{1}{m} \frac{V_{BZ}}{(2\pi)^3}$$

Estimate of the decay constant for pressure calculation.

We have that the hopping constant is given by

$$t = t_0 e^{-a/\ell}$$

the hybridization is given by

$$g = t_0 e^{-d_z/\ell}$$

the ratio of these two quantities is then

$$\frac{t}{g} = e^{d_z - a/\ell}$$

taking the log

$$\log \frac{t}{g} = \frac{d_z - a}{\ell}$$

so we have

$$\frac{1}{\ell} = \frac{1}{d_z - a} \log \frac{t}{g}$$

there is the strong assumption that the hopping and hybridization integrals are of the same order of magnitude. Then we can say that under pressure the hopping parameters are modified as follows:

$$t = t_0 e^{-a/\ell} \rightarrow t_0 e^{-(a-\varepsilon a)/\ell} = t e^{\varepsilon a/\ell}$$

while the out of plane frequencies are modified as

$$\omega_0 \rightarrow \omega_0 (1 + \varepsilon)$$

Estimation of the effective mass

$$\left(\frac{n}{m}\right)_{exp,c} = \frac{2}{m_c} \frac{V_{BZ,occ}}{(2\pi)^3}$$

$$\left(\frac{n}{m}\right)_{exp,ab} = \frac{2}{m_{ab}} \frac{V_{BZ,occ}}{(2\pi)^3}$$

then

$$\frac{\left(\frac{n}{m}\right)_{exp,c}}{\left(\frac{n}{m}\right)_{exp,ab}} = \frac{m_{ab}}{m_c}$$

we also have

$$m_{ab} = \frac{\hbar^2}{a_{xy}^2} \frac{1}{3t_1 + 9t_2}$$

and

$$m_c = \frac{\hbar^2}{a_z^2} \frac{1}{2t_z}$$

then we have

$$\frac{\left(\frac{n}{m}\right)_{exp,c}}{\left(\frac{n}{m}\right)_{exp,ab}} = \frac{a_z^2}{a_{xy}^2} \frac{2t_z}{3t_1 + 9t_2}$$

we have

$$\frac{\left(\frac{n}{m}\right)_{exp,c}}{\left(\frac{n}{m}\right)_{exp,ab}} \sim 0.002$$

and

$$\frac{a_z}{a_{xy}} \sim 3$$

so in the end we get

$$\begin{aligned} \frac{0.002}{9} \frac{(3t_1 + 9t_2)}{2} &= t_z \\ &= 0.01 \frac{a_{xy}^2}{a_z^2} (3t_1 + 9t_2) \end{aligned}$$

this leads to a hopping of the order of 0.1meV, where we expected a hopping of 0.1eV. This method, however, was consistent in extracting the hopping of the Cr compound in plane, where Now we try another method

$$\left(\frac{n}{m}\right)_{zz} = -\frac{1}{V} \sum_{\mathbf{q}} \frac{\partial f}{\partial \epsilon_{\mathbf{q}}} v_{\mathbf{k}z}^2$$

at temperatures below the degeneracy temperature we have

$$\begin{aligned} \left(\frac{n}{m}\right)_{zz} &= \frac{1}{a_{xy}^2 a_z} \frac{1}{N} \sum_{\mathbf{k}} \delta(\epsilon_{\mathbf{k}}) v_{\mathbf{k}z}^2 \\ &= \frac{2\nu_0 \left(\frac{t_{\perp} a_z}{\hbar}\right)^2}{a_{xy}^2 a_z} \\ &= 2\nu_0 \left(\frac{t_{\perp}}{\hbar}\right)^2 \frac{a_z}{a_{xy}^2} \\ \left(\frac{n}{m}\right)_{exp,c} &= \frac{2(2t_z) a_z^2}{\hbar^2} \frac{V_{BZ,occ}}{(2\pi)^3} \end{aligned}$$

so this would match if

$$\frac{V_{BZ,occ}}{(2\pi)^3} = \frac{\nu_0}{2} \frac{t_{\perp}}{a_z a_{xy}^2}$$

where we once again used

$$\frac{1}{N} \sum_{\mathbf{k}} \left[2 \frac{t_{\perp} a_z}{\hbar} \sin(k_z a_z) \right]^2 \delta(\epsilon_{\mathbf{k}}) = \nu_0 a_z \int_{-\pi/a_z}^{\pi/a_z} \frac{dk_z}{2\pi} \left[2 \frac{t_{\perp} a_z}{\hbar} \sin(k_z a_z) \right]^2 = 2\nu_0 \left(\frac{t_{\perp} a_z}{\hbar}\right)^2$$

If instead we calculate the

$$\begin{aligned} \left(\frac{n}{m}\right)_{xx} &= \frac{1}{a_{xy}^2 a_z} \frac{1}{N} \sum_{\mathbf{k}} \delta(\epsilon_{\mathbf{k}}) v_{\mathbf{k}x}^2 \\ &= \frac{2\nu_0 \left(\frac{t_{\perp} a_z}{\hbar}\right)^2}{a_{xy}^2 a_z} \\ &= 2\nu_0 \left(\frac{t_{\perp}}{\hbar}\right)^2 \frac{a_z}{a_{xy}^2} \\ \frac{1}{N} \sum_{\mathbf{k}} \delta(\epsilon_{\mathbf{k}}) v_{\mathbf{k}x}^2 &= \frac{1}{N} \sum_{\mathbf{k}} \delta(\epsilon_{\mathbf{k}}) \partial_x \epsilon_{\mathbf{k}} \partial_x \epsilon_{\mathbf{k}} \\ &= \frac{1}{N} \sum_{\mathbf{k}} \theta(-\epsilon_{\mathbf{k}}) \partial_x \partial_x \epsilon_{\mathbf{k}} \\ &= -\frac{1}{N} \sum_{\mathbf{k}} \partial_x \theta(-\epsilon_{\mathbf{k}}) \partial_x \epsilon_{\mathbf{k}} \\ &= \frac{1}{N} \sum_{\mathbf{k}} \delta(\epsilon_{\mathbf{k}}) \partial_x \epsilon_{\mathbf{k}} \partial_x \epsilon_{\mathbf{k}} \end{aligned}$$

Taking

$$\begin{aligned} &= \frac{1}{N} \sum_{\mathbf{k}} \theta(-\epsilon_{\mathbf{k}}) \partial_x \partial_x \epsilon_{\mathbf{k}} \\ &= \frac{1}{N} \sum_{\mathbf{k}} \theta(-\epsilon_{\mathbf{k}}) \partial_x \partial_x \epsilon_{\mathbf{k}} \end{aligned}$$

CALCULATION OF THE TRANSPORT RATE FROM ACOUSTIC PHONONS

The calculation is pretty much the same as for the einstein mode with

$$\tau_{tr}^{-1} = \frac{4\pi}{\hbar\beta} \int_0^\infty d\nu \frac{\alpha_{tr}^2 F(\nu)}{\nu} \left[\frac{\beta\nu}{2} \text{csch}\left(\frac{\beta\nu}{2}\right) \right]^2$$

with the main difference that now

$$\alpha_{tr}^2 F(\omega) = 2\lambda_A \left(\frac{\omega}{\omega_D} \right)^4 \theta(\omega - \omega_D)$$

this can be derived from

$$\frac{\sum_{\mathbf{k}'} |g(\mathbf{k}, \mathbf{k}')|^2 (v_{\mathbf{k}x} - v_{\mathbf{k}'x})^2 \delta(\epsilon_{\mathbf{k}}) \delta(\nu + \epsilon_{\mathbf{k}'}) \mathcal{B}^-(\mathbf{k} - \mathbf{k}', \nu)}{\sum_{\mathbf{k}} 2v_{\mathbf{k}z}^2 \left(-\frac{\partial f}{\partial \epsilon_{\mathbf{k}}} \right)} = \alpha_{tr\mathbf{k}}^2 F(\nu)$$

and

$$\sum_{\mathbf{k}} \alpha_{tr\mathbf{k}}^2 F(\nu) = \alpha_{tr}^2 F(\nu)$$

$$\begin{aligned} B(\mathbf{k}, \nu) &= \frac{\pi\hbar}{2M\omega_{\mathbf{k}}} (\delta(\nu - \omega_{\mathbf{k}}) - \delta(\nu + \omega_{\mathbf{k}})) \\ &= B^-(\mathbf{k}, \nu) - B^+(\mathbf{k}, \nu) \end{aligned}$$

so that we have

$$\frac{\sum_{\mathbf{k}\mathbf{k}'} |M(\mathbf{k}, \mathbf{k}')|^2 (v_{\mathbf{k}x} - v_{\mathbf{k}'x})^2 \delta(\epsilon_{\mathbf{k}}) \delta(\epsilon_{\mathbf{k}'}) \delta(\nu - \omega_{\mathbf{k}-\mathbf{k}'})}{\sum_{\mathbf{k}} 2v_{\mathbf{k}z}^2 \left(-\frac{\partial f}{\partial \epsilon_{\mathbf{k}}} \right)} = \alpha_{tr}^2 F(\nu)$$

where we defined

$$\frac{\pi |g(\mathbf{k}, \mathbf{k}')|^2}{2M\omega_{\mathbf{k}-\mathbf{k}'}} = |M(\mathbf{k}, \mathbf{k}')|^2$$

(importantly $g(\mathbf{k}, \mathbf{k}') \sim \text{constant}$ with units E/x for an optical mode and $g(\mathbf{k}, \mathbf{k}') \sim \alpha |\mathbf{k} - \mathbf{k}'|$ for an acoustic mode with $\alpha \sim \text{constant}$ with units of energy, note that the definition is a bit awkward since the $|g(\mathbf{k}, \mathbf{k}')|^2$ comes from the vertex of the interaction, while the rest comes from the free spectral function of the phonons) and approximated

$$\delta(\nu + \epsilon_{\mathbf{k}'}) \sim \delta(\epsilon_{\mathbf{k}'})$$

since we are expecting an integration range that is much smaller than the bandwidth of the electrons. We will come back to this calculation, for now, consider

$$\begin{aligned} \tau_{tr}^{-1} &= \frac{8\pi\lambda_A}{\hbar\beta} \int_0^{\omega_D} d\nu \frac{1}{\nu} \left(\frac{\nu}{\omega_D} \right)^4 \left[\frac{\beta\nu}{2} \text{csch}\left(\frac{\beta\nu}{2}\right) \right]^2 \\ &= \frac{8\pi\lambda_A}{\hbar\beta} \frac{16}{\beta^4} \frac{\beta}{2} \frac{2}{\beta} \int_0^{\omega_D} \frac{\beta}{2} d\nu \frac{2}{\beta\nu} \left(\frac{\beta\nu}{2\omega_D} \right)^4 \left[\frac{\beta\nu}{2} \text{csch}\left(\frac{\beta\nu}{2}\right) \right]^2 \\ &= \frac{8\pi\lambda_A}{\hbar\beta} \frac{16}{\beta^4} \int_0^{\omega_D} \frac{\beta}{2} d\nu \frac{\beta\nu}{2} \left(\frac{\beta\nu}{2\omega_D} \right)^4 \left[\text{csch}\left(\frac{\beta\nu}{2}\right) \right]^2 \\ &= \frac{8\pi\lambda_A}{\hbar\beta} \frac{16}{\beta^4} \int_0^{\frac{\beta\omega_D}{2}} dx x \left(\frac{x}{\omega_D} \right)^4 [\text{csch}(x)]^2 \\ &= \frac{8\pi\lambda_A}{\hbar\beta} \left(\frac{2}{\beta\omega_D} \right)^4 \int_0^{\frac{\beta\omega_D}{2}} dx x^5 [\text{csch}(x)]^2 \end{aligned}$$

We have that as $\beta\omega_D \rightarrow 0$

$$\left(\frac{2}{\beta\omega_D}\right)^4 \int_0^{\frac{\beta\omega_D}{2}} dx \, x^5 [\text{csch}(x)]^2 \rightarrow \frac{1}{4}$$

so overall we have

$$\tau_{tr}^{-1} \rightarrow \frac{2\pi\lambda_A}{\hbar\beta}$$

Spectral function for the phonons

we first use that for acoustic phonons we have

$$\begin{aligned} \frac{\pi |g(\mathbf{k}, \mathbf{k}')|^2}{2M\omega_{\mathbf{k}-\mathbf{k}'}} &= \frac{\pi g^2 |\mathbf{k} - \mathbf{k}'|^2}{2Mc |\mathbf{k} - \mathbf{k}'|} \\ &= \frac{\pi g^2 |\mathbf{k} - \mathbf{k}'|}{2Mc} \end{aligned}$$

$$\begin{aligned} \alpha_{tr}^2 F(\nu) &= \frac{\sum_{\mathbf{k}\mathbf{k}'} |M(\mathbf{k}, \mathbf{k}')|^2 (v_{\mathbf{k}x} - v_{\mathbf{k}'x})^2 \delta(\epsilon_{\mathbf{k}}) \delta(\epsilon_{\mathbf{k}'}) \delta(\nu - \omega_{\mathbf{k}-\mathbf{k}'})}{\sum_{\mathbf{k}} 2v_{\mathbf{k}z}^2 \left(-\frac{\partial f}{\partial \epsilon_{\mathbf{k}}}\right)} \\ &= |M(\mathbf{k}, \mathbf{k}')|^2 \end{aligned}$$

$\alpha^2 F$ function for 2d

$$\alpha_{tr}^2 F(\omega) = 2\lambda_A \left(\frac{\omega}{\omega_D}\right)^4 \theta(\omega - \omega_D)$$

we calculate the following quantity for a circular fermi surface

$$\frac{\sum_{\mathbf{k}\mathbf{k}'} |M(\mathbf{k}, \mathbf{k}')|^2 (v_{\mathbf{k}x} - v_{\mathbf{k}'x})^2 \delta(\epsilon_{\mathbf{k}}) \delta(\epsilon_{\mathbf{k}'}) \delta(\nu - \omega_{\mathbf{k}-\mathbf{k}'})}{\sum_{\mathbf{k}} 2v_{\mathbf{k}z}^2 \left(-\frac{\partial f}{\partial \epsilon_{\mathbf{k}}}\right)} = \alpha_{tr}^2 F(\nu)$$

where we define

$$\begin{aligned} \lambda_A &= 2 \int_0^\infty d\nu \frac{\alpha_{tr}^2 F(\nu)}{\nu} \\ &= \frac{\sum_{\mathbf{k}\mathbf{k}'} \frac{2}{\omega_{\mathbf{k}-\mathbf{k}'}} |M(\mathbf{k}, \mathbf{k}')|^2 (v_{\mathbf{k}x} - v_{\mathbf{k}'x})^2 \delta(\epsilon_{\mathbf{k}}) \delta(\epsilon_{\mathbf{k}'})}{\sum_{\mathbf{k}} 2v_{\mathbf{k}z}^2 \left(-\frac{\partial f}{\partial \epsilon_{\mathbf{k}}}\right)} \end{aligned}$$

for a circular fermi surface

$$\frac{1}{\nu_0} \sum_{\mathbf{k}\mathbf{k}'} |M(\mathbf{k}, \mathbf{k}')|^2 \frac{(v_{\mathbf{k}x} - v_{\mathbf{k}'x})^2}{2v_F^2} \delta(\epsilon_{\mathbf{k}}) \delta(\epsilon_{\mathbf{k}'}) \delta(\nu - \omega_{\mathbf{k}-\mathbf{k}'}) = \alpha_{tr}^2 F(\nu)$$

using

$$v_{\mathbf{k}x} - v_{\mathbf{k}'x} = \frac{1}{m} (\mathbf{k} - \mathbf{k}')_x$$

$$\frac{1}{\nu_0} \sum_{\mathbf{k}\mathbf{k}'} |M(\mathbf{k}, \mathbf{k}')|^2 \frac{(\mathbf{k} - \mathbf{k}')^2}{2v_F^2 m^2} \delta(\epsilon_{\mathbf{k}}) \delta(\epsilon_{\mathbf{k}'}) \delta(\nu - \omega_{\mathbf{k}-\mathbf{k}'}) = \alpha_{tr}^2 F(\nu)$$

in D we have

$$\frac{1}{Dm^2\nu_0} \sum_{\mathbf{k}\mathbf{k}'} |M(\mathbf{k}, \mathbf{k}')|^2 \frac{(\mathbf{k} - \mathbf{k}')^2}{2v_F^2} \delta(\epsilon_{\mathbf{k}}) \delta(\epsilon_{\mathbf{k}'}) \delta(\nu - \omega_{\mathbf{k}-\mathbf{k}'}) = \alpha_{tr}^2 F(\nu)$$

changing variables and using

$$|M(\mathbf{k}, \mathbf{k}')|^2 = \frac{\alpha^2 (\mathbf{k} - \mathbf{k}')^2}{M\omega_{\mathbf{k}-\mathbf{k}'}}$$

we have

$$\frac{1}{Dm^2\nu_0} \frac{\alpha^2}{Mc} \sum_{\mathbf{q}\mathbf{k}'} \frac{q^3}{2v_F^2} \delta(\epsilon_{\mathbf{q}-\mathbf{k}'}) \delta(\epsilon_{\mathbf{k}'}) \delta(\nu - cq) = \alpha_{tr}^2 F(\nu)$$

where we used $\omega_{\mathbf{q}} = cq$. Chanign variables

$$\alpha_{tr}^2 F(\nu) = \frac{1}{Dm^2\nu_0} \frac{\alpha^2}{Mc} \sum_{\mathbf{q}\mathbf{k}} \frac{q^3}{2v_F^2} \delta(\epsilon_{\mathbf{k}+\mathbf{q}}) \delta(\epsilon_{\mathbf{k}}) \delta(\nu - cq)$$

we can integrate as follows

$$\begin{aligned} \alpha_{tr}^2 F(\nu) &= \frac{1}{Dm^2\nu_0} \frac{\alpha^2}{Mc} \int \frac{d\mathbf{q}}{(2\pi)^D} \frac{d\mathbf{k}}{(2\pi)^D} \frac{q^3}{2v_F^2} \delta(\epsilon_{\mathbf{k}+\mathbf{q}}) \delta(\epsilon_{\mathbf{k}}) \delta(\nu - cq) \\ &= \frac{1}{Dm^2\nu_0} \frac{\alpha^2}{Mc} \int \frac{d\mathbf{q}}{(2\pi)^D} \frac{d\mathbf{k}}{(2\pi)^D} \frac{q^3}{2v_F^2} \delta(\epsilon_{\mathbf{k}+\mathbf{q}}) \delta\left(\frac{k^2 - k_F^2}{2m}\right) \delta(\nu - cq) \\ &= \frac{1}{Dm^2\nu_0} \frac{\alpha^2}{Mc} \int \frac{q^{D-1} dq d\Omega_q}{(2\pi)^D} \frac{k^{D-1} dk d\Omega_k}{(2\pi)^D} \frac{q^3}{2v_F^2} \delta(\epsilon_{\mathbf{k}+\mathbf{q}}) \delta\left(\frac{k^2 - k_F^2}{2m}\right) \delta(\nu - cq) \end{aligned}$$

Density of states in D dimensions

$$\begin{aligned} \nu_0 &= \frac{1}{\hbar^D} \int \frac{d^D \mathbf{k}}{(2\pi)^D} \delta(\epsilon_{\mathbf{k}}) = \frac{2m}{\hbar^D} \int \frac{k^{D-1} dk d\Omega_k}{(2\pi)^D} \delta(k^2 - k_F^2) \\ &= \frac{2m}{2\hbar^D k_F} \int \frac{k^{D-1} dk d\Omega_k}{(2\pi)^D} \delta(k - k_F) \\ &= \frac{mk_F^{D-2}}{\hbar^D} \int \frac{d\Omega_k}{(2\pi)^D} \end{aligned}$$

The solid angle in D dimensions is given by

$$\begin{aligned} \frac{1}{(2\pi)^D} \int d\Omega_k &= \frac{1}{(2\pi)^D} \frac{2\pi^{\frac{D}{2}}}{\Gamma(\frac{D}{2})} \\ &= \bar{S}_D \end{aligned}$$

$$\nu_0 = \frac{mk_F^{D-2}}{\hbar^D} \bar{S}_D$$

Related to the low density limit of a hopping model

$$\begin{aligned}\epsilon_{\mathbf{k}} &= -2t \sum_{i=1 \dots D} \cos k_i a - \mu \\ &\sim -2t \left(D - \sum_{i=1 \dots D} \frac{(k_i a)^2}{2} \right) - \mu \\ &= \frac{k^2}{2m_{eff}} - (\mu + 2tD)\end{aligned}$$

where

$$m_{eff} = \frac{1}{2ta^2}$$

as such, we have

$$k_F = \sqrt{2m_{eff} (\mu + 2tD)}$$

or

$$k_F = \sqrt{\frac{\mu + 2tD}{ta^2}}$$

$$\begin{aligned}\alpha_{tr}^2 F(\nu) &= \frac{1}{Dm^2 \nu_0} \frac{\alpha^2}{Mc} \int \frac{q^{D-1} dq d\Omega_q}{(2\pi)^D} \frac{k^{D-1} dk d\Omega_k}{(2\pi)^D} \frac{q^3}{2v_F^2} \delta \left(\frac{k^2 + q^2 - 2qk \cos \theta - k_F^2}{2m} \right) \delta \left(\frac{k^2 - k_F^2}{2m} \right) \delta(\nu - cq) \\ &= \frac{1}{Dm^2 \nu_0} \frac{\alpha^2}{Mc} \int \frac{q^{D-1} dq d\Omega_q}{(2\pi)^D} \frac{k^{D-1} dk d\Omega_k}{(2\pi)^D} \frac{q^3}{2v_F^2} \delta \left(\frac{k_F^2 + q^2 - 2qk \cos \theta - k_F^2}{2m} \right) \delta \left(\frac{k^2 - k_F^2}{2m} \right) \delta(\nu - cq) \\ &= \frac{1}{Dm^2} \frac{\alpha^2}{Mc} \int \frac{q^{D-1} dq d\Omega_q}{(2\pi)^D} \frac{q^3}{2v_F^2} \delta \left(\frac{k_F^2 + q^2 - 2qk_F \cos \theta - k_F^2}{2m} \right) \delta(\nu - cq)\end{aligned}$$

then we have

$$\alpha_{tr}^2 F(\nu) = \frac{1}{Dm^2} \frac{\alpha^2}{Mc} \int \frac{dq d\Omega_q}{(2\pi)^D} \frac{q^{D+2}}{2v_F^2} \delta \left(\frac{q^2 - 2qk_F \cos \theta}{2m} \right) \delta(\nu - cq)$$

Then we have

$$\alpha_{tr}^2 F(\nu) = \frac{1}{Dm^2} \frac{\alpha^2}{Mc} \left(\frac{2m}{2k_F} \right) \int_0^{k_F} \frac{dq d\Omega_q}{(2\pi)^D} \frac{q^{D+1}}{2v_F^2} \delta \left(\frac{q}{2k_F} - \cos \theta \right) \delta(\nu - cq)$$

which gives

$$\omega = cq$$

$$\begin{aligned}\alpha_{tr}^2 F(\nu) &= \frac{1}{Dm^2} \frac{\alpha^2}{Mc} \left(\frac{2m}{2k_F} \right) \int_0^{\omega_F} \frac{1}{c} \frac{d\omega d\Omega_q}{(2\pi)^D} \frac{1}{2v_F^2} \left(\frac{\omega}{c} \right)^{D+1} \delta \left(\frac{\omega}{2ck_F} - \cos \theta \right) \delta(\nu - \omega) \\ &= \frac{1}{Dm^2} \frac{\alpha^2}{Mc} \left(\frac{m}{2cv_F^2 k_F} \right) \left(\frac{\nu}{c} \right)^{D+1} \int_0^{\omega_F} \frac{d\omega d\Omega_q}{(2\pi)^D} \delta \left(\frac{\nu}{2ck_F} - \cos \theta \right) \delta(\nu - \omega)\end{aligned}$$

$$\alpha_{tr}^2 F(\nu) = \frac{1}{Dm^2} \frac{\alpha^2}{Mc} \left(\frac{m}{2cv_F^2 k_F} \right) \left(\frac{\nu}{c} \right)^{D+1} \int_0^{\omega_F} \frac{d\omega d\Omega_q}{(2\pi)^D} \delta \left(\frac{\nu}{2ck_F} - \cos \theta \right) \delta(\nu - \omega)$$

For the case $D \geq 3$

$$\begin{aligned} \frac{1}{(2\pi)^D} \int_0^{2\pi} \left[\int_0^\pi \delta \left(\frac{\nu}{2ck_F} - \cos \varphi_1 \right) \sin^{D-2}(\varphi_1) \sin^{D-3}(\varphi_2) \cdots \sin(\varphi_{D-2}) d\varphi_1 d\varphi_2 \cdots d\varphi_{D-2} \right] d\varphi_{D-1} \\ = \overline{S_{D-2}} \frac{1}{2\pi} \int d\varphi_1 \delta \left(\frac{\nu}{2ck_F} - \cos \varphi_1 \right) \sin^{D-2}(\varphi_1) \\ = \overline{S_{D-2}} \frac{1}{2\pi} \int \sin(\varphi_1) d\varphi_1 \delta \left(\frac{\nu}{2ck_F} - \cos \varphi_1 \right) \sin^{D-3}(\varphi_1) \\ = \overline{S_{D-2}} \frac{1}{2\pi} \int_{-1}^1 du \delta \left(\frac{\nu}{2ck_F} - u \right) (1-u^2)^{\frac{D-3}{2}} \\ \overline{S_{D-1}} \frac{1}{2\pi} \theta \left(1 - \left| \frac{\nu}{2ck_F} \right| \right) \left(1 - \left(\frac{\nu}{2ck_F} \right)^2 \right)^{\frac{D-3}{2}} \\ \overline{S_{D-1}} \frac{1}{2\pi} \theta(2ck_F - |\nu|) \left(1 - \left(\frac{\nu}{2ck_F} \right)^2 \right)^{\frac{D-3}{2}} \end{aligned}$$

which gives

$$\begin{aligned} \alpha_{tr}^2 F(\nu) &= \frac{1}{Dm^2} \frac{\alpha^2}{Mc} \left(\frac{m}{2cv_F^2 k_F} \right) \left(\frac{\nu}{c} \right)^{D+1} \left[\frac{\overline{S_{D-1}}}{2\pi} \theta(2ck_F - |\nu|) \right] \left(1 - \left(\frac{\nu}{2ck_F} \right)^2 \right)^{\frac{D-3}{2}} \\ &= \frac{1}{Dm^2} \frac{\alpha^2}{Mc} \left(\frac{m^3 k_F^{D-2}}{2c} \right) \left(\frac{\nu}{ck_F} \right)^{D+1} \left[\frac{\overline{S_{D-1}}}{2\pi} \theta(2ck_F - |\nu|) \right] \left(1 - \left(\frac{\nu}{2ck_F} \right)^2 \right)^{\frac{D-3}{2}} \\ &= \frac{1}{2cD} \frac{\alpha^2}{Mc} (mk_F^{D-2}) \left(\frac{\nu}{ck_F} \right)^{D+1} \left[\frac{\overline{S_{D-1}}}{2\pi} \theta(2ck_F - |\nu|) \right] \left(1 - \left(\frac{\nu}{2ck_F} \right)^2 \right)^{\frac{D-3}{2}} \\ &= \frac{2^D}{2\pi D} \frac{\alpha^2 \nu_0}{Mc^2} \frac{\overline{S_{D-1}}}{\overline{S_D}} \left(\frac{\nu}{2ck_F} \right)^{D+1} \theta(2ck_F - |\nu|) \left(1 - \left(\frac{\nu}{2ck_F} \right)^2 \right)^{\frac{D-3}{2}} \end{aligned}$$

if we had

$$\lambda_A = \frac{2^{D-1}}{2\pi D} \frac{\alpha^2 \nu_0}{Mc^2} \frac{\overline{S_{D-1}}}{\overline{S_D}}$$

where

$$[\alpha] = \frac{E}{qx} = E$$

we would get

$$\alpha_{tr}^2 F(\nu) = 2\lambda_A \left(\frac{\nu}{2ck_F} \right)^{D+1} \left(1 - \left(\frac{\nu}{2ck_F} \right)^2 \right)^{\frac{D-3}{2}} \theta(2ck_F - |\nu|)$$

note that this is dimensionless as it should to be consistent with all the previous results on the dimensionless coupling and the scattering rate.

also we have for $D = 2$

$$\begin{aligned}
\frac{1}{(2\pi)^2} \int_0^{2\pi} \delta\left(\frac{\nu}{2ck_F} - \cos \varphi_1\right) d\varphi_1 &= \frac{1}{(2\pi)^2} \int_0^\pi \delta\left(\frac{\nu}{2ck_F} - \cos \varphi_1\right) d\varphi_1 \\
&\quad + \frac{1}{(2\pi)^2} \int_{-\pi}^0 \delta\left(\frac{\nu}{2ck_F} - \cos \varphi_1\right) d\varphi_1 \\
&= \frac{1}{(2\pi)^2} \int_0^\pi \delta\left(\frac{\nu}{2ck_F} - \cos \varphi_1\right) d\varphi_1 \\
&\quad - \frac{1}{(2\pi)^2} \int_{-\pi}^0 \delta\left(\frac{\nu}{2ck_F} - \cos(-\varphi_1)\right) d(-\varphi_1) \\
&= \frac{1}{(2\pi)^2} \int_0^\pi \delta\left(\frac{\nu}{2ck_F} - \cos \varphi_1\right) d\varphi_1 \\
&\quad - \frac{1}{(2\pi)^2} \int_\pi^0 \delta\left(\frac{\nu}{2ck_F} - \cos \varphi_1\right) d\varphi_1 \\
&= \frac{2}{(2\pi)^2} \int_0^\pi \delta\left(\frac{\nu}{2ck_F} - \cos \varphi_1\right) d\varphi_1
\end{aligned}$$

Now we focus on

$$\begin{aligned}
\int_0^\pi \delta\left(\frac{\nu}{2ck_F} - \cos \varphi_1\right) d\varphi_1 &= \int_0^\pi \frac{\delta\left(\cos^{-1} \frac{\nu}{2ck_F} - \varphi_1\right)}{\sin\left(\cos^{-1} \frac{\nu}{2ck_F}\right)} d\varphi_1 \\
&= \frac{\theta(2ck_F - |\nu|)}{\sin\left(\cos^{-1} \frac{\nu}{2ck_F}\right)} \\
&= \frac{\theta(2ck_F - |\nu|)}{\sqrt{1 - \left(\frac{\nu}{2ck_F}\right)^2}}
\end{aligned}$$

So over all, we have

$$\begin{aligned}
\alpha_{tr}^2 F(\nu) &= \frac{1}{Dm^2} \frac{\alpha^2}{Mc} \left(\frac{m}{2cv_F^2 k_F}\right) \left(\frac{\nu}{c}\right)^3 \frac{2}{(2\pi)^2} \frac{\theta(2ck_F - |\nu|)}{\sqrt{1 - \left(\frac{\nu}{2ck_F}\right)^2}} \\
&= \frac{1}{2cD} \frac{\alpha^2}{Mc} (mk_F^{D-2}) \left(\frac{\nu}{ck_F}\right)^{D+1} \left[\frac{2}{(2\pi)^2} \frac{\theta(2ck_F - |\nu|)}{\sqrt{1 - \left(\frac{\nu}{2ck_F}\right)^2}} \right]
\end{aligned}$$

Using $D = 2$

$$\alpha_{tr}^2 F(\nu) = \frac{\alpha^2}{4Mc^2} (m) \left(\frac{\nu}{ck_F}\right)^3 \left[\frac{2}{(2\pi)^2} \frac{\theta(2ck_F - |\nu|)}{\sqrt{1 - \left(\frac{\nu}{2ck_F}\right)^2}} \right]$$

or

$$\alpha_{tr}^2 F(\nu) = \frac{4\alpha^2}{Mc^2} \left(\frac{m}{(2\pi)^2} \right) \left(\frac{\nu}{2ck_F} \right)^3 \left[\frac{\theta(2ck_F - |\nu|)}{\sqrt{1 - \left(\frac{\nu}{2ck_F} \right)^2}} \right]$$

since

$$\nu_0 = \frac{mk_F^{D-2}}{\hbar^D} \bar{S}_D$$

where

$$\begin{aligned} \bar{S}_D &= \frac{1}{(2\pi)^D} \frac{2\pi^{\frac{D}{2}}}{\Gamma\left(\frac{D}{2}\right)} \\ &= \frac{1}{(2\pi)^2} \frac{2\pi}{\Gamma(1)} \\ &= \frac{1}{2\pi} \end{aligned}$$

$$\alpha_{tr}^2 F(\nu) = \frac{4\alpha^2}{Mc^2} \frac{\nu_0}{2\pi} \left(\frac{\nu}{2ck_F} \right)^3 \left[\frac{\theta(2ck_F - |\nu|)}{\sqrt{1 - \left(\frac{\nu}{2ck_F} \right)^2}} \right]$$

if we had

$$\begin{aligned} \lambda_A &= \frac{2\alpha^2}{Mc^2} \frac{\nu_0}{2\pi} \\ &= \frac{1}{\pi} \frac{\alpha^2 \nu_0}{Mc^2} \end{aligned}$$

then

$$\alpha_{tr}^2 F(\nu) = 2\lambda_A \left(\frac{\nu}{2ck_F} \right)^3 \left[\frac{\theta(2ck_F - |\nu|)}{\sqrt{1 - \left(\frac{\nu}{2ck_F} \right)^2}} \right]$$

2d transport rate

we start from

$$\tau_{tr}^{-1} = \frac{4\pi}{\hbar\beta} \int_0^\infty d\nu \frac{\alpha_{tr}^2 F(\nu)}{\nu} \left[\frac{\beta\nu}{2} \text{csch}\left(\frac{\beta\nu}{2}\right) \right]^2$$

introducing the function we calculated above, we have

$$\begin{aligned}
\tau_{tr}^{-1} &= \frac{8\pi\lambda_A}{\hbar\beta} \int_0^{\omega_D} d\nu \frac{1}{\nu} \frac{\left(\frac{\nu}{\omega_D}\right)^3}{\sqrt{1 - \left(\frac{\nu}{\omega_D}\right)^2}} \left[\frac{\beta\nu}{2} \text{csch}\left(\frac{\beta\nu}{2}\right) \right]^2 \\
&= \frac{8\pi\lambda_A}{\hbar\beta} \frac{8}{\beta^3} \int_0^{\omega_D} \frac{\beta}{2} d\nu \frac{2}{\beta\nu} \frac{\left(\frac{\beta\nu}{2\omega_D}\right)^3}{\sqrt{1 - \left(\frac{2}{\beta}\right)^2 \left(\frac{\beta\nu}{2\omega_D}\right)^2}} \left[\frac{\beta\nu}{2} \text{csch}\left(\frac{\beta\nu}{2}\right) \right]^2 \\
&= \frac{8\pi\lambda_A}{\hbar\beta} \frac{8}{\beta^3} \int_0^{\omega_D} \frac{\beta}{2} d\nu \frac{\beta\nu}{2} \frac{\left(\frac{\beta\nu}{2\omega_D}\right)^3}{\sqrt{1 - \left(\frac{2}{\beta}\right)^2 \left(\frac{\beta\nu}{2\omega_D}\right)^2}} \left[\text{csch}\left(\frac{\beta\nu}{2}\right) \right]^2 \\
&= \frac{8\pi\lambda_A}{\hbar\beta} \frac{8}{\beta^3} \int_0^{\frac{\beta\omega_D}{2}} dx x \frac{\left(\frac{x}{\omega_D}\right)^3}{\sqrt{1 - \left(\frac{2}{\beta}\right)^2 \left(\frac{x}{\omega_D}\right)^2}} [\text{csch}(x)]^2 \\
&= \frac{8\pi\lambda_A}{\hbar\beta} \left(\frac{2}{\beta\omega_D}\right)^3 \int_0^{\frac{\beta\omega_D}{2}} dx \frac{x^4}{\sqrt{1 - \left(\frac{2}{\beta\omega_D}\right)^2 x^2}} [\text{csch}(x)]^2
\end{aligned}$$

We have that as $\beta\omega_D \rightarrow \infty$ we have

$$\tau_{tr}^{-1} = \frac{8\pi\lambda_A}{\hbar\beta} \left(\frac{2}{\beta\omega_D}\right)^3 \int_0^\infty dx x^4 [\text{csch}(x)]^2$$

so that

$$\tau_{tr}^{-1} \sim T^4$$

If we take the opposite limit, we have again

$$\frac{8\pi\lambda_A}{\hbar\beta} \left(\frac{2}{\beta\omega_D}\right)^3 \int_0^{\frac{\beta\omega_D}{2}} dx \frac{x^4}{\sqrt{1 - \left(\frac{2}{\beta\omega_D}\right)^2 x^2}} [\text{csch}(x)]^2$$

If we change variables

$$\frac{8\pi\lambda_A}{\hbar\beta} \left(\frac{\beta\omega_D}{2}\right)^2 \int_0^1 dx \frac{x^4}{\sqrt{1 - x^2}} \left[\text{csch}\left(\frac{\beta\omega_D}{2} x\right) \right]^2$$

in the limit where $\beta \rightarrow 0$ we have

$$\begin{aligned}
&\frac{8\pi\lambda_A}{\hbar\beta} \left(\frac{\beta\omega_D}{2}\right)^2 \int_0^1 dx \frac{x^4}{\sqrt{1 - x^2}} \frac{1}{\left[\frac{\beta\omega_D}{2} x\right]^2} \\
&= \frac{8\pi\lambda_A}{\hbar\beta} \int_0^1 dx \frac{x^2}{\sqrt{1 - x^2}}
\end{aligned}$$

with

$$\int_0^1 dx \frac{x^2}{\sqrt{1 - x^2}} = \frac{\pi}{4}$$

so overall this behaves as a constant at high temperature, giving rise to a T-linear behavior with

$$\tau_{tr}^{-1} \sim T$$

Acoustic phonon dimensionless coupling

First we verify

$$\lambda_A = \frac{2^{D-1}}{2\pi D} \frac{\alpha^2 \nu_0}{Mc^2} \frac{\overline{S_{D-1}}}{\overline{S_D}}$$

With

$$\frac{1}{(2\pi)^D} \frac{2\pi^{\frac{D}{2}}}{\Gamma\left(\frac{D}{2}\right)} = \bar{S}_D$$

so we have

$$\begin{aligned} \frac{\overline{S_{D-1}}}{\overline{S_D}} &= \frac{\frac{1}{(2\pi)^{D-1}} \frac{2\pi^{\frac{D-1}{2}}}{\Gamma\left(\frac{D-1}{2}\right)}}{\frac{1}{(2\pi)^D} \frac{2\pi^{\frac{D}{2}}}{\Gamma\left(\frac{D}{2}\right)}} \\ &= (2\pi) \frac{1}{\sqrt{\pi}} \frac{\Gamma\left(\frac{D}{2}\right)}{\Gamma\left(\frac{D-1}{2}\right)} \end{aligned}$$

For $D = 2$ we have

$$(2\pi) \frac{1}{\sqrt{\pi}} \frac{\Gamma(1)}{\Gamma\left(\frac{1}{2}\right)} = 4$$

so we have

$$\begin{aligned} \lambda_A &= \frac{2}{4\pi} \frac{\alpha^2 \nu_0}{Mc^2} 4 \\ &= \frac{2}{\pi} \frac{\alpha^2 \nu_0}{Mc^2} \end{aligned}$$

comparing, we found before

$$\lambda_A = \frac{1}{\pi} \frac{\alpha^2 \nu_0}{Mc^2}$$

the extra factor of 2 comes from the fact that the D -dim integration has the range $[0, \pi]$ while the 2-dim integration has the range $[0, 2\pi]$.

We calculate the coupling for 3d and then 2d explicitly to verify that we didn't miss anything

$$\lambda_A = \frac{\sum_{\mathbf{k}\mathbf{k}'} \frac{2}{\omega_{\mathbf{k}-\mathbf{k}'}} \left| M(\mathbf{k}, \mathbf{k}') \right|^2 (v_{\mathbf{k}x} - v_{\mathbf{k}'x})^2 \delta(\epsilon_{\mathbf{k}}) \delta(\epsilon_{\mathbf{k}'})}{\sum_{\mathbf{k}} 2v_{\mathbf{k}z}^2 \left(-\frac{\partial f}{\partial \epsilon_{\mathbf{k}}} \right)}$$

We have then

$$\lambda_A = \frac{\sum_{\mathbf{k}\mathbf{k}'} \frac{2}{\omega_{\mathbf{k}-\mathbf{k}'}} \frac{\alpha^2 (\mathbf{k}-\mathbf{k}')^2}{M\omega_{\mathbf{k}-\mathbf{k}'}} (v_{\mathbf{k}x} - v_{\mathbf{k}'x})^2 \delta(\epsilon_{\mathbf{k}}) \delta(\epsilon_{\mathbf{k}'})}{\sum_{\mathbf{k}} 2v_{\mathbf{k}z}^2 \left(-\frac{\partial f}{\partial \epsilon_{\mathbf{k}}} \right)}$$

Cancelling the twos

$$\lambda_A = \frac{\alpha^2}{Mc^2} \frac{\sum_{\mathbf{k}\mathbf{k}'} (v_{\mathbf{k}x} - v_{\mathbf{k}'x})^2 \delta(\epsilon_{\mathbf{k}}) \delta(\epsilon_{\mathbf{k}'})}{\sum_{\mathbf{k}} v_{\mathbf{k}z}^2 \left(-\frac{\partial f}{\partial \epsilon_{\mathbf{k}}} \right)}$$

then we have

$$\lambda_A = \frac{1}{D} \frac{\alpha^2}{Mc^2} \frac{1}{v_F^2 m^2} \frac{1}{\nu_0} \sum_{\mathbf{k}, \mathbf{q}} q^2 \delta(\epsilon_{\mathbf{k}}) \delta(\epsilon_{\mathbf{k}+\mathbf{q}})$$

converting to integrals

$$\lambda_A = \frac{1}{D} \frac{\alpha^2}{Mc^2} \frac{1}{v_F^2 m^2} \frac{1}{\nu_0} \int \frac{q^{D-1} dq d\Omega_q}{(2\pi)^D} \frac{k^{D-1} dk d\Omega_k}{(2\pi)^D} q^2 \delta(\epsilon_{\mathbf{k}}) \delta(\epsilon_{\mathbf{k}+\mathbf{q}})$$

Then we have

$$\lambda_A = \frac{1}{D} \frac{\alpha^2}{Mc^2} \frac{1}{v_F^2 m^2} \int \frac{q^{D+1} dq d\Omega_q}{(2\pi)^D} \delta(\epsilon_{\mathbf{k}+\mathbf{q}})$$

then, expanding the delta function

$$\lambda_A = \frac{1}{D} \frac{\alpha^2}{Mc^2} \frac{1}{v_F^2 m^2} \int \frac{q^{D+1} dq d\Omega_q}{(2\pi)^D} \delta\left(\frac{k_F^2 + q^2 - 2qk \cos \theta - k_F^2}{2m}\right)$$

simplifying a bit

$$\lambda_A = \frac{1}{D} \frac{\alpha^2}{Mc^2} \frac{1}{v_F^2 m^2} \frac{2m}{2k_F} \int q^D \frac{dq d\Omega_q}{(2\pi)^D} \delta\left(\frac{q}{2k_F} - \cos \theta\right)$$

Performing the angular integration in $D > 2$, we have

$$\lambda_A = \frac{1}{D} \frac{\alpha^2}{Mc^2} \frac{1}{v_F^2 m^2} \frac{2m}{2k_F} \frac{\bar{S}_{D-1}}{2\pi} \int q^D dq \theta \left(1 - \frac{q}{2k_F}\right) \left(1 - \left(\frac{q}{2k_F}\right)^2\right)^{\frac{D-3}{2}}$$

Then changing variables we have

$$\lambda_A = \frac{1}{D} \frac{\alpha^2}{Mc^2} \frac{1}{v_F^2 m^2} \frac{2m}{2k_F} \frac{\bar{S}_{D-1}}{2\pi} \frac{(2k_F)^{D+1}}{2\pi} \int_0^1 dx x^D (1-x^2)^{\frac{D-3}{2}}$$

Simplifying a bit

$$\lambda_A = \frac{1}{D} \frac{\alpha^2}{Mc^2} m \frac{1}{k_F^3} \frac{\bar{S}_{D-1}}{2\pi} \frac{(2k_F)^{D+1}}{2\pi} \int_0^1 dx x^D (1-x^2)^{\frac{D-3}{2}}$$

Then we have

$$\lambda_A = \frac{1}{D} 2^D \frac{\alpha^2}{Mc^2} \frac{\bar{S}_{D-1}}{2\pi} \frac{mk_F^{D-2}}{2\pi} \int_0^1 dx x^D (1-x^2)^{\frac{D-3}{2}}$$

Since we have

$$\nu_0 = \frac{mk_F^{D-2}}{\hbar^D} \bar{S}_D$$

the dimensionless coupling

$$\lambda_A = \frac{1}{D} 2^D \frac{\alpha^2}{Mc^2} \frac{\overline{S_{D-1}}}{\overline{S_D}} \frac{\nu_0}{2\pi} \int_0^1 dx x^D (1-x^2)^{\frac{D-3}{2}}$$

Performing the integral, we have

$$\lambda_A = \frac{1}{D} 2^D \frac{\alpha^2}{Mc^2} \frac{\overline{S_{D-1}}}{\overline{S_D}} \frac{\nu_0}{2\pi} \frac{\sqrt{\pi} 2^{-D} \Gamma\left(\frac{D-1}{2}\right)}{\Gamma\left(\frac{D}{2}\right)}$$

which yields

$$\lambda_A = \frac{1}{D} 2^D \frac{\alpha^2}{Mc^2} \frac{\overline{S_{D-1}}}{\overline{S_D}} \frac{\nu_0}{2\pi} \frac{\sqrt{\pi} 2^{-D} \Gamma\left(\frac{D-1}{2}\right)}{\Gamma\left(\frac{D}{2}\right)}$$

using

$$\frac{1}{(2\pi)} \frac{\overline{S_{D-1}}}{\overline{S_D}} = \frac{1}{\sqrt{\pi}} \frac{\Gamma\left(\frac{D}{2}\right)}{\Gamma\left(\frac{D-1}{2}\right)}$$

we have

$$\lambda_A = \frac{1}{D} \frac{\alpha^2}{Mc^2} \nu_0$$

Redefinitions after the calculation

we had the phonon spectral function given by

$$\alpha_{tr}^2 F(\nu) = 2\lambda'_A \left(\frac{\nu}{2ck_F}\right)^{D+1} \left(1 - \left(\frac{\nu}{2ck_F}\right)^2\right)^{\frac{D-3}{2}} \theta(2ck_F - |\nu|)$$

with

$$\lambda'_A = \frac{2^{D-1}}{2\pi D} \frac{\alpha^2 \nu_0}{Mc^2} \frac{\overline{S_{D-1}}}{\overline{S_D}}$$

Howeverm we now know that

$$\lambda'_A = \frac{2^{D-1}}{2\pi} \frac{\overline{S_{D-1}}}{\overline{S_D}} \lambda_A$$

and

$$\frac{1}{(2\pi)} \frac{\overline{S_{D-1}}}{\overline{S_D}} = \frac{1}{\sqrt{\pi}} \frac{\Gamma\left(\frac{D}{2}\right)}{\Gamma\left(\frac{D-1}{2}\right)}$$

so that

$$\lambda'_A = 2^{D-1} \frac{1}{\sqrt{\pi}} \frac{\Gamma\left(\frac{D}{2}\right)}{\Gamma\left(\frac{D-1}{2}\right)} \lambda_A$$

For $D = 3$ we have

$$\begin{aligned}\lambda'_A &= 4 \frac{1}{\sqrt{\pi}} \frac{\Gamma\left(\frac{3}{2}\right)}{\Gamma\left(\frac{2}{2}\right)} \lambda_A \\ &= 2\lambda_A\end{aligned}$$

with

$$\Gamma\left(\frac{3}{2}\right) = \frac{\sqrt{\pi}}{2}$$

So with the result from the previous section:

$$\lambda_A = \frac{1}{3} \frac{\alpha^2}{Mc^2} \nu_0$$

we have

$$\alpha_{tr}^2 F(\nu) = \frac{4}{3} \frac{\alpha^2}{Mc^2} \nu_0 \left(\frac{\nu}{2ck_F} \right)^4 \theta(2ck_F - |\nu|)$$

and if someone called

$$\frac{2}{3} \frac{\alpha^2}{Mc^2} \nu_0$$

the dimensionless coupling, the definitions would agree.

Explicit calculation for 2D

we start form

$$\lambda_A = \frac{1}{D} \frac{\alpha^2}{Mc^2} \frac{1}{v_F^2 m^2} \frac{2m}{2k_F} \int q^D \frac{dq d\Omega_q}{(2\pi)^D} \delta\left(\frac{q}{2k_F} - \cos\theta\right)$$

we have

$$\lambda_A = \frac{1}{D} \frac{\alpha^2}{Mc^2} \frac{1}{k_F^2} \frac{m}{k_F} \int q^2 \frac{dq d\theta}{(2\pi)^2} \delta\left(\frac{q}{2k_F} - \cos\theta\right)$$

using

$$\int_0^\pi \delta\left(\frac{q}{2k_F} - \cos\varphi_1\right) d\varphi_1 = \frac{\theta\left(1 - \frac{q}{2k_F}\right)}{\sqrt{1 - \left(\frac{q}{2k_F}\right)^2}}$$

we have

$$\lambda_A = \frac{1}{2} \frac{\alpha^2}{Mc^2} \frac{1}{k_F^2} \frac{2m}{k_F} \int q^2 \frac{dq}{(2\pi)^2} \frac{\theta\left(1 - \frac{q}{2k_F}\right)}{\sqrt{1 - \left(\frac{q}{2k_F}\right)^2}}$$

now we perform the momentum integration

$$\int q^2 \frac{dq}{(2\pi)^2} \frac{\theta\left(1 - \frac{q}{2k_F}\right)}{\sqrt{1 - \left(\frac{q}{2k_F}\right)^2}} = (2k_F)^3 \int_0^1 dx \frac{x^2}{\sqrt{1 - x^2}}$$

using our previous result

$$\int_0^1 dx \frac{x^2}{\sqrt{1 - x^2}} = \frac{\pi}{4}$$

we then obtain

$$\lambda_A = \frac{1}{2} \frac{\alpha^2}{Mc^2} \frac{1}{k_F^2} \frac{2m}{k_F} (2k_F)^3 \frac{\pi}{4} \frac{1}{(2\pi)^2}$$

Overall

$$\lambda_A = 2 \frac{\alpha^2}{Mc^2} m \pi \frac{1}{(2\pi)^2}$$

converting to the density of states in 2D

$$\lambda_A = \frac{\alpha^2 \nu_0}{Mc^2}$$

Redefinitions

we used to have for 2D

$$\lambda_A = \frac{2\alpha^2}{Mc^2} \frac{\nu_0}{2\pi}$$

then

$$\alpha_{tr}^2 F(\nu) = 2\lambda'_A \left(\frac{\nu}{2ck_F}\right)^3 \left[\frac{\theta(2ck_F - |\nu|)}{\sqrt{1 - \left(\frac{\nu}{2ck_F}\right)^2}} \right]$$

then we have

$$\begin{aligned} \lambda'_A &= \frac{2\alpha^2}{Mc^2} \frac{\nu_0}{2\pi} \\ &= \frac{2}{2\pi} \lambda_A \\ &= \frac{1}{\pi} \lambda_A \end{aligned}$$

To me, it seems natural to define the dimensionless coupling as

$$\lambda_A = \frac{2}{D} \frac{\alpha^2 \nu_0}{Mc^2}$$

Wich would entail that for 2D, the spectral function would actually be

$$\alpha_{tr}^2 F(\nu) = \frac{2}{\pi} \lambda_A \left(\frac{\nu}{2ck_F} \right)^3 \left[\frac{\theta(2ck_F - |\nu|)}{\sqrt{1 - \left(\frac{\nu}{2ck_F} \right)^2}} \right]$$

Now, since the higher dimensional analogue was

$$\begin{aligned} \lambda'_A &= \frac{2^{D-1}}{2\pi D} \frac{\alpha^2 \nu_0}{Mc^2} \frac{\overline{S_{D-1}}}{\overline{S_D}} \\ &= \frac{2^{D-2}}{2\pi} \frac{\overline{S_{D-1}}}{\overline{S_D}} \lambda_A \\ &= \frac{2^{D-2}}{2\pi} (2\pi) \frac{1}{\sqrt{\pi}} \frac{\Gamma(\frac{D}{2})}{\Gamma(\frac{D-1}{2})} \lambda_A \\ &= \frac{2^{D-2}}{\sqrt{\pi}} \frac{\Gamma(\frac{D}{2})}{\Gamma(\frac{D-1}{2})} \lambda_A \end{aligned}$$

with

$$\alpha_{tr}^2 F(\nu) = 2\lambda_A \frac{2^{D-2}}{\sqrt{\pi}} \frac{\Gamma(\frac{D}{2})}{\Gamma(\frac{D-1}{2})} \left(\frac{\nu}{2ck_F} \right)^{D+1} \left(1 - \left(\frac{\nu}{2ck_F} \right)^2 \right)^{\frac{D-3}{2}} \theta(2ck_F - |\nu|)$$

This would be the more general version of this spectral function in any dimension. Again, the particular cases of interest are $D = 2$

$$\alpha_{tr}^2 F(\nu) = \frac{2}{\pi} \lambda_A \left(\frac{\nu}{2ck_F} \right)^3 \left[\frac{\theta(2ck_F - |\nu|)}{\sqrt{1 - \left(\frac{\nu}{2ck_F} \right)^2}} \right]$$

and $D = 3$

$$\alpha_{tr}^2 F(\nu) = 2\lambda_A \left(\frac{\nu}{2ck_F} \right)^4 \theta(2ck_F - |\nu|).$$

Revised transport rate in 2D

With the new extra factor of $1/\pi$ the transport rate will be given by

$$\tau_{tr}^{-1} = \frac{8\pi\lambda_A}{\hbar\beta} \left(\frac{2}{\beta\omega_D} \right)^3 \frac{1}{\pi} \int_0^{\frac{\beta\omega_D}{2}} dx \frac{x^4}{\sqrt{1 - \left(\frac{2}{\beta\omega_D} \right)^2 x^2}} [\text{csch}(x)]^2$$

In the high temperature limit we have

$$\begin{aligned}
\tau_{tr}^{-1} &= \frac{8\pi\lambda_A}{\hbar\beta} \left(\frac{2}{\beta\omega_D} \right)^{-2} \frac{1}{\pi} \int_0^1 dx \frac{x^4}{\sqrt{1-x^2}} \left[\operatorname{csch} \left(\frac{\beta\omega_D}{2} x \right) \right]^2 \\
&\sim \frac{8\pi\lambda_A}{\hbar\beta} \left(\frac{2}{\beta\omega_D} \right)^{-2} \frac{1}{\pi} \int_0^1 dx \frac{x^4}{\sqrt{1-x^2}} \left[\left(\frac{\beta\omega_D}{2} x \right) \right]^{-2} \\
&= \frac{8\pi\lambda_A}{\hbar\beta} \frac{1}{\pi} \int_0^1 dx \frac{x^2}{\sqrt{1-x^2}} \\
&= \frac{8\pi\lambda_A}{\hbar\beta} \frac{1}{\pi} \frac{\pi}{4} \\
&= \frac{2\pi\lambda_A}{\hbar\beta}.
\end{aligned}$$

similar to the 3D case. In the low temperature limit, we have

$$\begin{aligned}
\tau_{tr}^{-1} &= \frac{8\pi\lambda_A}{\hbar\beta} \left(\frac{2}{\beta\omega_D} \right)^3 \frac{1}{\pi} \int_0^\infty dx x^4 [\operatorname{csch}(x)]^2 \\
&\sim T^4
\end{aligned}$$

Making the integral better behaved

We

start

with

$$\frac{1}{\pi} \int_0^1 dx \frac{x^4}{\sqrt{1-x^2}} \left[\operatorname{csch} \left(\frac{\beta\omega_D}{2} x \right) \right]^2$$

with the change of variables

$$x = \sin \theta$$

$$dx = \cos \theta d\theta$$

we get

$$\frac{1}{\pi} \int_0^{\pi/2} \cos \theta d\theta \frac{\sin^4 \theta}{\cos \theta} \left[\operatorname{csch} \left(\frac{\beta\omega_D}{2} \sin \theta \right) \right]^2$$

cancelling stuff

$$\frac{1}{\pi} \int_0^{\pi/2} d\theta \sin^4 \theta \left[\operatorname{csch} \left(\frac{\beta\omega_D}{2} \sin \theta \right) \right]^2$$